

Complete Master Compendium

Controller-Aware Spectral Graph Theory for Damping Margins in IBR Grids

Internal research consolidation

Compiled June 9, 2026

Abstract

This compendium is intentionally redundant. It preserves the full IEEE manuscript, the unified theory note with the Gemini extensions, the diagrams, and the validation reports. This front matter explains how the final merged PDF is organized and appends the raw gate ledgers. The source papers themselves are merged after this front matter as complete PDFs, so their equations and figures are not rewritten or dropped.

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How to Read This Compendium

The final merged artifact has four layers:

1. this front matter and artifact appendix;
2. the unified theory note with calibrated Gemini additions;
3. the full IEEE Transactions manuscript with all equations, tables, figures, derivation appendices, gates, and reviewer checklists;
4. raw validation reports and compact CSV ledgers.

The document is deliberately not concise. Its role is to preserve the entire research state before splitting the material into submission-sized papers.

Recovered Content Manifest

Source	Preserved content
<code>internal_unified_theory/main.pdf</code>	Gemini extensions after correction: protected resolvent sensitivity, flow-feasible retrofitting candidate, damping-operator hierarchy, and explicit novelty boundaries.
<code>paper_ieee_transactions/main.pdf</code>	Full prior manuscript: all equations, the second-order correction, rational graph-filter identity, joint inertia–damping perturbation, ANDES IEEE39 sections, NEP bridge, Braess mechanism, hidden margin, weak nodes/links, planning scores, line reinforcement, appendices, and all manuscript figures.
<code>outputs/phase0d_nep_contour_solver</code>	Non-circular Beyn/NEP bridge validation against ANDES, including Mix60 control-family critical mode at 13.742 Hz.
<code>outputs/phase_braess_nep</code>	Resonant Braess mechanism test and classification: strict resonant minority cases plus negative controls.
<code>outputs/phase_inertia_bridge_gate</code>	Gate for inertia sensitivity over the controller-aware bridge.
<code>outputs/phase_coordination_lever_gate</code>	Reduced-model gate for coordinated topology/damping/inertia packages and the absence of robust nonlinear synergy.
<code>outputs/phase_gamma_robustness_gate</code>	Robustness of the protected-resolvent discrepancy under physical and random diagonal protection profiles.
<code>outputs/final_claim_audit</code>	Final claim-readiness ledger and status labels.

1 Explicit Gemini Addendum: Flow-Feasible Graph-Filter Shaping

The newest Gemini proposal is worth preserving because it points to a clean conference-paper split:

Flow-Feasible Graph-Filter Shaping: Coordinating Topology, Damping, and Virtual Inertia in Non-Normal IBR Grids.

The proposal is included here, but with the same claim discipline used throughout the project. The useful idea is not that every formula in the raw proposal is already a theorem. The useful idea is that the local protected resolvent provides a common currency for topology, damping, and virtual inertia when the state matrix is non-normal.

1.1 Gemini G1: Bi-Orthogonal Resolvent Sensitivity

Let

$$H(j\omega) = C(j\omega I - A_{\text{sys}})^{-1}B, \quad g_{\text{prot}} = \sigma_1(H(j\omega^*)),$$

where ω^* is the protected-resonance peak. Let u^* and v^* be the left and right singular vectors of $H(j\omega^*)$ associated with g_{prot} . Define

$$\phi^{\text{in}} = (j\omega^* I - A_{\text{sys}})^{-1}Bv^*, \quad \phi^{\text{out}} = (j\omega^* I - A_{\text{sys}})^{-H}C^H u^*.$$

For a smooth real parameter p in A_{sys} , the first-order sensitivity of the protected log-gain $\mathcal{S} = \log g_{\text{prot}}$ is

$$\boxed{\frac{\partial \mathcal{S}}{\partial p} = \frac{1}{g_{\text{prot}}} \text{Re} \left[(\phi^{\text{out}})^H \frac{\partial A_{\text{sys}}}{\partial p} \phi^{\text{in}} \right]}.$$

Status. This is standard singular-value perturbation applied to the power-system resolvent. The contribution is the use of this expression as a planning exchange rate across grid levers in an IBR damping-margin setting, not the invention of singular-value perturbation.

1.2 Gemini G2: Spatial Decoupling, Corrected

The raw Gemini note states a strong damping-placement rule of the form $\partial \mathcal{S} / \partial D_i \propto -|\phi_i^{\text{out}}|^2$. That rule is *not* generally valid for non-normal systems. In the general case the correct local damping sensitivity is a bi-orthogonal product. For a second-order state $x = [\theta^T, \omega^T]^T$ with local damping entering the ω_i equation as $-\Delta D_i / M_i$, one obtains

$$\boxed{\frac{\partial \mathcal{S}}{\partial D_i} = -\frac{1}{g_{\text{prot}} M_i} \text{Re} \left[\overline{\phi_{\omega_i}^{\text{out}}} \phi_{\omega_i}^{\text{in}} \right]}.$$

The simplified $-|\phi_i|^2$ expression is recovered only in aligned or nearly normal cases where input and output spatial profiles coincide. The actual non-normal insight is stronger:

the location that receives forcing and the location where response should be damped need not be the same bus.

For a line $e = (i, j)$ with incidence vector $b_e = e_i - e_j$ and stiffness change w_e , the corresponding topology sensitivity has the form

$$\boxed{\frac{\partial \mathcal{S}}{\partial w_e} = -\frac{1}{g_{\text{prot}}} \text{Re} \left[(\phi_{\omega}^{\text{out}})^H M^{-1} b_e b_e^T \phi_{\theta}^{\text{in}} \right]},$$

up to the precise state ordering and sign convention used in the assembled A_{sys} . This is the graph-filter version of a weak-link score: a line is dangerous or useful according to how it couples the forcing profile to the response profile, not only according to Fiedler participation or stiffness gain.

Status. The certificate-discrepancy gates support that the modal damping margin S_ζ and protected resolvent S_g disagree robustly under several protection profiles. The mechanism of the discrepancy is not fully explained: global non-normality and global commutator descriptors were weak predictors. The spatial-decoupling theorem is therefore a strong candidate, not yet a closed validated theorem over ANDES.

1.3 Gemini G3: Flow-Feasible Dynamic Braess

A line reinforcement is a *flow-feasible dynamic Braess* candidate if, at a given operating point, it satisfies the three local conditions

$$\frac{\partial \nu_c}{\partial w_e} > 0, \quad \frac{\partial \mathcal{S}}{\partial w_e} > 0, \quad |f_0 + \Delta f(w_e, P)| \leq \bar{f}.$$

The first condition says that the line improves a frequency or stiffness proxy. The second says that the protected dynamic gain worsens. The third prevents a purely dynamic claim from ignoring steady-state flow feasibility.

The local retrofit problem suggested by Gemini can be written as

$$\begin{aligned} \min_{z=(\Delta w, \Delta D, \Delta M)} \quad & c^T z \\ \text{s.t.} \quad & \mathcal{S}_0 + \nabla \mathcal{S}^T z \leq \mathcal{S}_{\text{target}}, \\ & F_0 + J_F z \leq \bar{F}, \\ & z_{\min} \leq z \leq z_{\max}. \end{aligned}$$

This is not yet a global optimality theorem. It is a local planning layer that can be tested against full NEP or ANDES eigensolves after a candidate package is selected.

1.4 Algorithm G1: Flow-Feasible Resolvent Retrofitting

1. Build or import A_{sys} , protected input/output matrices B, C , flow limits $\bar{F}$, and action costs c .
2. Find ω^* , g_{prot} , and singular vectors u^*, v^* of $H(j\omega)$.
3. Compute ϕ^{in} and ϕ^{out} from the two resolvent solves.
4. Generate exchange rates $\partial \mathcal{S} / \partial w_e$, $\partial \mathcal{S} / \partial D_i$, and $\partial \mathcal{S} / \partial M_i$.
5. Compute flow penalties from J_F or from an AC/DC sensitivity model.
6. Solve the local retrofit LP for a sparse package $z^* = (\Delta w^*, \Delta D^*, \Delta M^*)$.
7. Validate the selected package against the NEP/ANDES poles. If the critical singular vectors or pole family switch, rerun the local model.

1.5 Figures Needed for This Spin-Off Paper

The Gemini note proposes three figures. They should be generated only after the corresponding gates are passed.

1. **Spatial decoupling heatmap:** forcing bus/location versus damping placement bus/location from ϕ^{in} and ϕ^{out} .

2. **Flow-feasible Braess witness:** a line that improves the stiffness proxy, keeps flows legal, but worsens the protected resolvent or damping margin.
3. **Cost-synergy frontier:** topology-only, damping/inertia-only, and coordinated packages. Existing gates support $w + D$ and $w + M$ as reduced-model coordination candidates, but do not support a strong nonlinear synergy claim.

1.6 Claim Boundary for the Gemini Direction

This direction is promising because it converts weak nodes and weak links from single-score graph objects into input-output objects. The claim should be:

Protected-resolvent sensitivities define a common local exchange rate for topology, damping, and inertia actions in non-normal IBR grids, and early reduced-model gates show robust disagreement with modal damping-margin rankings.

It should *not* yet be:

The optimal damping bus is always the output hot spot, or coordinated packages are always synergistic, or the LP is globally optimal.

2 Figure and Diagram Recovery Appendix

2.1 Manuscript Figures

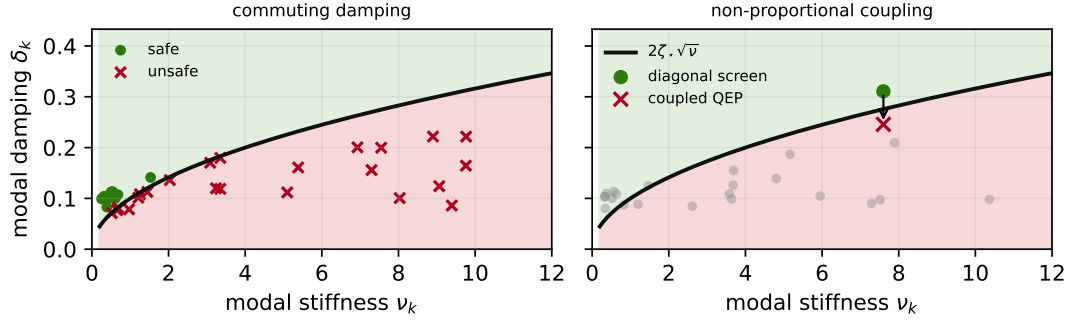


Figure 1: Damping-region geometry from the full manuscript.

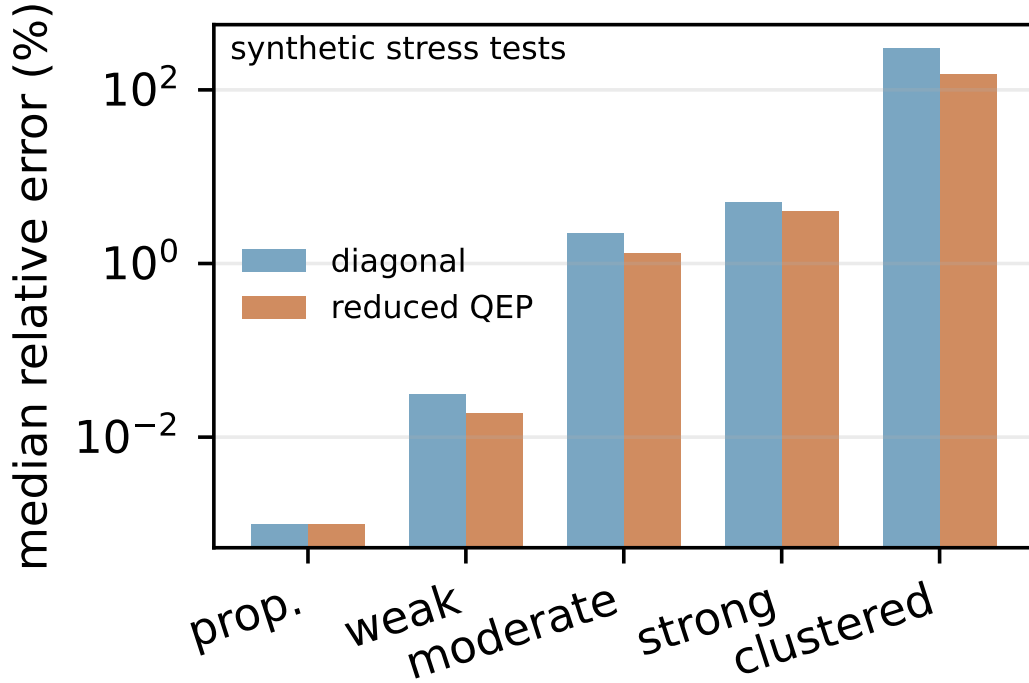


Figure 2: Estimator error stress test from the full manuscript.

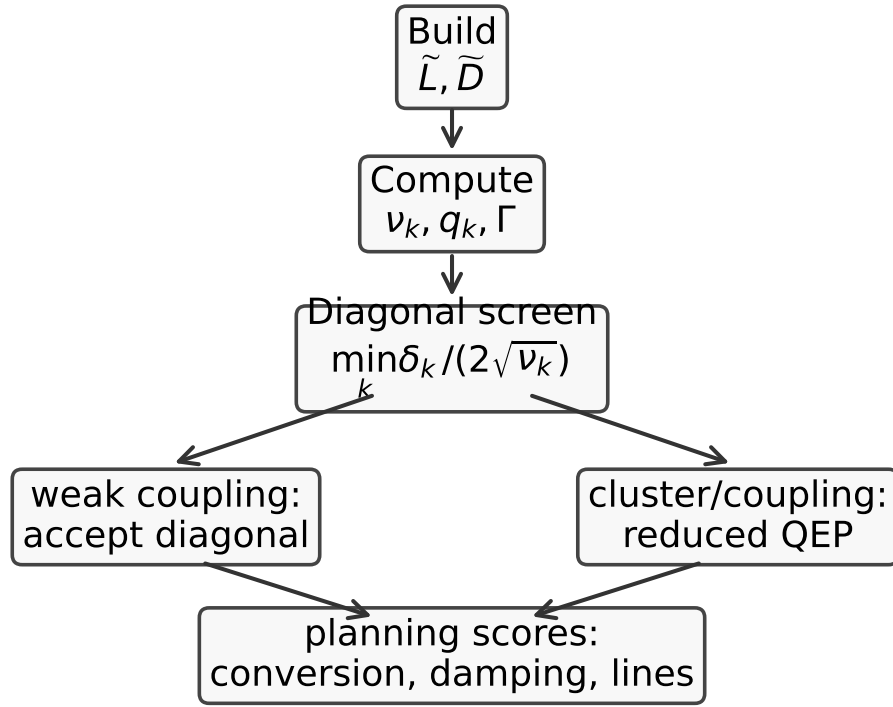


Figure 3: Adaptive workflow: diagonal screen, correction, and fallback.

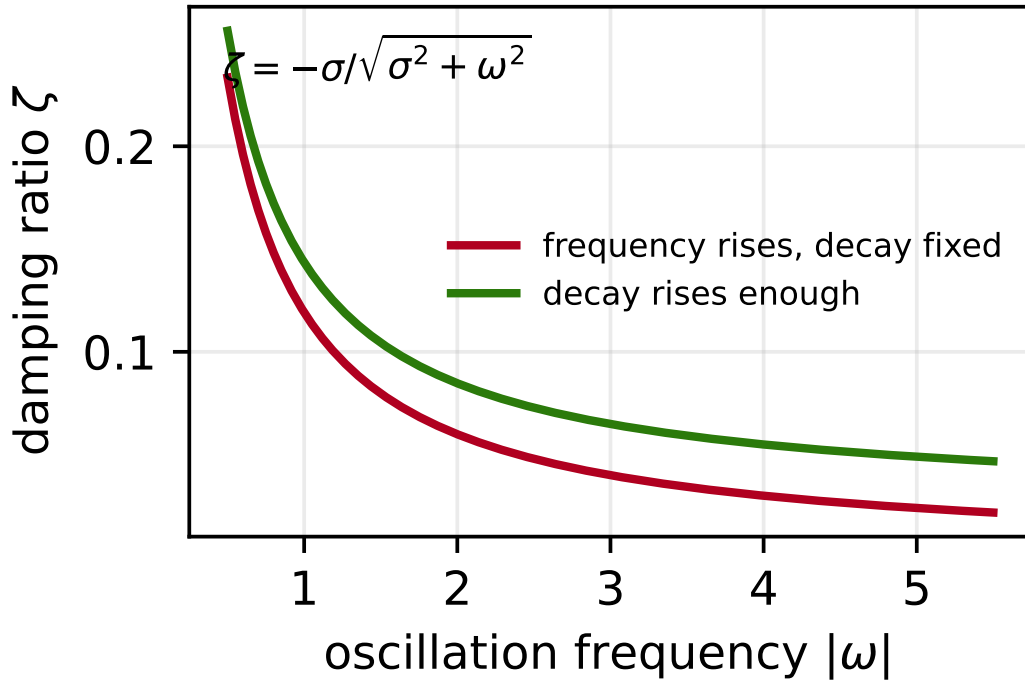


Figure 4: Frequency-only versus damping-aware line reinforcement tradeoff.

IEEE 39-bus Mix60 SG-to-IBR replacement map

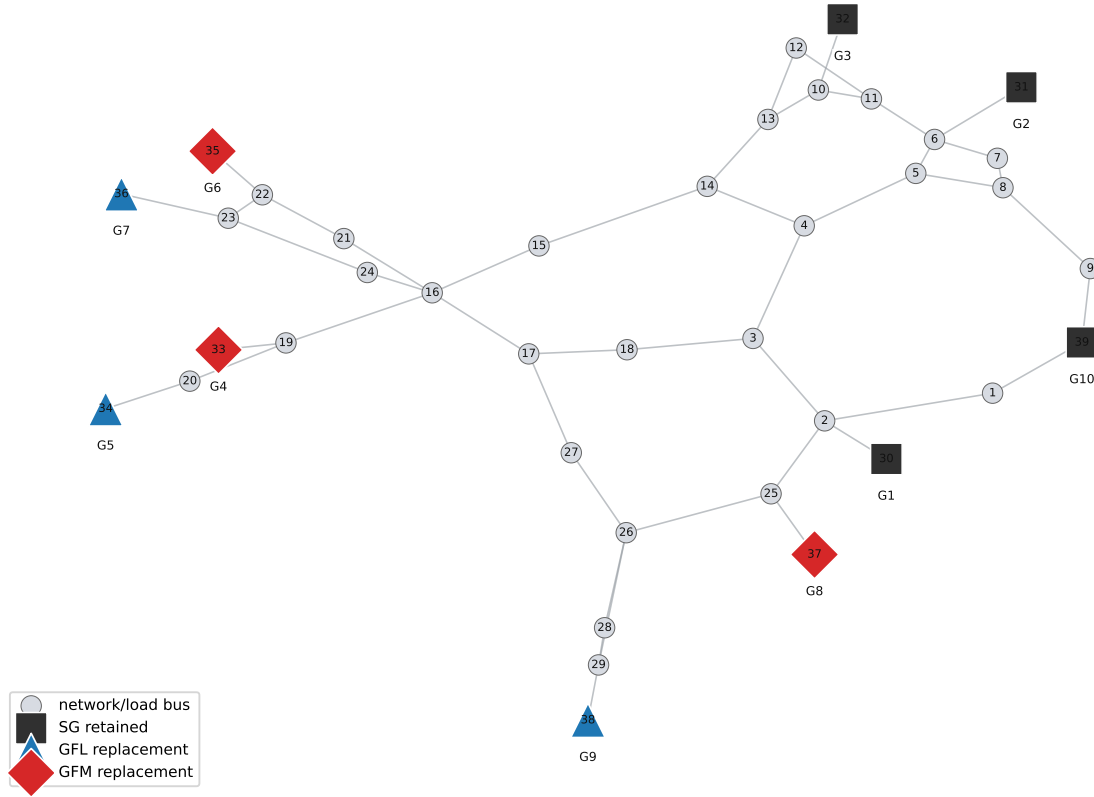


Figure 5: IEEE39 SG-to-IBR replacement map.

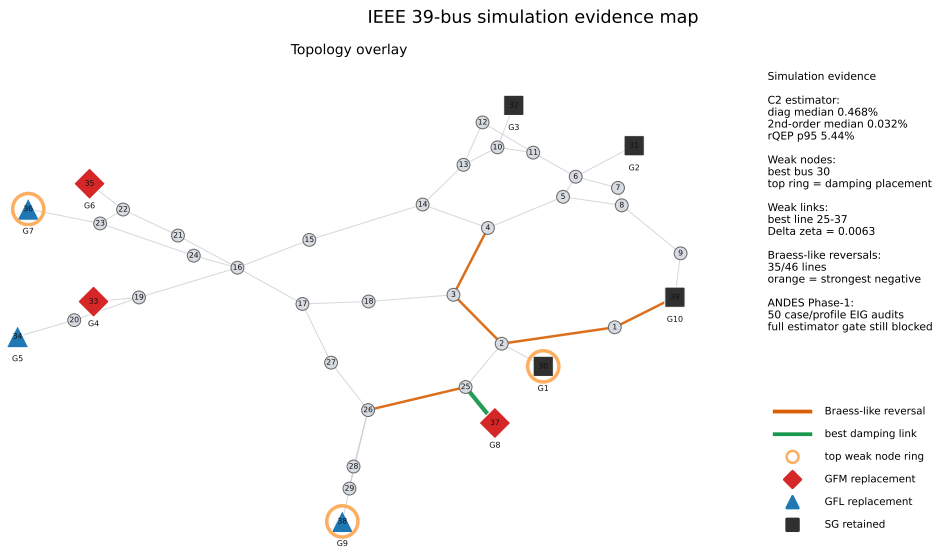


Figure 6: IEEE39 result map and validation status overview.

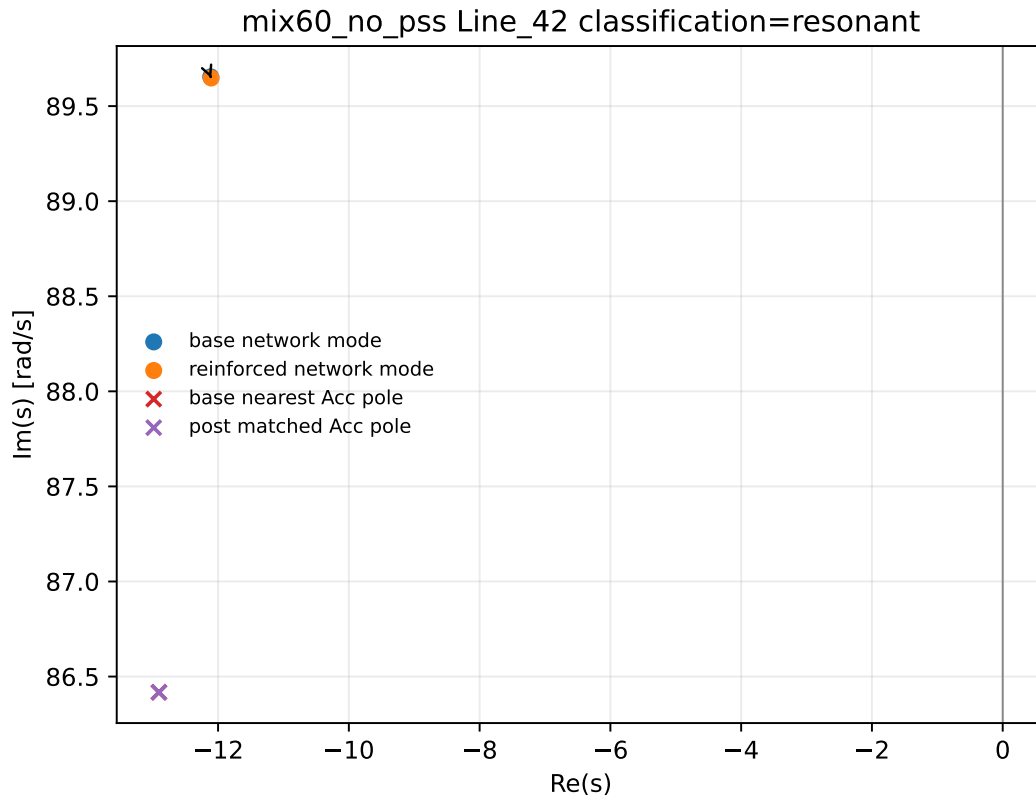


Figure 7: Control-resonant Braess example in the complex plane.

2.2 Output Figures

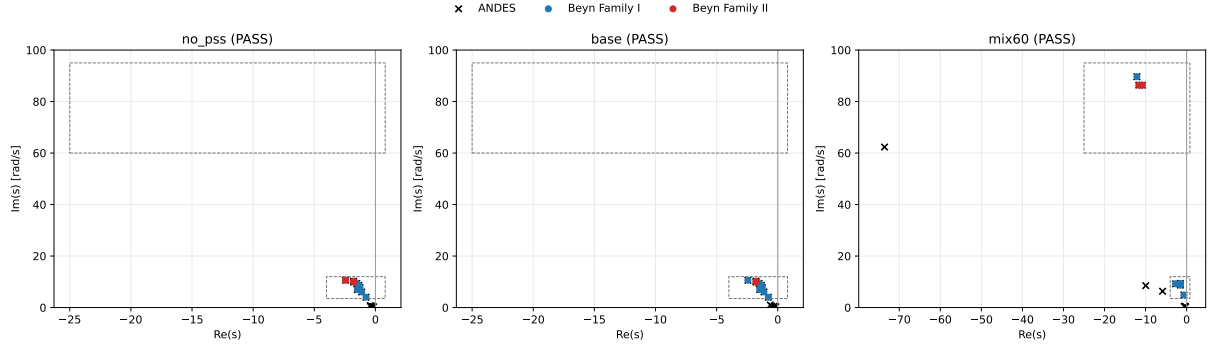


Figure 8: Phase-0D NEP contour validation: ANDES poles, NEP zeros, control poles, and contours.

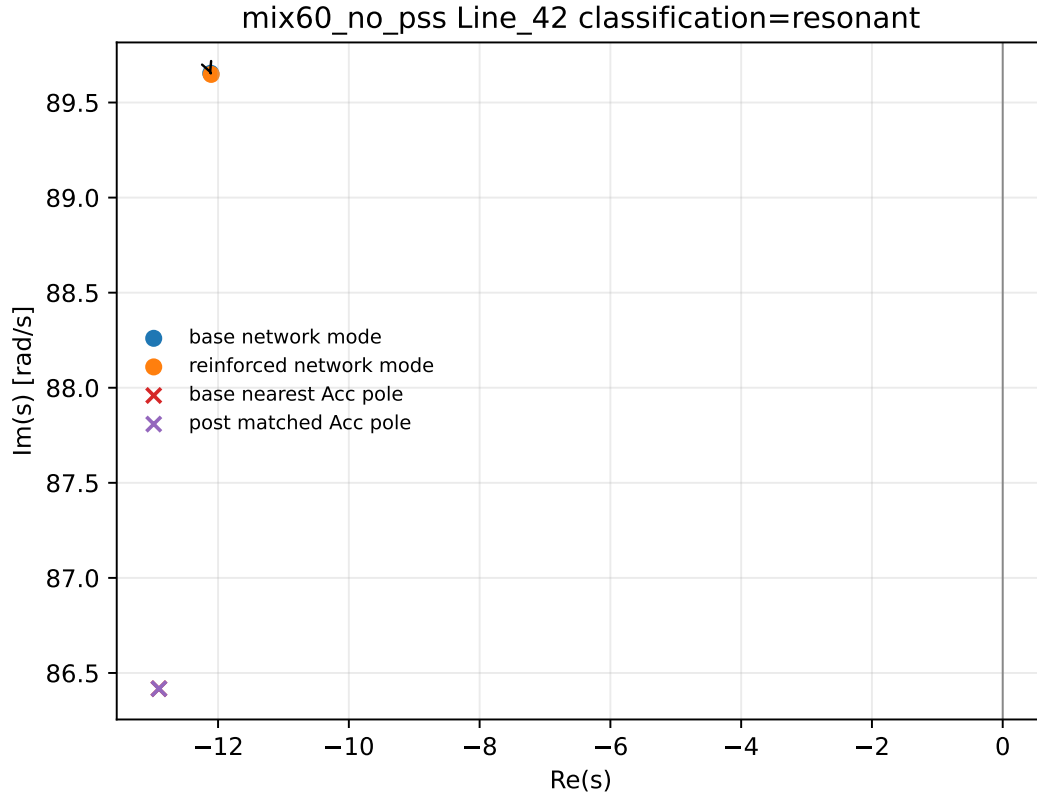


Figure 9: Phase Braess-NEP resonant line-reinforcement witness.

IEEE 39-bus Mix60 SG-to-IBR replacement map

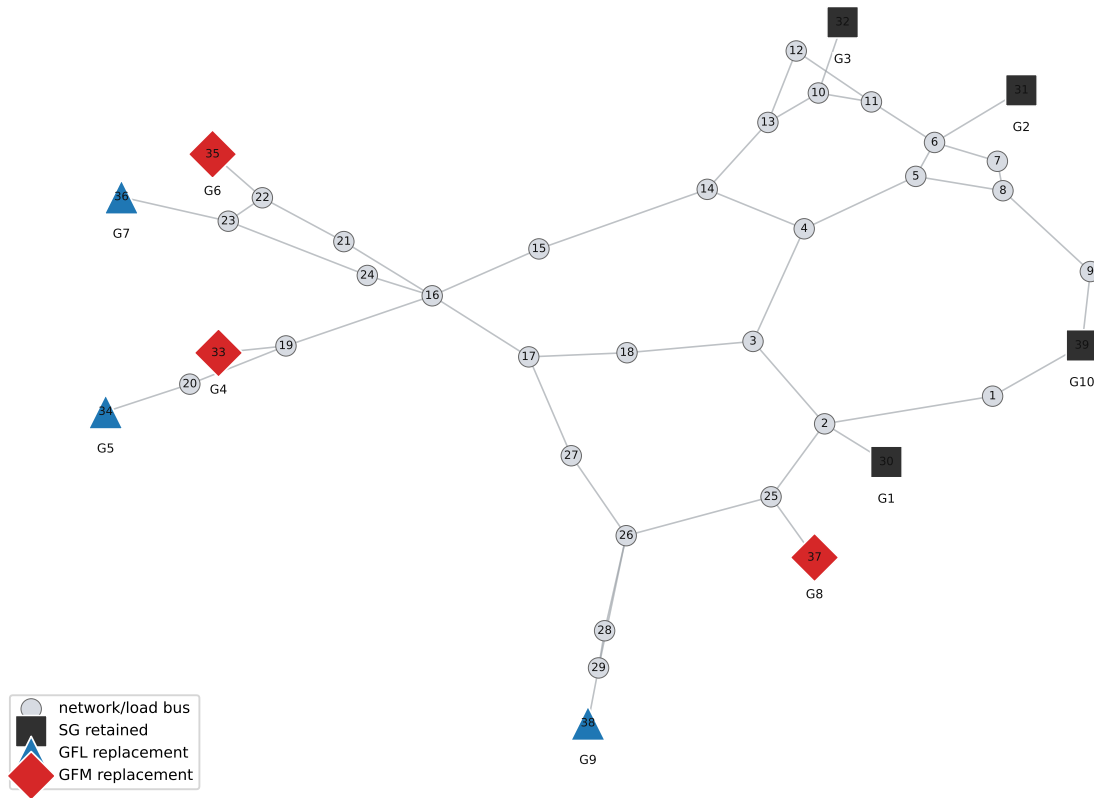


Figure 10: Mix60 IBR map generated during the ANDES bridge phase.

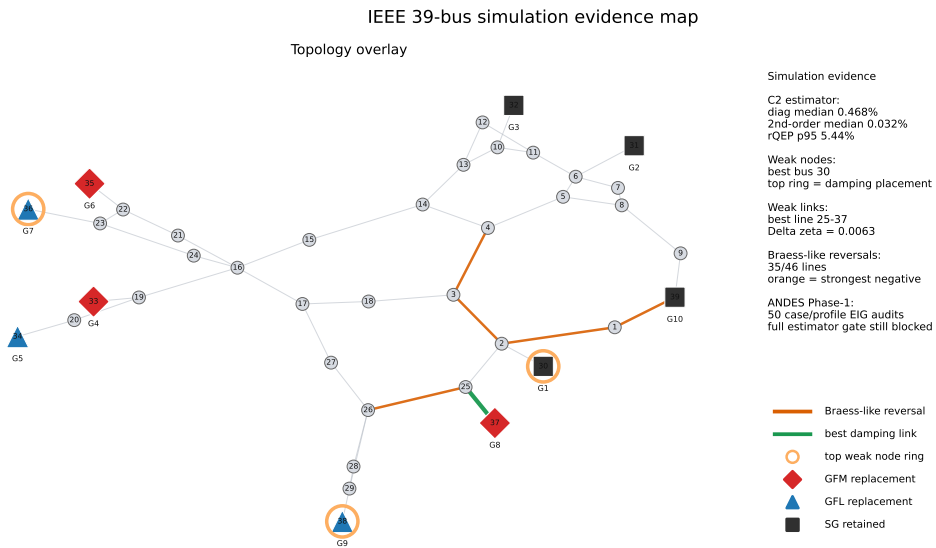


Figure 11: IEEE39 result-map artifact.

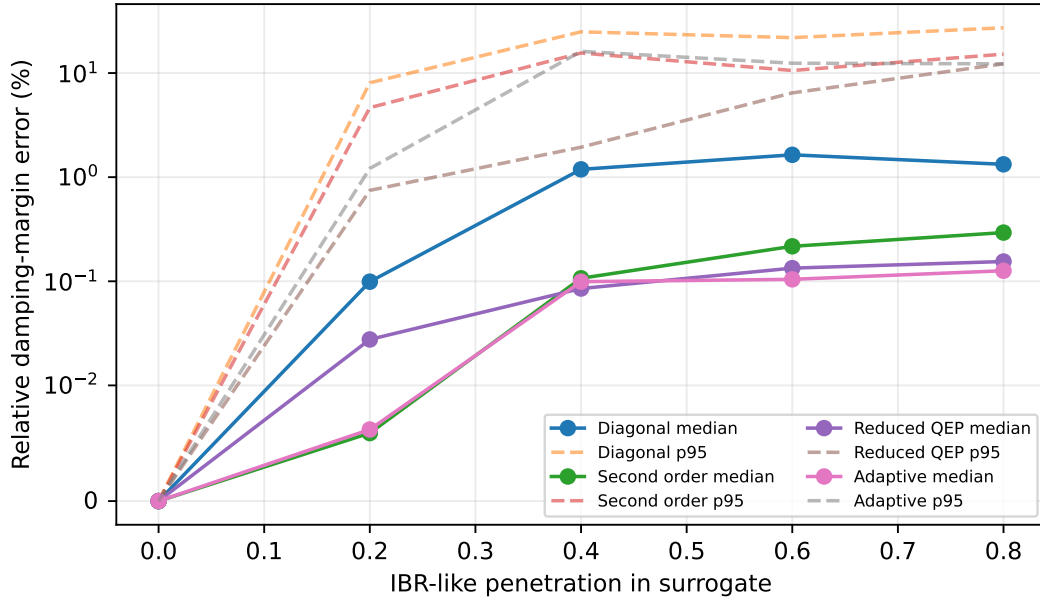


Figure 12: Estimator errors by IBR penetration in the rational-filter artifact set.

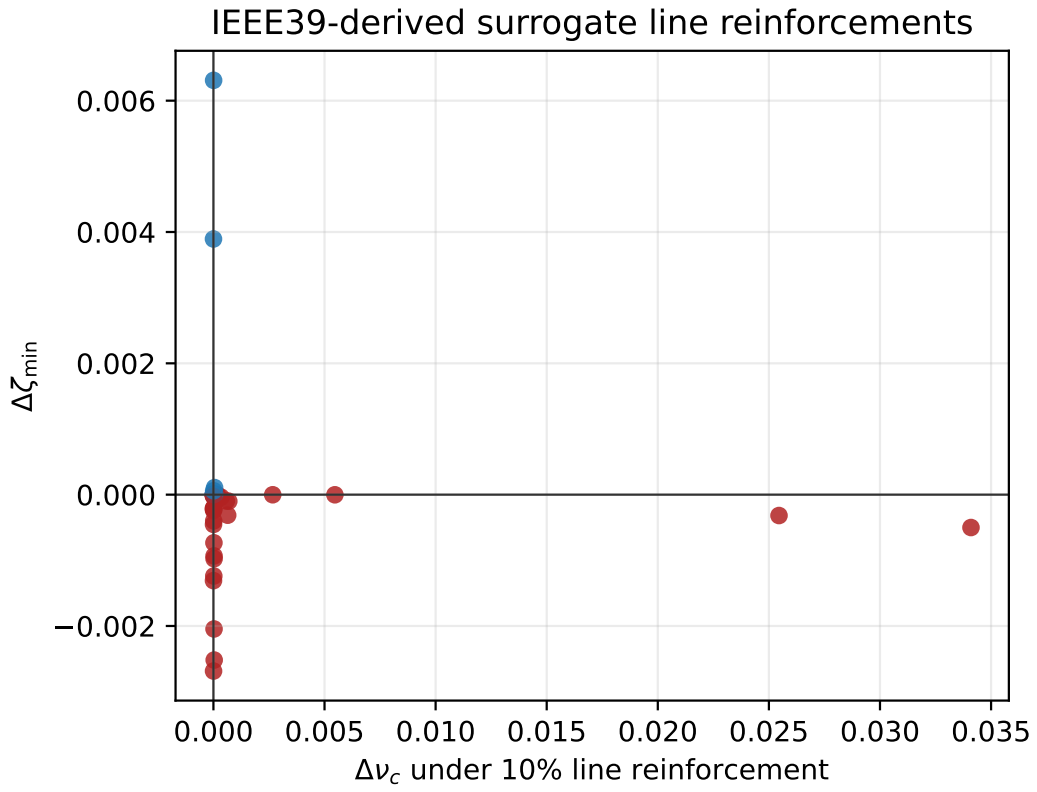


Figure 13: Line frequency score versus damping score in the rational-filter artifact set.

3 Raw Gate Reports and Result Ledgers

3.1 Phase-0D NEP Contour Solver Report

```
# Phase-0D Rational NEP Contour Solver

Gate status: **PASS**

This audit solves the exact first-order Schur nonlinear eigenproblem
'S_e(s)=sI-A_ee-A_ec(sI-A_cc)^(-1)A_ce' with Beyn contour integration.
No fixed-point iteration is used for acceptance. The full ANDES eigensolve
is the reference.

## Non-Circularity Audit
- Beyn zeros are first extracted from the contour moments 'AO' and 'A1' and the reduced eigenproblem.
- The local polish starts only from each Beyn Ritz value and solves for the smallest eigenvalue of 'S_e(s)' to
  vanish.
- The polish function receives no ANDES pole, mode index, frequency, or damping target. ANDES eigenvalues are used
  only after extraction for validation and count diagnostics.
- Therefore a zero can be produced from the Jacobian/Schur NEP even if the corresponding ANDES pole is not supplied
  to the solver.

## Contours
- 'network_margin': Re=[-4.0, 0.8], Im=[3.5, 12.0] rad/s (0.557-1.91 Hz), 512 quadrature nodes.
- 'pll_control': Re=[-25.0, 0.8], Im=[60.0, 95.0] rad/s (9.55-15.1 Hz), 512 quadrature nodes.

## Required Gate Checks
### no_pss
- Accepted: **True**
- Argument-count checks pass: **True**
- States: retained=20, condensed=110
- Beyn zeros accepted: 9
- Sigma/D_eff at s=0: scalar D norm=0, real Fro=9.968, imag Fro=0, solve=solve, cond=8.94e+04.
- Sigma/D_eff at matched critical pole: real Fro=4.053, imag Fro=2.178. Phase-0B critical-pole reference: real Fro
  =4.053, imag Fro=2.178.
- ANDES critical pole: -1.34635 + j8.61104, f=1.3705 Hz, zeta=0.15447.
- NEP-Beyn matched pole: -1.34635 + j8.61104, f=1.3705 Hz, zeta=0.15447, family=I_network, pi_c=0.680.
- Critical errors: frequency=4.861e-14%, damping=6.289e-13%, relative residual=5.467e-17.
- Nearest Acc pole to matched NEP zero: -0.44778 + j0.584957, distance=8.08.
- Top condensed-state reconstruction: LAG_y TGOV1N TGOV1 1:3.536; LAG_y TGOV1N TGOV1 8:2.216; LAG_y TGOV1N TGOV1
  9:1.144; LAG_y TGOV1N TGOV1 3:1.122; LA_y IEEEEX1 9:1.025; LA_y IEEEEX1 7:0.7354
- Contour counts:
  - network_margin: extracted=9, rank=14, argument-count=9.000+j3.51e-06, Acc poles inside=0, argument+Acc=9.000,
    ANDES poles inside=9, count error=6.42e-05, pass=True.
  - pll_control: extracted=0, rank=20, argument-count=-0.000+j1.56e-06, Acc poles inside=0, argument+Acc=-0.000,
    ANDES poles inside=0, count error=2.19e-06, pass=True.
- Zeros table: 'outputs\phase0d_nep_contour_solver\nopss_zeros.csv'
- Contour diagnostics: 'outputs\phase0d_nep_contour_solver\nopss_contour_counts.csv'

### base
- Accepted: **True**
- Argument-count checks pass: **True**
- States: retained=20, condensed=140
- Beyn zeros accepted: 9
- Sigma/D_eff at s=0: scalar D norm=0, real Fro=9.968, imag Fro=0, solve=pinv, cond=7.71e+18.
- Sigma/D_eff at matched critical pole: real Fro=4.053, imag Fro=2.178. Phase-0B critical-pole reference: real Fro
  =4.053, imag Fro=2.178.
- ANDES critical pole: -1.34601 + j8.61148, f=1.3706 Hz, zeta=0.15443.
- NEP-Beyn matched pole: -1.34601 + j8.61148, f=1.3706 Hz, zeta=0.15443, family=I_network, pi_c=0.653.
- Critical errors: frequency=1.62e-13%, damping=1.06e-12%, relative residual=2.981e-17.
- Nearest Acc pole to matched NEP zero: -0.495739 + j0.71038, distance=7.95.
- Top condensed-state reconstruction: LAG_y TGOV1N TGOV1 1:3.536; LAG_y TGOV1N TGOV1 8:2.215; LAG_y TGOV1N TGOV1
  9:1.145; LAG_y TGOV1N TGOV1 3:1.123; LA_y IEEEEX1 9:0.9703; LA_y IEEEEX1 7:0.7134
- Contour counts:
  - network_margin: extracted=9, rank=14, argument-count=9.000+j4.14e-06, Acc poles inside=0, argument+Acc=9.000,
    ANDES poles inside=9, count error=6.54e-05, pass=True.
  - pll_control: extracted=0, rank=20, argument-count=-0.000+j1.56e-06, Acc poles inside=0, argument+Acc=-0.000,
    ANDES poles inside=0, count error=2.19e-06, pass=True.
- Zeros table: 'outputs\phase0d_nep_contour_solver\base_zeros.csv'
- Contour diagnostics: 'outputs\phase0d_nep_contour_solver\base_contour_counts.csv'
```

```

### mix60
- Accepted: **True**
- Argument-count checks pass: **True**
- States: retained=11, condensed=167
- Beyn zeros accepted: 7
- Sigma/D_eff at s=0: scalar D norm=0, real Fro=4.94, imag Fro=0, solve=pinv, cond=3.85e+20.
- Sigma/D_eff at matched critical pole: real Fro=0.05444, imag Fro=0.5387. Phase-OB critical-pole reference: real
  Fro=2.216, imag Fro=0.82.
- ANDES critical pole: -10.7514 + j86.3419, f=13.742 Hz, zeta=0.12357.
- NEP-Beyn matched pole: -10.7514 + j86.3419, f=13.742 Hz, zeta=0.12357, family=II_control, pi_c=0.853.
- Critical errors: frequency=8.475e-11%, damping=2.479e-09%, relative residual=8.606e-11.
- Nearest Acc pole to matched NEP zero: -10.5593 + j86.246, distance=0.215.
- Top condensed-state reconstruction: ae PLL1 G7:14.18; ae PLL1 G5:13.83; Qsen_y REGF1 G6:7.926; Qsig_y REGF1 G6
  :3.478; am PLL1 G7:3.266; am PLL1 G5:3.185
- Contour counts:
  - network_margin: extracted=4, rank=9, argument-count=4.000+j-6.17e-06, Acc poles inside=0, argument+Acc=4.000,
    ANDES poles inside=4, count error=2.87e-05, pass=True.
  - pll_control: extracted=3, rank=11, argument-count=-0.000+j1.14e-06, Acc poles inside=3, argument+Acc=3.000,
    ANDES poles inside=3, count error=1.89e-06, pass=True.
- Zeros table: 'outputs\phase0d_nep_contour_solver\mix60_zeros.csv'
- Contour diagnostics: 'outputs\phase0d_nep_contour_solver\mix60_contour_counts.csv'

## Interpretation
Phase-OD passed for all three cases. The rational Schur NEP contour solver can be used as the bridge for later
validation phases.

```

3.2 Phase Braess-NEP Report

```

# Phase Braess-NEP Report

Gate/verdict: **PASS_RESONANT_FOUND**
Random seed: '20260608'
Git commit: '46f48b23d2ca168678e843f32d844eae5f4d93e5'

## Method
Each line reinforcement reduces the physical ANDES 'r' and 'x' fields by '1/(1+eps)' and reruns power flow, full
eigensolve, and the Phase-OD Beyn Schur-NEP contour solver. The finite-difference reference is the NEP/ANDES
pole movement, not a scalar fixed-point model.

The first-order Schur-NEP sensitivity is decomposed as:
- A: retained omega-row / delta-column stiffness proxy;
- B: residual retained-network/modal-rotation proxy;
- C: condensed-control self-energy contribution from the Schur term;
with the control-frequency signature reported as the fraction of the denominator coming from 'A_ec (sI-A_cc)^(-2)
A_ce'.

A resonant Braess classification requires a network-family damping decrease, decreasing distance to a matched 'A_cc
' pole, dominant negative C contribution, relative pole distance below 10%, and a nearest 'A_cc' pole whose
eigenvector is PLL-dominated. Otherwise the reversal is not claimed resonant.

## Summary
- Cases tested: mix60, mix60_soft_gfl, mix60_no_pss
- Line-mode perturbations completed: 460
- Margin reversals: 348
- Network-family mode reversals: 232
- Resonant reversals: 40
- Smooth analytic-vs-NEP median relative error: 0.000807729516004942

## Clearest Case
- Case/line/mode-rank: 'mix60_no_pss' 'Line_42' (20-34), mode 2
- Classification: **resonant**
- Margin zeta: 0.123566 -> 0.123558 (Delta=-7.985e-06)
- Network zeta: 0.133884 -> 0.133816 (FD derivative=-6.860e-03)
- Distance to matched Acc pole: 3.33227 -> 3.3279 (Delta=-4.370e-03)
- Relative Acc distance: 3.683e-02 -> 3.679e-02; nearest Acc PLL-dominated=True
- Base nearest Acc top states: ae PLL1 G9:0.5613; ae PLL1 G5:0.5097; ae PLL1 G7:0.4108; Qsen_y REGF1 G4:0.2273;
  Qsen_y REGF1 G6:0.1969; S1_y REECA1 1:0.1706
- A dzeta=-1.091e-09, B dzeta=7.974e-09, C dzeta=-6.874e-03

```

```

- Analytic total dzeta=-6.874e-03, FD network dzeta=-6.860e-03, relative error=2.015e-03
- Control denominator fraction=9.594e-01

## Verdict
At least one line reinforcement produced a network-family damping decrease, moved the tracked mode closer to a
  matched condensed-control pole, and was dominated by the self-energy term. The resonant Braess mechanism is
  observed in this benchmark.

Artifacts:
- CSV: 'outputs\phase_braess_nep\phase_braess_nep_line_decomposition.csv'
- JSON: 'outputs\phase_braess_nep\phase_braess_nep_status.json'
- Figure: 'outputs\phase_braess_nep\fig_braess_nep_s_plane.png'

```

3.3 Phase Inertia Bridge Gate Report

```

# Phase Gate: Inertia Assignment over Controller-Aware Bridge

This gate asks whether inertia-placement diagnostics survive when evaluated over the
controller-aware Schur NEP bridge. The script perturbs only documented inertia
parameters present in the case files. In the current IEEE39 IBR cases, REGF1 does
not expose a clear 'M', 'H', or 'Tj' virtual-inertia parameter, so GFM virtual-inertia
assignment remains blocked unless a physical parameter mapping is supplied.

- Commit: '46f48b23d2ca168678e843f32d844eae5f4d93e5'
- Random seed: '20260608'
- eps: '0.01'

## Case Results

### no_pss

- Critical full mode: zeta=0.154474, freq=1.37049 Hz, family=I_network, pi_c=0.6804.
- Critical bridge mode: zeta=0.154474, freq=1.37049 Hz, family=I_network, pi_c=0.6804.
- Gate-1 bridge-vs-ANDES: **PASS**; sign_rate=1.000, top3_overlap=1.000, finite_signal=10/10.
- Gate-1 local NEP sensitivity: sign_rate_vs_full_matched=1.000, top3_overlap_vs_full_global=1.000,
  median_rel_error_vs_full_matched=0.0254.
- Gate-2 substitutability: **NETWORK_MODE_SCOPE**. Critical mode is network-family; structural substitutability can
  be tested on retained network modes.
- Gate-3 inertia sign reversal: **SIGN_REVERSAL_OBSERVED_ON_FULL_ANDES**; negative_full=5/10.
- Virtual-inertia validation: **BLOCKED_NO_EXPLICIT_GFM_VIRTUAL_INERTIA_COLUMN**.

### base

- Critical full mode: zeta=0.154429, freq=1.37056 Hz, family=I_network, pi_c=0.6526.
- Critical bridge mode: zeta=0.154429, freq=1.37056 Hz, family=I_network, pi_c=0.6526.
- Gate-1 bridge-vs-ANDES: **PASS**; sign_rate=1.000, top3_overlap=1.000, finite_signal=10/10.
- Gate-1 local NEP sensitivity: sign_rate_vs_full_matched=1.000, top3_overlap_vs_full_global=1.000,
  median_rel_error_vs_full_matched=0.0254.
- Gate-2 substitutability: **NETWORK_MODE_SCOPE**. Critical mode is network-family; structural substitutability can
  be tested on retained network modes.
- Gate-3 inertia sign reversal: **SIGN_REVERSAL_OBSERVED_ON_FULL_ANDES**; negative_full=5/10.
- Virtual-inertia validation: **BLOCKED_NO_EXPLICIT_GFM_VIRTUAL_INERTIA_COLUMN**.

### mix60

- Critical full mode: zeta=0.123566, freq=13.7417 Hz, family=II_control, pi_c=0.8532.
- Critical bridge mode: zeta=0.123566, freq=13.7417 Hz, family=II_control, pi_c=0.8532.
- Gate-1 bridge-vs-ANDES: **WEAK_SIGNAL**; sign_rate=1.000, top3_overlap=1.000, finite_signal=0/4.
- Gate-1 local NEP sensitivity: sign_rate_vs_full_matched=1.000, top3_overlap_vs_full_global=1.000,
  median_rel_error_vs_full_matched=0.00017.
- Gate-2 substitutability: **NOT_APPLICABLE_TO_CONTROL_CRITICAL**. Critical bridge mode is control-family.
  Projecting the network-inertia shunt residual onto this control-family mode would not test the original red-
  inertia classifier.
- Gate-3 inertia sign reversal: **NO_SIGN_REVERSAL_OBSERVED**; negative_full=0/4.
- Virtual-inertia validation: **BLOCKED_NO_EXPLICIT_GFM_VIRTUAL_INERTIA_COLUMN**.

## Verdict

- Overall status: **CONSERVATIVE_SCOPE**

```

- Abstract implication: Do not claim validated virtual-inertia assignment over the controller-aware bridge. Report GENROU.M sensitivity checks and keep GFM virtual inertia as blocked pending a physical parameter mapping.

What this gate did not test

- It did not validate a real GFM virtual-inertia knob because the tested REGF1 sheets do not expose one.
- It did not prove that the red-inertia substitutability classifier applies to control-family critical modes.
- It did not replace the registered estimator/planning validation over the NEP bridge.

3.4 Phase Coordination Lever Gate Report

Gate 1: Coordination Value of Planning Levers

This gate tests whether coordinated packages of topology, damping, and inertia outperform the best single lever under equal normalized cost and DC flow feasibility. The ground truth is the complete eigensolve of the tested second-order dynamic model.

- Commit: '46f48b23d2ca168678e843f32d844eae5f4d93e5'
- Random seed: '20260609'
- Costs: 'c_w=1.0', 'c_D=2.0', 'c_M=8.0'
- Equal-cost budget: '1.0'
- Combined-package search: top-6 screened candidates per lever
- Cases passing base filters: '62' (60 synthetic, 2 IEEE-topology)
- Median topology-action feasibility ratio: '1.000'

Aggregate Table

metric	pair	eps	n	median SynNorm	% SynNorm < -0.10	median CV	% CV < 0
S_zeta	w+D	0.05	62	0.007668	0.0%	0.0173	31.1%
S_zeta	w+D	0.10	62	0.02207	0.0%	0.0173	31.1%
S_zeta	w+D	0.20	62	0.04991	4.8%	0.0173	31.1%
S_zeta	w+M	0.05	62	-0.002966	11.3%	-0.001815	50.8%
S_zeta	w+M	0.10	62	-0.00512	22.6%	-0.001815	50.8%
S_zeta	w+M	0.20	62	0.02065	17.7%	-0.001815	50.8%
S_zeta	D+M	0.05	62	0.004641	0.0%	0.03641	21.0%
S_zeta	D+M	0.10	62	0.01641	6.5%	0.03641	21.0%
S_zeta	D+M	0.20	62	0.03434	4.8%	0.03641	21.0%
S_g	w+D	0.05	62	0.0007638	0.0%	-0.05294	72.1%
S_g	w+D	0.10	62	0.04141	0.0%	-0.05294	72.1%
S_g	w+D	0.20	62	0.1427	0.0%	-0.05294	72.1%
S_g	w+M	0.05	62	0.0003818	12.9%	-0.07363	67.2%
S_g	w+M	0.10	62	0.2555	11.3%	-0.07363	67.2%
S_g	w+M	0.20	62	0.5423	11.3%	-0.07363	67.2%
S_g	D+M	0.05	62	0.02298	1.6%	-0.006759	51.6%
S_g	D+M	0.10	62	0.08476	0.0%	-0.006759	51.6%
S_g	D+M	0.20	62	0.1519	1.6%	-0.006759	51.6%

Metric Agreement

- Pair-case comparisons: '184'
- Same CV sign for modal and resolvent margins: '50.0%'
- Same single-lever ranking for modal and resolvent margins: '28.3%'

Verdict

- Status: **COORDINATION_VALUE_OBSERVED**
- Implication: Equal-cost mixed packages beat the best single lever in a majority of cases for at least one metric/pair.
- Rows with CV majority support: '4'
- Rows with nonlinear-synergy majority support: '0'

Scope

- This is not a full ANDES IBR validation. IEEE cases are used as topology/flow templates.
- The resolvent margin uses identity protection scalings because no protection matrices were provided.
- The equal-cost package search is favorable but screened: it evaluates combinations among the top single-lever candidates.

3.5 Phase Gamma Robustness Gate Report

```
# Gate Gamma Robustness

This gate repeats the equal-cost coordination test using the same networks,
actions, costs, and seeds as Gate 1, but changes the protected-resolvent
input/output weights.

- Commit: '46f48b23d2ca168678e843f32d844eae5f4d93e5'
- Random seed: '20260609'
- Cases: '62'
- Gamma profiles: '12'
- Random diagonal Gamma samples: '8'

## Gamma Robustness Table

| Gamma | kind | ranking agreement S_zeta vs S_g | CV sign agreement | % CV<0 w+D | % CV<0 w+M | % CV<0 D+M |
|----|----|----|----|----|----|----|
| frequency_output | physical | 29.0% | 50.0% | 72.1% | 67.2% | 51.6% |
| identity | physical | 29.0% | 50.0% | 72.1% | 67.2% | 51.6% |
| inertia_output | physical | 21.0% | 49.5% | 73.8% | 73.8% | 48.4% |
| random_diag_00 | random | 19.4% | 52.7% | 67.2% | 78.7% | 43.5% |
| random_diag_01 | random | 27.4% | 49.5% | 63.9% | 75.4% | 37.1% |
| random_diag_02 | random | 27.4% | 56.5% | 67.2% | 68.9% | 25.8% |
| random_diag_03 | random | 32.3% | 57.1% | 62.3% | 68.9% | 22.6% |
| random_diag_04 | random | 16.1% | 50.5% | 67.2% | 73.8% | 35.5% |
| random_diag_05 | random | 29.0% | 52.7% | 70.5% | 73.8% | 32.3% |
| random_diag_06 | random | 14.5% | 58.7% | 65.6% | 78.7% | 24.2% |
| random_diag_07 | random | 27.4% | 50.0% | 70.5% | 68.9% | 35.5% |
| type_input_low_inertia | physical | 22.6% | 48.4% | 73.8% | 70.5% | 48.4% |

## Descriptor Correlations

Correlations use case-level descriptor values and certificate-disagreement scores.

| Gamma | descriptor | target | n | Pearson r | Spearman r |
|----|----|----|----|----|----|
| frequency_output | henrici_departure | ranking_disagreement | 62 | 0.0573 | 0.0695 |
| frequency_output | commutator_chi | ranking_disagreement | 62 | 0.216 | 0.216 |
| identity | henrici_departure | ranking_disagreement | 62 | 0.0573 | 0.0695 |
| identity | commutator_chi | ranking_disagreement | 62 | 0.216 | 0.216 |
| inertia_output | henrici_departure | ranking_disagreement | 62 | -0.164 | -0.192 |
| inertia_output | commutator_chi | ranking_disagreement | 62 | 0.142 | 0.125 |
| random_diag_00 | henrici_departure | ranking_disagreement | 62 | 0.21 | 0.249 |
| random_diag_00 | commutator_chi | ranking_disagreement | 62 | 0.00526 | 0.0183 |
| random_diag_01 | henrici_departure | ranking_disagreement | 62 | -0.237 | -0.221 |
| random_diag_01 | commutator_chi | ranking_disagreement | 62 | 0.204 | 0.217 |
| random_diag_02 | henrici_departure | ranking_disagreement | 62 | -0.031 | 0.00101 |
| random_diag_02 | commutator_chi | ranking_disagreement | 62 | 0.135 | 0.128 |
| random_diag_03 | henrici_departure | ranking_disagreement | 62 | 0.0985 | 0.133 |
| random_diag_03 | commutator_chi | ranking_disagreement | 62 | 0.275 | 0.291 |
| random_diag_04 | henrici_departure | ranking_disagreement | 62 | 0.255 | 0.265 |
| random_diag_04 | commutator_chi | ranking_disagreement | 62 | 0.0593 | 0.0147 |
| random_diag_05 | henrici_departure | ranking_disagreement | 62 | -0.00578 | -0.00596 |
| random_diag_05 | commutator_chi | ranking_disagreement | 62 | 0.213 | 0.214 |
| random_diag_06 | henrici_departure | ranking_disagreement | 62 | 0.0467 | 0.0576 |
| random_diag_06 | commutator_chi | ranking_disagreement | 62 | 0.23 | 0.229 |
| random_diag_07 | henrici_departure | ranking_disagreement | 62 | -0.241 | -0.195 |
| random_diag_07 | commutator_chi | ranking_disagreement | 62 | -0.0485 | -0.0273 |
| type_input_low_inertia | henrici_departure | ranking_disagreement | 62 | -0.128 | -0.14 |
| type_input_low_inertia | commutator_chi | ranking_disagreement | 62 | 0.141 | 0.112 |

## Verdict

- Status: **DISCREPANCY_ROBUST_WEAK_INTERPRETABILITY**
- Implication: The certificate discrepancy survives physical Gamma choices, but descriptor correlations are weak.
- Discrepancy robust over physical Gamma choices: 'True'
```

- Coordination robust over physical Gamma choices for w+D and w+M: 'True'
- Max absolute Spearman descriptor/ranking-disagreement correlation: '0.216'

Scope

- Gamma matrices are fixed at the base case for each network; actions change the system, not the protection standard.
- The frequency-output profile is intentionally equivalent to identity because Gate 1 already measured omega output
- This is still a reduced dynamic-model gate, not a full ANDES IBR validation.

3.6 Final Claim Readiness Report

Final Claim-Readiness Audit

This report is generated from versioned validation outputs. It does not rerun simulations.

Bottom Line

The paper is ready as a rigorous methodological/master-thesis manuscript package. It is not yet ready to claim final IEEE Transactions empirical validation on full calibrated IBR dynamics.

Claim Status

C1 damping-region coordinate system

- Verified: Exact in commuting/proportional damping by simultaneous diagonalization; figure shows non-proportional screen failure mode.
- Control/comparison: Commuting case versus non-proportional coupled-QEP case.
- Not proven: The region language itself is not proven novel; outside commuting damping it is a screen, not a certificate.
- Status: theoretical/diagnostic
- Confidence: high for the mathematics; medium for novelty and planning value

C2 closed-form second-order damping-margin correction

- Verified: On the IEEE39-derived surrogate, median relative error improves from 0.468% diagonal to 0.0321% second order; reduced PLL Monte Carlo reported in the manuscript improves 2.16% to 0.09% median error.
- Control/comparison: Full QEP of the surrogate; diagonal all-mode screen; reduced QEP fallback.
- Not proven: Not yet validated against full calibrated ANDES IBR poles; second order does not dominate large reduced QEP in tails.
- Status: verified on reduced models; benchmark pending
- Confidence: high for reduced models; medium for full IBR generalization

C2 adaptive workflow diagonal -> second order -> reduced QEP

- Verified: Tail behavior requires fallback: second-order p95 is 9.91%, reduced-QEP p95 is 5.44%, adaptive p95 is 7.43% on the IEEE39-derived surrogate.
- Control/comparison: Always-diagonal, always-second-order, and reduced-QEP estimates.
- Not proven: No universal trigger threshold is proved; false-safe behavior must be measured on calibrated ANDES IBR cases.
- Status: diagnostic workflow
- Confidence: medium

C3 SG-to-IBR conversion ranking

- Verified: 5/5 generated SG-to-IBR ANDES cases load, solve power flow, and run eigenanalysis; 1/5 is small-signal stable with generic uncalibrated gains. Phase-1 ran 50 case/profile audits; 1/5 cases have a stable controller-only profile and 4/5 have a stable diagnostic retained-SG-control-removal profile.
- Control/comparison: ANDES pflow/eigenanalysis feasibility, device inventory checks, controller-only profiles, and diagnostic retained-control removal.
- Not proven: No low-regret conversion ranking against full eigensolve has been demonstrated yet; stable cases are still calibration candidates, not final benchmark tuning.
- Status: case-construction and calibration audit
- Confidence: high for construction; low for planning superiority

C4 damping-aware weak links and Braess-like reversals

- Verified: On the IEEE39-derived surrogate, 35/46 line reinforcements increase critical modal stiffness while reducing damping margin; the best damping-aware reinforcement is line 25-37.
- Control/comparison: Frequency-only line effect versus finite post-action damping-margin change.
- Not proven: Not yet a full ANDES IBR topology claim; generic QEP sensitivity prior art must be distinguished.
- Status: surrogate planning diagnostic
- Confidence: medium

C5 inertia-placement sign reversal

- Verified: With fixed local damping-per-inertia, 5447/16229 perturbations reduced both the diagonal modal screen and the full-QEP damping margin; the strongest reported example matches the analytic sign against finite differences.
- Control/comparison: Fixed D/M nontrivial check versus fixed-D direct damping-dilution control.
- Not proven: Not validated on calibrated ANDES IBR dynamics; the theorem is a reduced-model modal sign condition, not a universal inertia-planning law.
- Status: verified reduced-model sign diagnostic
- Confidence: medium

C6 joint inertia-damping cross term

- Verified: The manuscript derives the local off-diagonal coupling term for simultaneous inertia and damping perturbations and shows that it reduces to the existing second-order damping correction when Delta M is zero.
- Control/comparison: Special-case reduction to the fixed-inertia second-order correction; separation of first-order diagonal shift from second-order modal coupling.
- Not proven: No Monte Carlo or ANDES validation is claimed for finite SG-to-IBR conversion accuracy; large conversions must be recomputed.
- Status: theoretical local mechanism
- Confidence: medium for derivation; low for finite-conversion prediction

ANDES IEEE39 benchmark readiness

- Verified: Packaged baseline has 9 positive-real non-oscillatory modes; removing IEEEEST gives 0 positive modes while keeping the 1.37 Hz oscillatory pair essentially unchanged. GFL/PLL cases exist: yes; GFM cases exist: yes. Phase-1 found 9/50 stable case/profile runs. The strict physical bridge audit status is BLOCKED (base critical frequency error 12.4%, damping-ratio error 100%).
- Control/comparison: Baseline versus removed-model-family diagnostics; IBR device inventory checks.
- Not proven: A defensible calibrated dynamic library and physically matched extraction of reduced L,M,D/control coupling from full ANDES DAE are still pending; the strict physical bridge audit remains blocked.
- Status: benchmark harness and calibration audit ready; final validation pending
- Confidence: high for harness status; low for final empirical claim

Submission Decision

Use the current manuscript for thesis defense, internal circulation, or a methods-oriented preprint. For an IEEE Transactions submission, complete the calibrated ANDES IBR benchmark and add full eigensolve comparisons for estimator accuracy, conversion ranking regret, and line-sensitivity finite differences.

4 Compact CSV Ledgers

4.1 Claim Audit CSV

claim,what_was_verified,control_or_comparison,not_proven,status,confidence
C1 damping-region coordinate system,Exact in commuting/proportional damping by simultaneous diagonalization; figure shows non-proportional screen failure mode.,Commuting case versus non-proportional coupled-QEP case.,\"The region language itself is not proven novel; outside commuting damping it is a screen, not a certificate.\".,theoretical/diagnostic,high for the mathematics; medium for novelty and planning value
C2 closed-form second-order damping-margin correction,\"On the IEEE39-derived surrogate, median relative error improves from 0.468% diagonal to 0.0321% second order; reduced PLL Monte Carlo reported in the manuscript improves 2.16% to 0.09% median error.\".,Full QEP of the surrogate; diagonal all-mode screen; reduced QEP fallback.,Not yet validated against full calibrated ANDES IBR poles; second order does not dominate large reduced QEP in tails.,verified on reduced models; benchmark pending,high for reduced models; medium for full IBR generalization
C2 adaptive workflow diagonal -> second order -> reduced QEP,\"Tail behavior requires fallback: second-order p95 is 9.91%, reduced-QEP p95 is 5.44%, adaptive p95 is 7.43% on the IEEE39-derived surrogate.\".,\"Always-diagonal, always-second-order, and reduced-QEP estimates.\".,No universal trigger threshold is proved; false-safe behavior must be measured on calibrated ANDES IBR cases.,diagnostic workflow,medium
C3 SG-to-IBR conversion ranking,\"5/5 generated SG-to-IBR ANDES cases load, solve power flow, and run eigenanalysis; 1/5 is small-signal stable with generic uncalibrated gains . Phase-1 ran 50 case/profile audits; 1/5 cases have a stable controller-only profile and 4/5 have a stable diagnostic retained-SG-control-removal profile.\".,\"ANDES pfflow/eigenanalysis feasibility, device inventory checks, controller-only profiles, and diagnostic retained-control removal.\".,\"No low-regret conversion ranking against full eigensolve has been demonstrated yet; stable cases are still calibration candidates, not final benchmark tuning.\".,case-construction and calibration audit,high for construction; low for planning superiority
C4 damping-aware weak links and Braess-like reversals,\"On the IEEE39-derived surrogate, 35/46 line reinforcements increase critical modal stiffness while reducing damping margin; the best damping-aware reinforcement is line 25-37.\".,Frequency-only line effect versus finite post-action damping-margin change.,Not yet a full ANDES IBR topology claim; generic QEP sensitivity prior art must be distinguished.,surrogate planning diagnostic,medium
C5 inertia-placement sign reversal,\"With fixed local damping-per-inertia, 5447/16229 perturbations reduced both the diagonal modal screen and the full-QEP damping margin; the strongest reported example matches the analytic sign against finite differences.\".,Fixed D/M nontrivial check versus fixed-D direct damping-dilution control.,\"Not validated on calibrated ANDES IBR dynamics; the theorem is a reduced-model modal sign condition, not a universal inertia-planning law.\".,verified reduced-model sign diagnostic,medium
C6 joint inertia-damping cross term,\"The manuscript derives the local off-diagonal coupling term for simultaneous inertia and damping perturbations and shows that it reduces to the existing second-order damping correction when Delta M is zero.,Special-case reduction to the fixed-inertia second-order correction; separation of first-order diagonal shift from second-order modal coupling.,No Monte Carlo or ANDES validation is claimed for finite SG-to-IBR conversion accuracy; large conversions must be recomputed.,theoretical local mechanism,medium for derivation; low for finite-conversion prediction
ANDES IEEE39 benchmark readiness,\"Packaged baseline has 9 positive-real non-oscillatory modes; removing IEEEEST gives 0 positive modes while keeping the 1.37 Hz oscillatory pair essentially unchanged. GF/PLL cases exist: yes; GFM cases exist: yes. Phase-1 found 9/50 stable case/profile runs. The strict physical bridge audit status is BLOCKED (base critical frequency error 12.4%, damping-ratio error 100%).\".,Baseline versus removed-model-family diagnostics; IBR device inventory checks.,\"A defensible calibrated dynamic library and physically matched extraction of reduced L,M,D/control coupling from full ANDES DAE are still pending; the strict physical bridge audit remains blocked.\".,benchmark harness and calibration audit ready; final validation pending,high for harness status; low for final empirical claim

4.2 Phase-0D Critical Validation CSV: No-PSS

andes_mode_index,andes_real,andes_imag,andes_freq_hz,andes_zeta,andes_top_states,nep_real,nep_imag,nep_freq_hz,nep_zeta,nep_family,nep_pi_c,nep_relative_residual,relative_freq_error,relative_zeta_error,damping_sign_correct,pole_distance,nearest_acc_real,nearest_acc_imag,nearest_acc_freq_hz,distance_to_nearest_acc,top_condensed_states
71,-1.34634537929171,8.611042315453796,1.3704899496779517,0.1544743228879957,omega GENROU 1:0.3684; delta GENROU 1:0.3423; omega GENROU 8:0.09074; delta GENROU 8:0.08036; omega GENROU 3:0.05055; delta GENROU 3:0.04642; e2q GENROU 1:0.02908; omega GENROU 9:0.02118,-1.346345379291718,8.611042315453792,1.370489949677951,0.15447432288799667,I_network,0.680403305071333,5.4665204419076374e-17,4.860552351599713e-16,6.288716003962517e-15,True,8.747542958250513e-15,-0.44777986616109694,0.5849571139737398,0.0930988161856899,8.076228305515192,LAG_y TGOV1N TGOV1 1:3.536; LAG_y TGOV1N TGOV1 8:2.216; LAG_y TGOV1N TGOV1 9:1.144; LAG_y TGOV1N TGOV1 3:1.122; LA_y IEEEX1 9:1.025; LA_y IEEEX1 7:0.7354

4.3 Phase-0D Critical Validation CSV: Base

andes_mode_index,andes_real,andes_imag,andes_freq_hz,andes_zeta,andes_top_states,nep_real,nep_imag,nep_freq_hz,nep_zeta,nep_family,nep_pi_c,nep_relative_residual,relative_freq_error,relative_zeta_error,damping_sign_correct,pole_distance,nearest_acc_real,nearest_acc_imag,nearest_acc_freq_hz,distance_to_nearest_acc,top_condensed_states
71,-1.346013072429833,8.61148480240536,1.3705603736635468,0.15442935855929482,omega GENROU 1:0.3682; delta GENROU 1:0.3422; omega GENROU 8:0.09063; delta GENROU 8:0.08027; omega GENROU 3:0.05055; delta GENROU 3:0.04643; e2q GENROU 1:0.03479; omega GENROU 9:0.02119,-1.3460130724298205,8.611484802405375,1.3705604519076374e-17,4.860552351599713e-16,6.288716003962517e-15,True,8.747542958250513e-15,1.6201008667097224e-15,1.0604065034002843e-14,True,1.888293221827313e-14,-0.4957392820467232,0.710380304412895,0.1130605369224376,7.946723715268416,LAG_y TGOV1N TGOV1 1:3.536; LAG_y TGOV1N TGOV1 8:2.215; LAG_y TGOV1N TGOV1 9:1.145; LAG_y TGOV1N TGOV1 3:1.123; LA_y IEEEX1 9:0.9703; LA_y IEEEX1 7:0.7134

4.4 Phase-0D Critical Validation CSV: Mix60

andes_mode_index,andes_real,andes_imag,andes_freq_hz,andes_zeta,andes_top_states,nep_real,nep_imag,nep_freq_hz,nep_zeta,nep_family,nep_pi_c,nep_relative_residual,relative_freq_error,relative_zeta_error,damping_sign_correct,pole_distance,nearest_acc_real,nearest_acc_imag,nearest_acc_freq_hz,distance_to_nearest_acc,top_condensed_states
46,-10.751357900248504,86.34193765604479,13.741746174091784,0.12356640434765424,am PLL1 G7:0.124; am PLL1 G5:0.1035; ae PLL1 G5:0.04002; ae PLL1 G7:0.03956; am PLL1 G9:0.00702; af_y PLL1 G7:0.00469; af_y PLL1 G5:0.00397; ae PLL1 G9:0.0024,-10.75135790052832,86.34193765611796,13.74174617410343,0.12356640435071799,II_control1,0.8532274444325421,8.606266985913992e-11,8.474756622316766e-13,2.4794309742639428e-11,True,2.8922600597485146e-10,-10.5593179330641,86.24595386311101,13.726469878989665,0.2146910282793857,ae PLL1 G7:14.18; ae PLL1 G5:13.83; Qsen_y REGF1 G6:7.926; Qsig_y REGF1 G6:3.478; am PLL1 G7:3.266; am PLL1 G5:3.185

4.5 Coordination Aggregate Table

metric,pair,eps,n_syn,median_syn_norm,pct_syn_norm_lt_minus_Op10,n_cv,median_cv,pct_cv_lt_0
S_zeta,w+D,0.05,62,-0.00768302773441172,0.0,61,0.01729617454846455,0.3114754098360656
S_zeta,w+D,0.1,62,0.022069472696906192,0.0,61,0.01729617454846455,0.3114754098360656
S_zeta,w+D,0.2,62,0.04991209754616945,0.04838709677419355,61,0.01729617454846455,0.3114754098360656
S_zeta,w+M,0.05,62,-0.002966170657092108,0.11290322580645161,61,-0.0018149072526383314,0.5081967213114754
S_zeta,w+M,0.1,62,-0.005119928887203324,0.22580645161290322,61,-0.0018149072526383314,0.5081967213114754
S_zeta,w+M,0.2,62,0.020647076852967912,0.1774193548387097,61,-0.0018149072526383314,0.5081967213114754
S_zeta,D+M,0.05,62,0.00464058018071738,0.0,62,0.03640714147183133,0.20967741935483872
S_zeta,D+M,0.1,62,0.016413039731715137,0.06451612903225806,62,0.03640714147183133,0.20967741935483872
S_zeta,D+M,0.2,62,0.0343226439295836,0.04838709677419355,62,0.03640714147183133,0.20967741935483872
S_g,w+D,0.05,62,0.000763786402409719,0.0,61,-0.052937125774324745,0.7213114754098361
S_g,w+D,0.1,62,0.04140606492944125,0.0,61,-0.052937125774324745,0.7213114754098361
S_g,w+D,0.2,62,0.14274149902187033,0.0,61,-0.052937125774324745,0.7213114754098361

```
S_g,w+M,0.05,62,0.0003818153534009315,0.12903225806451613,61,-0.07362958055494628,0.6721311475409836
S_g,w+M,0.1,62,0.25547186035613817,0.11290322580645161,61,-0.07362958055494628,0.6721311475409836
S_g,w+M,0.2,62,0.5423448925195331,0.11290322580645161,61,-0.07362958055494628,0.6721311475409836
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4.6 Gamma Aggregate Table

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```

Internal Unified Theory Note: Damping-Operator Hierarchy, Rational Self-Energy, and Flow-Feasible Retrofitting in IBR Grids

Project working document

June 2026

Abstract

This document is the internal map of the project. It is not a submission draft. It organizes the verified results, failed hypotheses, and open gates under one operator hierarchy. The central thesis is that the transition from synchronous grids to inverter-rich grids is a descent through a damping-operator hierarchy: proportional viscous damping, non-proportional viscous damping, and rational frequency-dependent damping induced by controller self-energy. Each descent breaks a classical guarantee and creates a planning risk. Level 0 gives normal modes and topology-invariant decay under proportional damping. Level 1 breaks classical modal separation; heterogeneous D/M creates modal coupling, dynamic Braess-like reversals, and non-monotone inertia effects. Level 2 condenses controller states into a rational self-energy

$$\Sigma(s) = D_0 + C(sI - A_{cc})^{-1}B,$$

which yields a nonlinear eigenvalue problem. This bridge was validated against full ANDES eigensolves on IEEE 39-bus cases, including a 13.7417 Hz PLL-family critical mode in Mix60. The document also integrates the newer resolvent-margin and coordinated-retrofitting gates. Those gates support a secondary but useful claim: modal damping-ratio and protected-resolvent certificates rank actions differently across several protection weightings, and equal-cost packages $w + D$ and $w + M$ often beat the best single lever under the protected resolvent. The mechanism of that certificate discrepancy is still open; simple global non-normality and commutator descriptors correlated weakly. Every claim below is labeled by validation status.

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1 Executive Thesis

The project can be read as one question:

What object preserves damping-margin mechanisms after IBR controls are condensed?

The answer changed as the work progressed. A scalar (L, M, D) bridge is useful for intuition but not enough for detailed IBR dynamics. The correct reduced object for the ANDES cases is a rational Schur self-energy or, equivalently, a retained-state nonlinear eigenvalue problem (NEP). Once that bridge exists, we can ask smaller planning questions: which line is weak, which node is weak, which damping/inertia/topology action moves the margin, and whether packages of actions beat single levers under flow feasibility.

1.1 The operator hierarchy

The useful organizing device is a three-level hierarchy of damping operators. Each level has a mathematical owner in the literature, a guarantee, and a planning risk when the grid leaves that level.

Level	Operator	Classical guarantee	Planning risk when violated
0	Proportional viscous damping, $D = \gamma M$ or Rayleigh/Caughey-class damping	Normal-mode separation; common decay for underdamped proportional modes; topology moves frequency more than decay	Frequency-only planning can look valid, but only inside the proportional floor
1	Constant non-proportional viscous damping, $[\tilde{L}, \tilde{D}] \neq 0$	Modal coordinates are no longer decoupled; off-diagonal modal damping moves poles	Line reinforcement or inertia can reduce damping ratio; Fiedler/frequency rankings can fail
2	Rational/nonviscous damping, $\Sigma(s) = D_0 + C(sI - A_{cc})^{-1}B$	Constant-coefficient QEP is no longer the reduced object; controller poles can become critical	Static bridge and fixed-point screens can miss control-family critical modes

This hierarchy is not a claim that Rayleigh damping, non-proportional modal perturbation, nonviscous damping, Schur complements, or contour eigensolvers are new. They are not. The project contribution, if it survives adversarial review, is the power-system formulation and validation: treating IBR grids as active nonviscous networked structures and using a controller-aware rational bridge to classify network-family and control-family margin mechanisms.

1.2 The material-active candidate framing

The most compact physical sentence is:

An IBR-dominated grid behaves like a viscoelastic structure made of active material: its controller self-energy can have oscillatory, band-active poles, so nonviscous modes can become the critical small-signal modes.

This is a candidate framing, not a closed novelty claim. Passive viscoelastic models have a long literature, including hereditary kernels and nonviscous damping models. The power-system observation is that a PLL or inverter-control kernel is not a passive relaxation kernel in the same sense. In Mix60, the critical pole is a 13.7417 Hz control-family mode with $\pi_c = 0.853$, close to a PLL-dominated condensed-control pole. That is the empirical anchor. The adversarial literature search for “active viscoelastic structures,” active nonviscous kernels, and nonviscous damping in power grids remains open.

1.3 What papers can come out

- **Paper 1: the controller-aware bridge.** This is the strongest validated paper. It reports the scalar bridge failure, the Schur NEP bridge, Beyn contour extraction, network/control-family classification, Mix60’s control-family critical mode, and the resonant Braess mechanism with negative control.
- **Paper 2: the hierarchy and passivity gate.** This is conditional. It becomes strong if a passivity/active-kernel test separates cases where the static bridge is safe from cases where Level 2 is required.
- **Paper 3: flow-feasible resolvent retrofitting.** This is a conference-scale candidate. The Gate 1 and Gamma gates show a real phenomenon: protected-resolvent and modal certificates recommend different actions, and $w + D$ or $w + M$ packages often beat single levers under the protected resolvent. The missing piece is a convincing local mechanism descriptor.
- **Thesis material.** The weak-node/weak-link maps, substitutability spectrum, inertia reversals, and graph-configuration optimization are useful thesis chapters. Several are formulated or surrogate-verified rather than full ANDES-validated.

2 Validation Ledger

Table 1 is the most important table in the document. It separates validated bridge results from formulated planning ideas and dead ends.

Table 1: Claim ledger. Status terms are intentionally blunt.

Claim	Evidence	Control or comparison	Status
Scalar (L, M, D) bridge is insufficient for detailed ANDES damping	In IEEE39 no-PSS/base/Mix60, local scalar damping from GENROU.D is zero while modal damping exists; Phase-0D reports scalar D norm zero and nonzero Schur damping diagnostics.	Full ANDES eigensolve and Schur diagnostics.	ANDES-validated failure mode
Controller-aware Schur NEP bridge recovers critical poles	Phase-0D: no-PSS, base, Mix60 pass. Worst reported errors: Mix60 frequency $8.475 \times 10^{-11}\%$, damping $2.479 \times 10^{-9}\%$.	Full ANDES eigensolve; contour argument counts; non-circular polish.	ANDES-validated
Mix60 critical mode is control-family	Pole $-10.7514 + j86.3419$, $f = 13.7417$ Hz, $\zeta = 0.12357$, $\pi_c = 0.853$; nearest A_{cc} pole distance 0.215; top reconstructed states PLL1/REGF1.	Scalar bridge and fixed-point Schur reduction missed or misidentified the mode.	ANDES-validated mechanism anchor
Resonant Braess mechanism exists in benchmark	Phase Braess-NEP: 40/348 strict resonant records. Clearest case Line 42 (20–34), self-energy term C dominates with control denominator fraction 0.9594 and median analytic-vs-NEP smooth error 8.08×10^{-4} .	Requires network-family damping decrease, closer A_{cc} pole, PLL-dominated nearest pole, and dominant self-energy term; soft-GFL negative control removes resonances.	ANDES/NEP mechanism validated, minority and local
Second-order modal correction improves diagonal screen in weak/mild regimes	Synthetic and reduced-PLL audits: weak surrogate improvement from diagonal to second order; reduced PLL reported $2.16\% \rightarrow 0.09\%$ median in prior cross-check.	Diagonal screen and reduced QEP fallback.	Numerically verified outside full ANDES; estimator validation pending

Claim	Evidence	Control or comparison	Status
Second-order correction always beats large reduced QEP	Refuted by cross-checks; QEP $r = 20$ can beat or match second order in tails.	Reduced QEP size sweep.	KILLED
Inertia can reduce damping margin	Fixed- D_i/M_i surrogate audit: 5447/16229 perturbations reduce both diagonal and full-QEP margin. ANDES GENROU.M gate: no-PSS/base have 5/10 negative GENROU.M perturbations.	Fixed- D_i versus fixed- D_i/M_i control; full ANDES eigensolve for GENROU.M cases.	Valid for mechanical inertia checks; virtual inertia GFM blocked
Virtual-inertia assignment works over controller-aware IBR bridge	REGF1 sheets expose no clear M , H , or T_j knob. Mix60 response to retained GENROU.M is weak signal.	Phase inertia bridge gate.	BLOCKED; do not claim
Certificate discrepancy S_ζ vs S_g	Gate 1: ranking agreement 28.3%. Gamma gate: physical profiles have agreement 21–29%.	Four physical Gamma choices and eight random diagonal profiles.	Robust phenomenon, mechanism weak
Coordinated packages beat single levers at equal cost	Gate 1/Gamma: under S_g , $w + D$ has $CV < 0$ in 72.1–73.8%, $w + M$ in 67.2–73.8%; $D + M$ not robust.	Equal-cost budget, DC flow feasibility, same seeds, Gamma profiles.	Reduced-model gate supports $w + D$ and $w + M$ only
Nonlinear synergy of levers	SynNorm < -0.10 never exceeds 50% in Gate 1.	Three eps sizes and all lever pairs.	KILLED as headline
Cross inertia-control term is empirically frequent in ANDES	Phase E0 found 0/13 measurable singleton cases.	ANDES bridge checks.	KILLED as empirical headline; remains local theory

3 Level 0: The Classical Floor

The starting point is the second-order model

$$M\ddot{x} + D\dot{x} + Lx = 0, \quad (1)$$

where $M \succ 0$ and $L = L^T \succeq 0$. Inertia-normalized coordinates $y = M^{1/2}x$ give

$$\ddot{y} + \tilde{D}\dot{y} + \tilde{L}y = 0, \quad \tilde{L} = M^{-1/2}LM^{-1/2}, \quad \tilde{D} = M^{-1/2}DM^{-1/2}. \quad (2)$$

If $D = \gamma M$, then $\tilde{D} = \gamma I$ and every eigenvector of $\tilde{L}$ is also an eigenvector of $\tilde{D}$. Writing $\tilde{L}q_k = \mu_k q_k$, each nonzero mode obeys

$$\ddot{z}_k + \gamma \dot{z}_k + \mu_k z_k = 0, \quad (3)$$

with poles

$$s_{k,\pm} = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\mu_k}}{2}. \quad (4)$$

For underdamped modes, $\mu_k > \gamma^2/4$, the real part is exactly

$$\text{Re}(s_{k,\pm}) = -\gamma/2. \quad (5)$$

Proposition 1 (Topology-invariant decay at the proportional floor) *Under $D = \gamma M$, every mode that remains underdamped has decay rate $-\gamma/2$, independent of topology.*

This is a floor theorem, not a project contribution. It is the same normal-mode logic behind classical proportional damping and Rayleigh damping. It explains why many graph-frequency intuitions feel safe in homogeneous systems. The escape channel is also clear: topology can move μ_k across the overdamped/underdamped boundary. The rigorous statement is not that topology never matters. It is that decay is invariant for modes that stay in the underdamped proportional class.

The project validation notes recorded numerical deviations at the level of 5.8×10^{-16} to 1.4×10^{-16} in proportional tests. Those numbers are useful sanity checks, not new theory.

4 Level 1: Non-Proportional Viscous Damping

Level 1 begins when $\tilde{D}$ and $\tilde{L}$ no longer share eigenvectors. In modal coordinates of $\tilde{L}$,

$$\ddot{z} + \Gamma \dot{z} + \Lambda z = 0, \quad \Gamma = Q^T \tilde{D} Q, \quad \Lambda = \text{diag}(\nu_k). \quad (6)$$

Write

$$\Gamma = \Delta + E, \quad \Delta = \text{diag}(\delta_k), \quad E_{kk} = 0. \quad (7)$$

The diagonal screen uses scalar modal polynomials

$$p_k(s) = s^2 + \delta_k s + \nu_k. \quad (8)$$

The actual QEP is coupled by sE . This is the first place where topology, damping, and inertia can interact in non-intuitive ways.

4.1 QEP sensitivities

For the QEP

$$P(s)v = (s^2 M + sD + L)v = 0, \quad (9)$$

assume a simple eigenvalue and use the complex-symmetric bilinear product $v^T(\cdot)v$ when M, D, L are real symmetric. Define

$$\xi = v^T(2sM + D)v. \quad (10)$$

Then the local parameter sensitivities are

$$\frac{\partial s}{\partial M_i} = -\frac{s^2 v_i^2}{\xi}, \quad (11)$$

$$\frac{\partial s}{\partial D_i} = -\frac{s v_i^2}{\xi}, \quad (12)$$

$$\frac{\partial s}{\partial w_e} = -\frac{(b_e^T v)^2}{\xi}, \quad (13)$$

where b_e is the signed incidence vector of line e in the selected coordinates. These formulas are standard perturbation theory. In the project they were used as a common local currency for topology, damping, and inertia. They were checked against central differences with relative errors around 10^{-8} in the three channels in the associated validation notes.

4.2 Why inertia can be deceptive

At the proportional point $D = \gamma M$, holding D fixed while perturbing M_i breaks proportionality. That direction is not the same as increasing both M_i and D_i proportionally. For an underdamped proportional mode, the directional derivative of the real part can move the pole right; in the project notes the exact local rate was written as a positive proportional quantity of the modal participation. The correct interpretation is damping dilution: if damping is held fixed and inertia is added, the ratio between energy storage and dissipation changes.

The stronger phrase “inertia is universally Braessian” is false and must not be used. Monte Carlo and fixed-ratio controls rejected it. The defensible statement is narrower:

Under heterogeneous damping, increasing inertia frequently but not universally reduces damping margin; at the proportional point, the harmful direction can be computed exactly when damping is held fixed.

The current full-ANDES evidence is also limited. In no-PSS and base IEEE39 checks, retained mechanical GENROU.M perturbations had 5/10 negative margin effects and the local NEP sensitivity matched full ANDES. In Mix60, the critical mode was control-family and the retained GENROU.M signal was too weak. No GFM virtual-inertia claim is available until a physical REGF1 inertia knob is identified.

4.3 The line Braess dichotomy

At Level 0, small-signal Braess-like damping degradation is blocked for underdamped proportional modes because line actions move modal frequency but not decay. At Level 1, off-diagonal modal damping allows a line action to move both the frequency and the decay rate. The damping-ratio derivative is

$$\frac{\partial \zeta}{\partial \rho} = -\frac{1}{|s|} \operatorname{Re} \left(\frac{\partial s}{\partial \rho} \right) + \frac{\operatorname{Re}(s)}{|s|^3} \operatorname{Re} \left(\bar{s} \frac{\partial s}{\partial \rho} \right), \quad (14)$$

so a line can improve a graph stiffness proxy while reducing damping ratio. The project found such reversals in surrogate and NEP audits. The resonant Level-2 version is stronger and appears below.

4.4 Second-order modal correction

Let s_0 be a diagonal-screen pole of the critical mode c :

$$p_c(s_0) = s_0^2 + \delta_c s_0 + \nu_c = 0. \quad (15)$$

To second order in E , eliminating the noncritical modal coordinates gives

$$\Delta s_c = -\frac{s_0^2}{2s_0 + \delta_c} \sum_{\ell \neq c} \frac{E_{c\ell} E_{\ell c}}{s_0^2 + \delta_\ell s_0 + \nu_\ell} + O(\|E\|^3), \quad (16)$$

up to sign convention depending on whether the eliminated term is moved to the left or right side. The useful object is the denominator:

$$H_{\ell\ell}(s_0) = \frac{1}{s_0^2 + \delta_\ell s_0 + \nu_\ell}. \quad (17)$$

Then the correction is a rational graph/modal filter:

$$\Delta s_c \propto [E H(\Lambda, \Delta; s_0) E]_{cc}. \quad (18)$$

This is not a new perturbation theorem. Its value in the project is interpretive: each term says which noncritical mode couples into the critical margin and how strongly its modal denominator amplifies the effect.

The empirical status is mixed in a useful way. The correction improves the diagonal screen in weak-to-mild non-proportional regimes and in a reduced PLL model. It does not replace a large reduced QEP in clustered or strongly coupled tails. Therefore the estimator is a workflow:

diagonal screen $\rightarrow$ second-order correction $\rightarrow$ reduced/full QEP or NEP fallback.

5 Level 2: Rational Nonviscous Bridge

5.1 Schur self-energy

Let the full linearized ANDES state matrix be partitioned as

$$A = \begin{bmatrix} A_{ee} & A_{ec} \\ A_{ce} & A_{cc} \end{bmatrix}, \quad (19)$$

where x_e are retained network/interface states and x_c are condensed controller, exciter, stabilizer, PLL, governor, and auxiliary states. The Schur NEP is

$$S_e(s) = sI - A_{ee} - A_{ec}(sI - A_{cc})^{-1}A_{ce}. \quad (20)$$

When the retained coordinates are reduced to a second-order form, the same structure can be written as

$$T(s) = s^2 M + s \Sigma(s) + \tilde{L}, \quad \Sigma(s) = D_0 + C(sI - A_{cc})^{-1}B. \quad (21)$$

This is the Level-2 object. It is a rational nonviscous damping operator. It is also the correct place to connect power-system IBR controls to the structural-dynamics literature on hereditary and nonviscous damping.

5.2 Beyn contour validation

The Phase-0D gate solved (20) by Beyn contour integration. No fixed-point iteration was used for acceptance. The contour solver first extracted zeros from contour moments and only afterward compared them with ANDES poles.

Case	retained	condensed	pole family	f Hz	ζ	critical error
no-PSS	20	110	network	1.3705	0.15447	$4.861 \times 10^{-14}\%$ freq
base	20	140	network	1.3706	0.15443	$1.62 \times 10^{-13}\%$ freq
Mix60	11	167	control	13.7417	0.12357	$8.475 \times 10^{-11}\%$ freq

For Mix60, the matched pole was

$$s_{\text{crit}} = -10.7514 + j86.3419, \quad (22)$$

with $\pi_c = 0.853$. The nearest condensed-control pole was at distance 0.215 and the reconstructed condensed states were dominated by PLL1 and REGF1 components. The scalar bridge and fixed-point reduction did not capture this mode. This is the strongest full-ANDES result in the project.

5.3 Active nonviscous material

Passive nonviscous damping kernels in structural dynamics are usually designed to dissipate energy. Their extra nonviscous modes are not expected to become the least-damped oscillatory modes in the same way a PLL-family mode can in an IBR grid. In contrast, the condensed IBR/control self-energy can have poles and frequency bands that act like active material. A concise way to write the candidate test is to define a passivity defect:

$$\Pi_{\text{pass}} = \max_{\omega \in \Omega} \left[-\lambda_{\min} \left(\frac{\Sigma(j\omega) + \Sigma(j\omega)^*}{2} \right) \right]_+. \quad (23)$$

The conjectured rule is:

Conjecture 1 (Passive-kernel screening safety) *If Π_{pass} is small on the relevant band and the Level-1 Caughey defect*

$$\chi_{\text{CK}} = \frac{\|\tilde{L}\tilde{D} - \tilde{D}\tilde{L}\|_F}{\|\tilde{L}\|_F \|\tilde{D}\|_F} \quad (24)$$

is small, then Level-0/1 screens are reliable for network-family modes. If Π_{pass} is large, control-family modes can become critical and the full rational bridge is required.

This is not yet a theorem. The Gamma gate already showed that simple global non-normality and global χ do not explain certificate discrepancy strongly. The passivity defect must therefore be tested locally at the critical mode and controller-pole band, not sold as an established result.

6 Resonant Braess and Hidden Control Margin

For a line reinforcement parameter ρ , the Schur NEP sensitivity is

$$\frac{\partial s}{\partial \rho} = -\frac{y^H S_\rho(s)x}{y^H S_s(s)x}, \quad (25)$$

where x and y are right and left null vectors of $S_e(s)$. The derivative $S_s(s)$ includes the resolvent-square term

$$A_{ec}(sI - A_{cc})^{-2}A_{ce}. \quad (26)$$

This term is the signature of frequency-dependent control self-energy.

The Phase Braess-NEP gate decomposed the damping-ratio movement into retained network terms and a condensed-control term. A reinforcement was called resonant only if it satisfied all of the following:

1. the tracked network-family damping ratio decreased;
2. the tracked mode moved closer to a matched A_{cc} pole;
3. the nearest condensed pole was PLL-dominated;
4. the self-energy term dominated the retained terms;
5. the effect survived finite-difference NEP/ANDES comparison.

The gate found 40 strict resonant records among 348 reversals. The clearest case was Mix60/no-PSS Line_42 (20–34), mode rank 2:

$$\zeta_{\text{margin}} : 0.123566 \rightarrow 0.123558, \quad \zeta_{\text{network}} : 0.133884 \rightarrow 0.133816,$$

with distance to the matched A_{cc} pole $3.33227 \rightarrow 3.3279$. The self-energy component was -6.874×10^{-3} , while the retained terms were orders of magnitude smaller, and the finite-difference derivative was -6.860×10^{-3} .

This result should be written as a mechanism result. It is not a claim that every line reinforcement or every IBR grid exhibits resonant Braess behavior. The soft-GFL negative control moved the system away from the resonance band and removed the strict resonant cases.

7 Protected Resolvent Certificates and Coordination

7.1 Two margins

The modal margin used throughout the earlier project is

$$S_\zeta = \log \frac{\zeta_{\text{req}}}{\zeta_{\text{crit}}}, \quad (27)$$

where lower is better and $S_\zeta \leq 0$ is safe. The protected-resolvent margin is

$$S_g = \log g_{\text{prot}}, \quad g_{\text{prot}} = \sup_{\omega \in \Omega} \sigma_1(\Gamma_y C(j\omega I - A)^{-1} B \Gamma_u). \quad (28)$$

The Gate 1 baseline used identity protection weights. The Gamma gate repeated the test with physically motivated and random diagonal protection weights.

7.2 Bi-orthogonal resolvent sensitivity

Let ω^* be a simple maximizer of (28), and let u^*, v^* be the leading left and right singular vectors of

$$H^* = \Gamma_y C(j\omega^* I - A)^{-1} B \Gamma_u.$$

Set

$$\phi^{\text{in}} = (j\omega^* I - A)^{-1} B \Gamma_u v^*, \quad (29)$$

$$\phi^{\text{out}} = (j\omega^* I - A)^{-H} C^H \Gamma_y^H u^*. \quad (30)$$

For a parameter p that changes A but not the protection weights, the local sensitivity is

$$\frac{\partial S_g}{\partial p} = \frac{1}{g_{\text{prot}}} \text{Re} \left[(\phi^{\text{out}})^H \frac{\partial A}{\partial p} \phi^{\text{in}} \right], \quad (31)$$

provided the peak and singular value are simple. This is standard singular-value perturbation applied to the resolvent. Its value is that all levers can be priced in the same margin units.

For the swing first-order matrix

$$A = \begin{bmatrix} 0 & I \\ -M^{-1}L & -M^{-1}D \end{bmatrix},$$

write $\phi = (\phi_\theta, \phi_\omega)$. Then local damping at bus i has

$$\frac{\partial A}{\partial D_i} = \begin{bmatrix} 0 & 0 \\ 0 & -M_i^{-1} e_i e_i^T \end{bmatrix}, \quad (32)$$

so

$$\frac{\partial S_g}{\partial D_i} = -\frac{1}{g_{\text{prot}} M_i} \text{Re} \left[\overline{\phi_{\omega,i}^{\text{out}}} \phi_{\omega,i}^{\text{in}} \right]. \quad (33)$$

This is the corrected version of the Gemini suggestion. It is not generally $-|\phi_i^{\text{out}}|^2$ unless the system is normal or the input and output spatial vectors align. The important point is the bi-orthogonal product. In a non-normal grid, the forcing vector and response vector can be spatially different. That is the mathematical source of spatial decoupling.

For a line $e = (i, j)$ with incidence b_e , L changes by $b_e b_e^T$, giving

$$\frac{\partial A}{\partial w_e} = \begin{bmatrix} 0 & 0 \\ -M^{-1} b_e b_e^T & 0 \end{bmatrix}, \quad (34)$$

and therefore

$$\frac{\partial S_g}{\partial w_e} = -\frac{1}{g_{\text{prot}}} \text{Re} \left[(\phi_\omega^{\text{out}})^H M^{-1} b_e b_e^T \phi_\theta^{\text{in}} \right]. \quad (35)$$

This formula explains why a line action is not judged only by flow loading or graph centrality. It is judged by the overlap between the output response path and the input forcing path through that edge.

7.3 Gate results

Gate 1 tested 60 connected Watts-Strogatz cases with $n = 5, \dots, 9$ and two IEEE topology templates, with DC flow feasibility enforced after topology changes. Costs were fixed at

$$c_w = 1, \quad c_D = 2, \quad c_M = 8.$$

Two subtests were run. Nonlinear synergy was measured by

$$\text{Syn}_\epsilon(a, b) = S(p + \epsilon a + \epsilon b) - S(p + \epsilon a) - S(p + \epsilon b) + S(p), \quad (36)$$

normalized by $|\Delta S(a)| + |\Delta S(b)| + \eta$. Equal-cost coordination was measured by

$$CV(a, b) = \Delta S_{\text{equalcost}}(a, b) - \min\{\Delta S_{\text{equalcost}}(a), \Delta S_{\text{equalcost}}(b)\}. \quad (37)$$

Negative CV means the package beats the best individual lever at the same total cost.

The nonlinear-synergy headline died. $\text{SynNorm} < -0.10$ never exceeded 50% of cases. But equal-cost coordination under S_g survived:

Gamma profile	$w + D$	$w + M$	$D + M$
identity	72.1%	67.2%	51.6%
inertia output	73.8%	73.8%	48.4%
type input low inertia	73.8%	70.5%	48.4%

Thus $w + D$ and $w + M$ are supported in this reduced-model gate; $D + M$ is not.

The certificate discrepancy also survived Gamma changes. Ranking agreement between S_ζ and S_g was 29.0% for identity, 21.0% for inertia-output, 22.6% for type-input-low-inertia, and 29.0% for the frequency-output control. Eight random positive diagonal Gamma samples did not remove the discrepancy.

The mechanism is not yet explained. The maximum absolute Spearman correlation between the tested global descriptors and ranking disagreement was only 0.216. The correct status is therefore:

Certificate discrepancy is robust to tested Gamma choices, but the global non-normality descriptors and alignment descriptors used so far do not explain it well.

7.4 Flow-feasible resolvent retrofitting algorithm

The Gemini proposal can be included as an algorithmic candidate, but only after correcting the sensitivity formulas as above and adding the status labels.

This algorithm is not yet validated on full ANDES IBR. It is the natural next step for the protected-resolvent paper.

8 Weak Nodes, Weak Links, and Substitutability

The weak-element language should be mode- and certificate-specific. A bus is not weak in the abstract. It is weak for a margin, a mode family, and a lever set.

Definition 1 (Local weak node) *For a margin S and admissible local action set $\mathcal{A}_i$ at bus i , define*

$$W_i(S) = \max_{a \in \mathcal{A}_i} [\nabla S^T a]_+. \quad (38)$$

The node is weak if its best admissible perturbation can increase the risk margin to first order.

Algorithm 1 Flow-feasible protected-resolvent retrofitting

Require: Dynamic model A , input/output matrices B, C , protection weights Γ_u, Γ_y , DC dispatch P , flow limits $\bar{f}$, action costs.

- 1: Compute g_{prot} , ω^* , and singular vectors u^*, v^* of $\Gamma_y C(j\omega I - A)^{-1} B \Gamma_u$.
- 2: Compute ϕ^{in} and ϕ^{out} .
- 3: For each bus, compute damping and inertia exchange rates using (31). For inertia, use the actual parameter map; do not substitute a virtual-inertia formula unless the controller exposes one.
- 4: For each line, compute the topology exchange rate (35) and the DC flow sensitivity/feasibility penalty.
- 5: Solve a local cost-constrained problem

$$\min_z c^T z \quad \text{s.t.} \quad S_g + \nabla S_g^T z \leq 0, \quad |f(w + \Delta w; P)| \leq \bar{f}, \quad z \in \mathcal{Z}.$$

- 6: Validate the proposed package by recomputing the full eigensolve and the protected resolvent. Reject if the local prediction is not matched.
-

Definition 2 (Repairability) *For the same node, define*

$$R_i(S) = \max_{a \in \mathcal{A}_i, c^T a \leq 1} [-\nabla S^T a]_+. \quad (39)$$

Weakness and repairability are different. A node can be exposed but expensive to repair, or easy to repair but not currently exposed.

For a line e , replace $\mathcal{A}_i$ by admissible line actions and include the DC feasibility constraint. A line reinforcement that improves a graph stiffness score but increases S_ζ or S_g is a dynamic Braess candidate. It is flow-feasible only if

$$|f(w + \Delta w; P)| \leq \bar{f}. \quad (40)$$

Substitutability compares lever subspaces. In the simplest matrix-geometric picture, existing-line changes live in

$$\mathcal{L}_{\text{existing}} = \text{span}\{b_e b_e^T : e \in E\},$$

new-line changes live in missing-edge directions, and shunt/inertia-like changes live in diagonal directions. A mode-specific substitutability score asks whether the desired pole motion can be achieved by cheap line directions or requires local equipment. This is a useful planning language. The current status is formulation plus surrogate audits, not full ANDES validation.

9 Open Gates

1. **Adversarial literature search for active nonviscous power grids.** Search terms must include active viscoelastic structures, non-dissipative nonviscous damping, active material damping kernels, and power-grid NEPs.
2. **Passivity-defect gate.** Compute (23) for no-PSS, base, Mix60, soft-GFL controls, and future calibrated IBR cases. Check whether it predicts when the scalar bridge fails.

3. **Local discrepancy mechanism.** The Gamma gate rejected global $d_F(A)$ and global χ as strong explanations. Test local descriptors: peak resolvent frequency, singular-vector spatial separation, critical-mode condition number, and local self-energy denominator fraction.
4. **Estimator validation over the NEP bridge.** Run diagonal, second-order, reduced-QEP, and full NEP/ANDES comparisons on calibrated IBR operating ranges.
5. **Planning validation.** Compare weak-node, weak-link, conversion, and coordinated-retrofitting rankings against full recomputation and against SCR, impedance, Fiedler, flow loading, damping participation, and published damping-index baselines.

10 What Is Actually New

The safest novelty statement is:

The project formulates detailed IBR small-signal damping margin analysis as a controller aware rational self-energy problem, validates the Schur NEP bridge against full ANDES eigensolves including a control-family critical mode, and uses that bridge to separate network-family and controller-family mechanisms.

The strongest current pieces are:

1. **Validated rational bridge.** The Phase-0D result is precise, non-circular, and full-ANDES-based.
2. **Control-family criticality.** Mix60's PLL-family 13.7417 Hz mode is the anchor that explains why static and fixed-point bridges fail.
3. **Resonant Braess mechanism.** Strict but minority cases show line reinforcement moving a network-family mode toward a PLL-dominated control pole.
4. **Certificate discrepancy.** S_ζ and S_g disagree under several Gamma choices; $w + D$ and $w + M$ packages have protected-resolvent coordination value. The mechanism remains open.
5. **Unified hierarchy.** The hierarchy itself is a synthesis, not a single theorem. Its value is that it prevents overclaim: each broken classical guarantee maps to a specific planning risk.

11 Submission Map

Paper 1: Controller-aware rational bridge

Core content: Level 2 bridge, Beyn contour validation, Mix60 control-family mode, resonant Braess mechanism, scalar/fixed-point failures, and negative controls. This is the nearest to a defensible submission.

Paper 2: Active nonviscous IBR grids

Core content: damping-operator hierarchy and active-material framing. This requires the passivity-defect gate and an adversarial literature search before it can be claimed.

Paper 3: Flow-feasible protected-resolvent retrofitting

Core content: bi-orthogonal resolvent sensitivity, spatial decoupling, flow-feasible retrofit LP, and Gamma-robust certificate discrepancy. This needs local mechanism descriptors and stronger comparisons against H2, SCR, impedance, Fiedler, and transmission-switching baselines.

Acknowledgment of Dead Ends

The project produced several attractive hypotheses that did not survive gates. They should remain in the internal ledger because they prevent future backsliding:

- Nonlinear three-lever synergy as a headline: killed by Gate 1.
- Universal inertia Braess: killed by fixed-ratio and Monte Carlo controls.
- Second order always beating large reduced QEP: killed by cross-checks.
- Inertia-control cross term as frequent ANDES mechanism: killed by 0/13 measurable cases.
- Virtual-inertia placement over the controller-aware bridge: blocked by missing physical GFM inertia parameter in REGF1 sheets.
- Global non-normality or global commutator defect as explanation for certificate discrepancy: weak in Gamma gate, not killed as a family but not currently supported.

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Controller-Aware Rational Self-Energy for Damping-Margin Analysis of IBR Grids

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Abstract—The replacement of synchronous generation by inverter-based resources changes not only grid inertia but also the location of damping itself. In detailed small-signal models, much of the observed damping can be supplied by controller, exciter, stabilizer, PLL, and inverter states rather than by a static nodal damping coefficient. This paper argues that the appropriate reduced object for such systems is therefore not a constant-coefficient scalar QEP, but a controller-aware rational self-energy,

$$\Sigma(s) = D_0 + C(sI - A_{cc})^{-1}B,$$

or, equivalently, a Schur nonlinear eigenvalue problem (NEP) for the retained network/interface states. The main full-ANDES result is a bridge audit on IEEE 39-bus cases: the static scalar bridge fails because $\text{GENROU.D}=0$ does not contain the measured modal damping, while the rational Schur NEP solved by Beyn contour integration extracts the same critical poles as the full ANDES eigensolve in no-PSS, packaged-base, and Mix60 SG-to-IBR cases. In Mix60, the bridge captures a 13.7 Hz PLL-family critical mode with $\pi_c = 0.853$ that is missed by both the scalar bridge and a fixed-point Schur iteration. This bridge separates network-family and control-family criticality and enables mechanism audits without discarding controller dynamics. Around network-family modes, the paper also develops a second-order modal perturbation and rational graph-filter interpretation of non-proportional damping. Synthetic and reduced-PLL tests show where this correction improves a diagonal damping-region screen and where reduced-QEP fallback remains necessary; full estimator and planning validation over the controller-aware bridge remain the next registered experiments. The bridge also reveals a modal-local control-resonant reinforcement reversal: in a minority of Mix60/no-PSS line tests, strengthening a line moves a network-family mode toward a PLL-dominated condensed-control pole and the self-energy term sets the damping decrease; the mechanism disappears under a soft-GFL negative control. Finally, an input–output extension derives a protected-resolvent graph-filter sensitivity that gives a common local exchange rate for topology, damping, inertia, and controller actions. Reduced-model gates show robust disagreement between modal damping rankings and protected-resolvent rankings and support coordinated $w + D$ and $w + M$ packages as candidates; those claims remain pending full ANDES flow-feasible validation.

Index Terms—Damping ratio, inverter-based resources, eigenvalue perturbation, phase-locked loop, quadratic eigenvalue problem, rational graph filter, graph signal processing, small-signal stability.

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This manuscript is a pre-submission research draft. The method, synthetic stress tests, reduced PLL tests, and an ANDES controller-aware bridge validation are reported explicitly; full estimator and planning validation over calibrated ANDES IBR operating ranges remains required before any final journal submission.

I. INTRODUCTION

Power systems are moving from synchronous-machine-dominated dynamics toward networks with high penetration of inverter-based resources (IBRs). The most visible effect of this transition is reduced inertia, but the more difficult small-signal effect is that damping is no longer proportional to inertia or to network stiffness. A synchronous generator contributes inertia and damping through electromechanical dynamics. A grid-following inverter contributes control damping through a phase-locked loop and current controller; a grid-forming inverter contributes droop or virtual-synchronous-machine behavior. These mechanisms are physically different and appear in the linearized model as heterogeneous, non-commuting dynamic coefficients.

The standard swing-equation intuition says that low-frequency inter-area modes are critical and that increasing network stiffness improves robustness. Those statements are useful but incomplete. The damping ratio of a mode is not simply a graph frequency. If a line reinforcement raises the oscillation frequency without increasing the modal decay rate enough, the damping ratio can decrease. Likewise, if damping is heterogeneous, the mode with lowest damping ratio need not be the Fiedler mode or the highest-frequency mode. Planning rules based only on λ_2 , Fiedler participation, or frequency sensitivities can therefore be misleading in an IBR-rich grid.

This paper asks a narrow question: what reduced spectral object preserves the damping-margin mechanisms of an IBR grid after controller states are condensed? The answer proposed here is a controller-aware rational self-energy. A static scalar damping coefficient is sufficient only when the condensed dynamics can be represented at the relevant frequency by a constant term. In the ANDES cases studied here, that condition fails: damping is supplied largely by controller and auxiliary states, so the reduced object must remain frequency dependent.

The central message is not that a new graph eigenvalue replaces small-signal analysis. The central message is that a rational Schur bridge preserves controller-mediated damping mechanisms that a static (L, M, D) bridge loses. Once that bridge is available, lower-order spectral perturbations can be used as interpretable screens, but they must be checked against the complete NEP or full eigensolve when controller resonances or clustered modes are present.

The paper makes three calibrated methodological contributions and two mechanism/bridge findings. The novelty claim is at the level of power-system formulation, interpretation, and validated mechanism, not at the level of inventing eigenvalue

perturbation theory or contour eigensolvers.

- It shows, on reproducible IEEE 39-bus ANDES cases, that a static scalar (L, M, D) bridge can fail because controller and auxiliary states carry the observed damping. The controller-aware rational Schur NEP is the reduced object that retains those effects.
- It solves that NEP by contour integration and classifies extracted zeros into network-family and control-family modes using condensed-state participation. This recovers the Mix60 13.7 Hz PLL-family critical pole missed by scalar and fixed-point reductions.
- It uses the damping-region plane (ν_k, δ_k) as the zero-order network picture. This is an interpretive coordinate system for heterogeneous IBR damping, not a claim that the proportional modal formula is new.
- It derives and tests a closed-form second-order correction for the critical damping-margin pole under off-diagonal modal damping. This correction explains, term by term, how non-proportional inverter damping couples the critical mode to other modes.
- It organizes the estimator as a diagnostic workflow: diagonal screen, second-order correction, and reduced-QEP fallback for clustered or strongly coupled regimes. The negative cases where second order worsens the tail are part of the method, because they justify the trigger; full-ANDES estimator validation remains a registered next experiment.
- It uses the bridge to identify a control-resonant reinforcement reversal: a line action can pull a network-family mode toward a PLL-dominated condensed-control pole, so the self-energy term drives a damping decrease. This is reported as a modal-local mechanism with a negative control, not as a universal Braess paradox.
- It derives a protected-resolvent graph-filter sensitivity for non-normal IBR grids. This input-output certificate is separate from the damping-ratio pole metric and exposes spatial decoupling between forcing and response. Current evidence supports it as a reduced-model planning candidate, not yet as a full-ANDES flow-feasible planner.

Unified spectral-attribution object. All pieces of the paper are organized by the same retained-network NEP,

$$T(s) = s^2 M + s \Sigma(s) + \tilde{L}, \quad \Sigma(s) = D_0 + C(sI - A_{cc})^{-1} B. \quad (1)$$

This operator is used as an attribution ledger. The network lever is $\tilde{L}$, the inertia or virtual-mass lever is M , and the controller lever is the rational self-energy $\Sigma(s)$. A damping-margin claim is therefore not accepted because one score is large; it must specify which lever moved the pole, which competing explanation was controlled, and whether the reduced prediction agrees with the full NEP or full ANDES eigensolve. In this ledger, the static damping-region screen is the zero-order network picture, the second-order correction is the local modal-coupling picture, the Schur self-energy is the controller-aware full bridge, and the reinforcement audit is a pole-sensitivity mechanism test. The possible inertia-control cross term is retained as local theory, but the initial Phase-E0 ANDES tests found 0/13 measurable singleton GENROU–

PLL cross cases; it is therefore not used as the empirical headline of this draft.

The present manuscript is deliberately calibrated. Proportional modal damping, Rayleigh sensitivity, QEP sensitivity, contour NEP solvers, Schur complements, and eigenvalue perturbation are established tools. The candidate contribution is to assemble them into a controller-aware damping-margin bridge for IBR grids, document why the static scalar bridge fails, and use the retained self-energy to expose network-family versus control-family mechanisms. Synthetic stress tests and a reduced PLL model are reported to identify how a cheap second-order estimator behaves. The IEEE 39-bus ANDES evidence validates the rational bridge and one bridge-enabled mechanism audit. It does not by itself validate the full estimator, conversion-ranking, or planning claims.

II. RELATED WORK

Small-signal power-system stability is a mature field; standard references include modal analysis of electromechanical oscillations and damping-ratio sensitivity [1], [2]. Spectral and system-theoretic approaches to power networks have clarified the role of Laplacian eigenvalues, inertia, and damping in low-inertia performance [3]–[5]. In the proportional or homogeneous case, modal frequencies scale as $\sqrt{\nu_k}$ and damping ratios take a closed form involving damping, inertia, and graph eigenvalues. Those results are essential background for this paper rather than its novelty.

IBR-dominated systems complicate this picture because their controls introduce additional states, time scales, and non-proportional damping. Recent small-signal studies of inverter networks emphasize the role of line dynamics, operating conditions, and converter-control parameters [6]–[8]. Stability-region and security-region viewpoints for converter-dominated systems are also active research topics [9], [10]. This activity creates both an opportunity and a risk: region language is useful, but broad claims of originality are unsafe unless the exact reduced spectral construction is differentiated carefully.

The proposed damping-region picture is inspired by consensus-region theory in multi-agent systems. In that literature, stability of a networked linear system can be tested by asking whether scaled nonzero Laplacian eigenvalues lie in a region determined by local agent dynamics and feedback gains [11]. Related networked-control books develop dynamic topologies, switching graphs, dwell-time logic, resilient consensus, delays, and event-triggered control [12], [13]. The present paper does not import those results wholesale. A power grid with IBR controls is not a first-order consensus system. The transfer is conceptual: define the region induced by local dynamics and use graph-modal coordinates to screen network-level stability.

Quadratic eigenvalue problems provide the natural algebraic language for second-order small-signal models [14]. If damping and stiffness commute, the QEP decouples into scalar modal equations. If not, modes are coupled through off-diagonal modal damping terms. Modal perturbation for non-proportionally damped systems is well established in structural dynamics [15], [16]; this paper does not claim that perturbation

theory itself is new. The proposed distinction is the power-system use of the second-order term as a damping-margin decomposition and as a rational graph filter of $(\tilde{L}, \tilde{D})$.

Planning-oriented damping metrics also have nearby prior art. Grid-forming inverter placement by composite frequency and damping indices has been studied on modified IEEE 39-bus systems [17]. Closed-form damping-ratio sensitivities based on QEP formulations have also been derived for power-system remedial actions [18]. The line-reinforcement and weak-node maps in this manuscript must therefore be read as applications of the proposed margin object and must be compared against these baselines. The validation platform used here is ANDES, an open-source symbolic-numeric power-system modeling tool with power flow, time-domain, and eigenvalue analysis [19], [20].

III. NETWORK MODEL AND DAMPING REGION

Consider a lossless small-signal power network linearized around an operating point:

$$M\ddot{\theta} + D\dot{\theta} + L\theta = p, \quad (2)$$

where $M = \text{diag}(M_i) \succ 0$, $D = \text{diag}(D_i) \succeq 0$, and L is the weighted Laplacian induced by the active-power Jacobian. In the usual approximation,

$$L_{ij} = -V_i V_j b_{ij} \cos(\theta_i^* - \theta_j^*), \quad i \neq j, \quad (3)$$

with diagonal entries chosen so that $L\mathbf{1} = 0$. The analysis is performed in inertia-normalized coordinates $\vartheta = M^{1/2}\theta$, giving

$$\ddot{\vartheta} + \tilde{D}\dot{\vartheta} + \tilde{L}\vartheta = M^{-1/2}p, \quad (4)$$

where

$$\tilde{L} = M^{-1/2}LM^{-1/2}, \quad \tilde{D} = M^{-1/2}DM^{-1/2}. \quad (5)$$

Let $\tilde{L}q_k = \nu_k q_k$, with $Q = [q_1, \dots, q_n]$ orthonormal. The zero eigenvalue associated with uniform angle shift is discarded. In modal coordinates $x = Q^T \vartheta$, the dynamics are

$$\ddot{x} + \Gamma \dot{x} + \Lambda x = Q^T M^{-1/2}p, \quad (6)$$

where $\Gamma = Q^T \tilde{D}Q$ and $\Lambda = \text{diag}(\nu_k)$. The diagonal entries

$$\delta_k = \Gamma_{kk} = q_k^T \tilde{D}q_k \quad (7)$$

are modal damping coefficients, while the off-diagonal entries $\Gamma_{\ell k}$ describe damping-induced modal coupling.

The exact small-signal poles are roots of the QEP

$$P(s)v = (s^2 I + s\tilde{D} + \tilde{L})v = 0. \quad (8)$$

Equivalently, they are eigenvalues of the first-order matrix

$$A = \begin{bmatrix} 0 & I \\ -M^{-1}L & -M^{-1}D \end{bmatrix}. \quad (9)$$

The damping margin is

$$\zeta_{\min} = \min_{s_j \in \mathcal{S}_{\text{osc}}} \frac{-\text{Re}(s_j)}{|s_j|}, \quad (10)$$

where $\mathcal{S}_{\text{osc}}$ contains complex oscillatory poles.

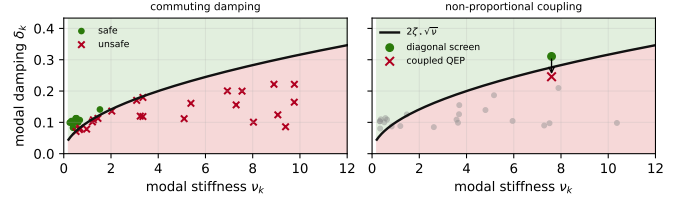


Fig. 1. Damping-region view. Left: in the commuting case, a target ζ_* becomes the exact boundary $\delta = 2\zeta_*\sqrt{\nu}$. Right: in non-proportional damping, a diagonal modal point can appear safe while QEP coupling moves the pole below the margin. The region is therefore a screen, not a certificate, outside the commuting regime.

A. Damping Region

If $\tilde{L}$ and $\tilde{D}$ commute, then Γ is diagonal and each nonzero mode solves

$$s^2 + \delta_k s + \nu_k = 0. \quad (11)$$

For an underdamped mode,

$$\zeta_k = \frac{\delta_k}{2\sqrt{\nu_k}}. \quad (12)$$

Thus a target damping margin ζ_* is equivalent to

$$\delta_k \geq 2\zeta_*\sqrt{\nu_k}, \quad k = 2, \dots, n. \quad (13)$$

Equation (13) defines the damping region shown in Fig. 1. Modes below the curve violate the target damping margin. Modes above it satisfy the target in the proportional case.

This region should not be overinterpreted. In the non-proportional case, Γ is not diagonal, and (12) is no longer exact. It remains a first-order approximation when damping is weak and modal frequencies are well separated. It can fail when low-damping modes are clustered or when $\Gamma_{\ell k}$ is large relative to the modal frequency separation. This observation motivates an adaptive estimator rather than a single formula.

Proposition 1 (Exact region in the commuting case): Suppose $\tilde{L}$ and $\tilde{D}$ are symmetric and share an orthonormal eigenbasis on the nonzero-angle subspace. Then the damping margin of (4) is

$$\zeta_{\min} = \min_{\nu_k > 0} \frac{\delta_k}{2\sqrt{\nu_k}}, \quad (14)$$

provided the limiting modes are underdamped. Moreover, a target margin ζ_* is satisfied if and only if every nonzero modal point obeys

$$\delta_k \geq 2\zeta_*\sqrt{\nu_k}. \quad (15)$$

The proof follows by diagonalizing both matrices in (4). It is stated explicitly because it fixes the boundary of the proposed framework. The region is exact only when damping is diagonal in the graph-modal basis. In every other case, the same inequality is a screen, not a certificate.

Proposition 2 (First-order modal-coupling warning): Let ν_k and ν_ℓ be two distinct inertial-Laplacian eigenvalues and let $\Gamma_{\ell k} \neq 0$. The sensitivity of modal damping to a perturbation that rotates q_k toward q_ℓ scales with

$$\frac{\Gamma_{\ell k}}{\nu_k - \nu_\ell}. \quad (16)$$

TABLE I

NEAREST PRIOR ART AND NOVELTY BOUNDARY. THE PAPER DOES NOT CLAIM NOVELTY FOR GENERIC PERTURBATION THEORY, QEP SENSITIVITY, OR DAMPING-RATIO PLACEMENT METRICS.

Reference area	What is already established	What is not claimed here	Candidate distinction here
Non-proportional modal perturbation [15], [16]	Perturbation methods and indices for non-classically damped second-order systems.	The perturbation expansion itself is not new.	Use the second-order term as a damping-margin decomposition for IBR grids and express it as a rational filter of $(\tilde{L}, \tilde{D})$.
GFM placement by stability indices [17]	Placement/ranking of grid-forming inverters on modified IEEE 39 cases using frequency and damping-related indices.	Ranking inverter placement by a damping metric is not new.	Compare conversion decisions using the proposed margin estimator and report regret against full eigensolve baselines.
QEP damping-ratio sensitivities [18]	Closed-form QEP sensitivities for damping-ratio remedial actions in synchronous-generator models.	QEP sensitivity formulas are not new.	Apply damping-ratio sensitivity to line reinforcement and IBR-oriented margin maps, with frequency-only ranking as a negative control.

Thus clustered graph frequencies and large off-diagonal modal damping are the cases in which the diagonal region is least reliable.

This warning is diagnostic, not a replacement for the full QEP. It explains why two networks with the same diagonal modal points can have different damping margins: the missing information is the coupling geometry in $Q^T \tilde{D} Q$, weighted by modal separation. This is the mechanism behind the adaptive estimator in the next section.

Remark 1 (Critical mode selection): For damping ratio, the critical graph mode is not generically the Fiedler mode. In proportional damping with homogeneous damping per inertia, the smallest damping ratio may be controlled by the largest ν_k because ζ_k scales like $1/\sqrt{\nu_k}$. In heterogeneous IBR cases, the critical mode is determined by the pair (ν_k, δ_k) and by modal coupling. Any benchmark that compares only the Fiedler-mode estimate against the full eigensolve is therefore using the wrong control.

IV. ADAPTIVE SPECTRAL MARGIN ESTIMATOR

The estimator uses the diagonal region as a cheap screen, a second-order perturbation as the closed-form correction, and the reduced QEP as a fallback. The diagonal all-mode estimate is

$$\hat{\zeta}_{\text{diag}} = \min_{k: \nu_k > 0} \frac{q_k^T \tilde{D} q_k}{2\sqrt{\nu_k}}. \quad (17)$$

This is not the Fiedler-mode estimate. It takes the minimum over all modes. In homogeneous damping the critical mode is often the largest-frequency mode; in heterogeneous damping it need not be.

A. Closed-Form Second-Order Correction

Let $\Gamma = \Delta + E$, where $\Delta = \text{diag}(\delta_k)$ and E contains the off-diagonal modal damping. The zero-order diagonal pole of the critical mode c is s_0 , where

$$s_0^2 + \delta_c s_0 + \nu_c = 0. \quad (18)$$

Let $p_\ell(s) = s^2 + \delta_\ell s + \nu_\ell$. Partition the modal coordinates into the critical coordinate z_c and all remaining coordinates z_R . To second order in the off-diagonal damping, the coupled modal equations are

$$p_c(s)z_c + sE_{cR}z_R = 0, \quad (19)$$

$$\text{diag}(p_\ell(s))z_R + sE_{Rc}z_c = 0, \quad (20)$$

where the E_{RR} block would contribute only higher-order terms to the scalar critical equation. Thus

$$z_R = -s \text{diag}(p_\ell(s))^{-1} E_{Rc} z_c + O(\|E\|^2). \quad (21)$$

Substitution into the critical equation gives

$$p_c(s) - s^2 E_{cR} \text{diag}(p_\ell(s))^{-1} E_{Rc} = 0 + O(\|E\|^3). \quad (22)$$

Writing $s = s_0 + \Delta s_c$ and evaluating the resolvent at the critical pole s_0 yields the second-order pole correction

$$\Delta s_c = \frac{s_0^2}{2s_0 + \delta_c} \sum_{\ell \neq c} \frac{E_{c\ell} E_{\ell c}}{s_0^2 + \delta_\ell s_0 + \nu_\ell}. \quad (23)$$

The corrected pole is $s_c^{(2)} = s_0 + \Delta s_c$ and the corrected margin is

$$\hat{\zeta}_2 = -\frac{\text{Re}(s_c^{(2)})}{|s_c^{(2)}|}. \quad (24)$$

Equation (23) is the central estimator in this manuscript. It is not a new perturbation theorem; it is a power-system use of second-order eigenvalue perturbation to quantify how non-proportional inverter damping moves the critical damping-margin pole away from the diagonal network prediction.

B. Joint Inertia–Damping Perturbation

The previous correction assumes that $\tilde{L}$ is fixed and that only off-diagonal modal damping perturbs the diagonal QEP. A finite SG-to-IBR conversion, however, changes both inertia and damping. Let a small conversion parameter ρ define

$$M(\rho) = M_0 + \rho \Delta M, \quad D(\rho) = D_0 + \rho \Delta D, \quad (25)$$

and define

$$\Lambda_1 = Q^T \frac{d\tilde{L}}{d\rho} Q, \quad \Gamma_1 = Q^T \frac{d\tilde{D}}{d\rho} Q \quad (26)$$

in the unperturbed inertial-Laplacian basis. The first-order shift of the critical diagonal pole is

$$s_1 = -\frac{\Lambda_{1,cc} + s_0 \Gamma_{1,cc}}{2s_0 + \delta_c}. \quad (27)$$

The local pole approximation has the form

$$s_c = s_0 + s_1 + s_{2,\text{coup}} + \text{diagonal self terms} + O(\rho^3). \quad (28)$$

The next equation isolates only the modal-coupling part of the second-order term; it is not the complete finite-conversion pole

shift. The off-diagonal second-order coupling contribution is obtained by replacing the perturbation sE in (23) with the combined channel

$$H_1(s_0) = s_0\Gamma_1 + \Lambda_1. \quad (29)$$

Thus

$$s_{2,\text{coup}} = \frac{1}{2s_0 + \delta_c} \sum_{\ell \neq c} \frac{N_{c\ell}(s_0)}{s_0^2 + \delta_\ell s_0 + \nu_\ell}, \quad (30)$$

$$N_{c\ell}(s_0) = \left(s_0 E_{c\ell}^{(D)} + E_{c\ell}^{(L)} \right) \left(s_0 E_{\ell c}^{(D)} + E_{\ell c}^{(L)} \right). \quad (31)$$

where $E^{(D)} = \text{offdiag}(\Gamma_1)$ and $E^{(L)} = \text{offdiag}(\Lambda_1)$. If $\Delta M = 0$, then $E^{(L)} = 0$ and (30) collapses to (23). Expanding the numerator exposes the non-additive term

$$s_0 \left(E_{c\ell}^{(D)} E_{\ell c}^{(L)} + E_{c\ell}^{(L)} E_{\ell c}^{(D)} \right), \quad (32)$$

which is the explicit red-control interaction: damping and inertia perturbations do not generally move the critical pole as two separable scalar effects.

Equation (30) is used here as a mechanism statement and as a local score for small virtual-inertia or droop retuning. It should not be presented as a reliable estimator for a large finite SG-to-IBR conversion. For such conversions, the adaptive estimator or the full model must be recomputed after each candidate action. The full second-order expansion also contains scalar self terms from the diagonal first-order shift s_1 ; (30) isolates the modal-coupling contribution that generalizes the rational-filter correction.

C. Rational Graph-Filter Form

The same correction can be written in graph-signal-processing language. A polynomial filter of the inertial Laplacian,

$$h(\tilde{L}) = \sum_{m=0}^{M_h} a_m \tilde{L}^m, \quad (33)$$

acts diagonally in the eigenbasis of $\tilde{L}$: $h(\tilde{L})q_k = h(\nu_k)q_k$. It can rescale modal components, but it cannot create the off-diagonal modal damping entries that are absent from a graph-frequency-only description. This is the structural reason why a polynomial filter of $\tilde{L}$ alone is not a damping-margin model for a non-proportionally damped IBR grid.

Define the pole-dependent rational filter

$$H(\Lambda; s_0) = \text{diag} \left(\frac{1}{s_0^2 + \delta_\ell s_0 + \nu_\ell} \right)_{\ell \neq c}, \quad (34)$$

with the critical index omitted from the diagonal action. Then (23) is equivalently

$$\Delta s_c = \frac{s_0^2}{2s_0 + \delta_c} [EH(\Lambda; s_0)E]_{cc}. \quad (35)$$

Proposition 3 (Rational pair-filter interpretation): The second-order margin correction is a rational filter of the dynamic operator pair $(\tilde{L}, \tilde{D})$ evaluated at the diagonal critical pole. It depends on $\tilde{L}$ through the modal frequencies ν_ℓ , on $\tilde{D}$ through δ_ℓ and $E = \text{offdiag}(Q^T \tilde{D} Q)$, and on the dynamics

Algorithm 1 Adaptive damping-margin estimator

- 1: Input: $\tilde{L}, M, \tilde{D}$, reduced size r , thresholds τ_η, τ_s
- 2: Build $\tilde{L} = M^{-1/2} L M^{-1/2}$ and $\tilde{D} = M^{-1/2} D M^{-1/2}$
- 3: Compute eigenpairs (ν_k, q_k) of $\tilde{L}$
- 4: Compute $\Gamma = Q^T \tilde{D} Q$ and $\hat{\zeta}_k = \Gamma_{kk} / (2\sqrt{\nu_k})$
- 5: Select the critical diagonal mode $c = \arg \min_k \hat{\zeta}_k$
- 6: Compute Δs_c from (23)
- 7: Select low-damping candidates $\mathcal{I}$ by sorting $\hat{\zeta}_k$
- 8: Expand $\mathcal{I}$ with nearby or strongly coupled modes
- 9: **if** $\eta(\mathcal{I}) < \tau_\eta$ and $|\Delta s_c|/|s_0| < \tau_s$ **then**
- 10: Return $\hat{\zeta}_2$ from (24)
- 11: **else**
- 12: Build $L_r = V^T \tilde{L} V$, $D_r = V^T \tilde{D} V$
- 13: Solve $(s^2 I_r + s D_r + L_r)w = 0$
- 14: Return the smallest damping ratio of the reduced roots
- 15: **end if**

through s_0 . It is therefore not reducible to a polynomial filter of $\tilde{L}$.

Proof: Equation (23) can be factored by collecting the modal denominators into the diagonal operator $H(\Lambda; s_0)$. The left and right multiplications by E inject and recover the non-proportional damping coupling. If $E = 0$, the filter output is zero and the proportional damping-region formula is exact. If $E \neq 0$, the correction weights each coupled mode by its resolvent-like separation from the critical pole. $\square$

This filter form is an interpretation of (23), not a second independent mechanism. Its value is that it states precisely what graph information is missing from $\tilde{L}$ alone: damping margin is governed by the pair $(\tilde{L}, \tilde{D})$ and by the pole at which the pair is evaluated.

The correction is used only when the perturbation is credible. If $|s_0^2 + \delta_\ell s_0 + \nu_\ell|$ is small for a neighboring mode, or if $|\Delta s_c|$ is comparable to $|s_0|$, the method should reject the correction and solve a reduced QEP instead.

Define a simple coupling score for a candidate mode set $\mathcal{I}$:

$$\eta(\mathcal{I}) = \max_{\substack{k, \ell \in \mathcal{I} \\ k \neq \ell}} \frac{|\Gamma_{\ell k}|}{|\sqrt{\nu_\ell} - \sqrt{\nu_k}| + \epsilon}, \quad (36)$$

where $\epsilon > 0$ prevents division by zero. Large η indicates that modal damping couples frequencies too strongly for a purely diagonal approximation. More elaborate diagnostics can include the coupling to nearby modes outside $\mathcal{I}$ and the residual of the reduced QEP.

Let $\mathcal{I}_r$ contain the modes with the smallest diagonal damping estimates and any neighboring modes triggered by (36). With $V = Q[:, \mathcal{I}_r]$, form

$$L_r = V^T \tilde{L} V, \quad D_r = V^T \tilde{D} V, \quad (37)$$

and solve

$$(s^2 I_r + s D_r + L_r)w = 0. \quad (38)$$

The adaptive estimate returns the diagonal estimate when the coupling diagnostic is small and the reduced-QEP estimate otherwise.

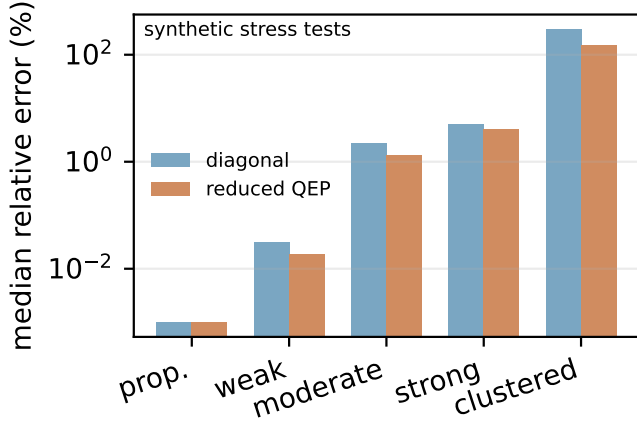


Fig. 2. Synthetic stress-test errors. The diagonal all-mode estimate is excellent in proportional and weak-damping cases, but not robust to strong non-proportional damping or clustered modal frequencies. The reduced QEP is a correction for those regimes, not a replacement for full validation.

D. Synthetic Stress Test

The purpose of the first numerical experiment is not to claim benchmark validity, but to identify mechanisms. Five ensembles were tested: proportional damping, weak heterogeneous damping, moderate heterogeneous damping, strong heterogeneous damping, and near-degenerate clustered modes. For each case the diagonal estimate, the second-order correction, and damping-selected reduced QEPs were compared against the full $2n \times 2n$ eigensolve.

Fig. 2 summarizes the median relative errors for the diagonal and reduced-QEP baselines. The same synthetic audit summarized in Table II shows that the second-order correction reduced the weak-IBR median error from 0.038% to 0.007%; in the underlying run, the 95th-percentile error also fell from 0.881% to 0.163%. In mild and strong heterogeneity, it improved the median error, but it did not reliably beat a larger reduced QEP and could worsen tail errors. This is exactly why the correction is embedded in a diagnostic workflow rather than used as a universal replacement for eigensolve-based validation.

This stress test supports the estimator’s logic. The contribution is not “always use the second-order formula” and not “the diagonal formula is always enough.” The contribution is the interpretable correction plus the diagnostic boundary between diagonal-safe, second-order-safe, and QEP-needed regimes.

E. Failure Mechanism of the Diagonal Screen

The reason for the regime dependence is visible in modal coordinates. Write

$$\Gamma = \Gamma_d + \Gamma_o, \quad (39)$$

where $\Gamma_d = \text{diag}(\delta_k)$ and Γ_o contains the off-diagonal modal damping. If $\Gamma_o = 0$, the QEP decouples. If Γ_o is small and modal frequencies are well separated, the pole perturbation induced by off-diagonal damping is small. If two modes are close, however, the same off-diagonal entry can rotate the critical subspace. A scalar estimate based on one unperturbed q_k then misses the actual least-damped combination of modes.

TABLE II
SYNTHETIC STRESS-TEST SUMMARY. VALUES ARE MEDIAN RELATIVE ERRORS AGAINST THE FULL FIRST-ORDER EIGENSOLVE. THE CLUSTERED CASE IS INTENTIONALLY PATHOLOGICAL AND EXPOSES THE FAILURE MECHANISM.

Regime	Diagonal	2nd order	QEP
Proportional damping	≈ 0	≈ 0	≈ 0
Weak heterogeneity	0.038%	0.007%	0.004–0.024%
Moderate heterogeneity	2.53%	0.65%	0.22–1.51%
Strong heterogeneity	5.16%	1.46%	1.24–3.90%
Clustered frequencies	308%	74%	9.8–160%

This mechanism explains why a single error number is not enough. A diagonal formula can be nearly exact in weak damping and badly wrong in clustered modes. A reduced QEP can improve clustered cases, but it is not magic: if the selected subspace omits a coupled neighbor, the reduced model still fails. The estimator must therefore report its own diagnostic label, not only a number.

F. Reduced PLL Check: Network Versus Control

To test whether the perturbation idea survives beyond the clean second-order surrogate, a reduced seven-state grid-following inverter model was evaluated. The three-node system contains one synchronous generator, one grid-forming inverter, and one grid-following inverter with PLL memory. The state vector is

$$x = [\theta_0, \theta_{pll}, \theta_2, \omega_0, \omega_{pll}, \omega_2, z_{pll}]^T, \quad (40)$$

where z_{pll} is the PLL integral state. The zero-order model sets the PLL memory coupling to zero, so the inverter behaves as a memory-free damping channel. Standard nonsymmetric eigenvalue perturbation then corrects the critical pole of this zero-order matrix.

In the canonical reduced case, the full critical pole is approximately $-0.034236 \pm j1.177983$. The zero-order pole is $-0.034827 + j1.178430$, and the second-order correction moves it to $-0.034154 + j1.177846$, reducing the complex-pole distance from 7.41×10^{-4} to 1.60×10^{-4} . Over 399 stable randomized reduced-PLL cases, the median damping-margin error falls from 2.16% at order zero to 0.09% at order two, while the 95th-percentile error falls from 29.2% to 12.9%.

This result supports the network-vs-control interpretation: the zero-order margin is the electromechanical network prediction, while the second-order term measures how PLL memory shifts the critical pole. It is not a substitute for ANDES validation. The model is reduced and deliberately transparent; current loops, limiters, algebraic network constraints, and vendor-grade IBR models remain outside this test.

V. ANDES IEEE 39-BUS SIMULATIONS

A. Baseline Synchronous Case

The first IEEE 39-bus experiment was run in ANDES 2.0.0 using the packaged `ieee39_full.xlsx` case. The case contains 39 buses, 46 lines, 10 GENROU machines, 10 IEEEEST/PSS devices, and no active REGCA1, REGCP1,

TABLE III
ANDES IEEE39 BASELINE SIMULATION SUMMARY.

Quantity	Value
Buses / lines	39 / 46
Synchronous machines	10 GENROU
IBR dynamic devices	0 REGCP1/REGF1/PLL1
Linearized eigenvalues	160
Oscillatory eigenvalues	64
Positive-real eigenvalues	9 non-oscillatory
Least-damped stable pole	$-1.3460 \pm j8.6115$
Damping ratio / frequency	0.1544 / 1.3706 Hz

TABLE IV
ANDES DIAGNOSTIC SIMULATIONS FOR POSITIVE-REAL MODES IN THE
PACKAGED IEEE39 DYNAMIC CASE.

Variant	Removed	Pos. modes	ζ_{osc}
Base	—	9	0.1544
No governor	TGOV1N	4	0.0568
No PSS	IEEEST	0	0.1545
No exciter/PSS	IEEEX1, IEEEST	0	0.1538
GENROU only	Controls	0	0.0606

REGF1, REGF2, or PLL1 devices. Thus it is a synchronous-generator benchmark, not a full IBR benchmark. Power flow and eigenanalysis completed. The linearized model has 160 eigenvalues, 64 oscillatory eigenvalues, two zeros, and nine positive-real non-oscillatory modes. The least-damped stable oscillatory pair is

$$s_{\star} = -1.3460 \pm j8.6115, \quad \zeta_{\star} = 0.1544, \quad (41)$$

with oscillation frequency 1.3706 Hz. The positive-real non-oscillatory modes are reported as a case-quality warning; they do not support a final practical stability claim.

B. Positive-Mode Diagnostic

A participation-factor audit was run to identify the source of those positive-real modes. The dominant positive modes participate primarily in TGOV1N, IEEEX1/IEEEST, and internal GENROU transient-voltage states rather than in electromechanical angle-frequency pairs. A control experiment removed model families from the packaged case. Removing only TGOV1N reduced the count from nine positive modes to four, while removing only IEEEST removed all positive-real modes; the least-damped oscillatory pair remained essentially unchanged at approximately $\zeta = 0.1545$ and 1.3705 Hz. Thus the positive modes appear to be a stabilizer-model artifact in the packaged dynamic case, not a network oscillation claim. The baseline is therefore useful for reproducible extraction tests, but the manuscript should not cite it as evidence of a physically unstable New England benchmark.

C. IEEE39-Derived Kron Surrogate

To exercise the proposed estimator on the IEEE 39 topology before a full IBR dynamic case is built, the solved network was converted into a generator-bus surrogate. The active-power stiffness Laplacian was assembled from the solved line series admittances and bus angles, then Kron-reduced from 39 buses

TABLE V
IEEE39-DERIVED SURROGATE ESTIMATOR ERRORS. VALUES ARE
RELATIVE ERROR AGAINST THE FULL QEP OF THE KRON-REDUCED
SURROGATE, NOT FULL IBR ANDES ERRORS.

Estimator	Median	95th pct.	Mean
Diagonal all-mode	0.468%	24.15%	4.18%
Second order	0.032%	9.91%	1.75%
Reduced QEP ($r = 6$)	0.037%	5.44%	1.09%
Adaptive	0.031%	7.43%	1.63%

to the 10 generator buses. IBR-like scenarios were generated by reducing effective inertia and introducing heterogeneous local damping at a penetration level ρ . The reference for this surrogate test is the full QEP of the Kron-reduced second-order model, not the full ANDES DAE.

Table V supports the same mechanism as the random ensembles: the second-order correction reduces the median error of the diagonal screen by about one order of magnitude, but the reduced QEP has the better tail. The adaptive workflow is therefore still necessary. In the representative $\rho = 0.6$ scenario, increasing local damping by 10% gives the largest margin improvement at generator bus 30 ($\Delta\zeta = 0.0112$), followed by buses 38 and 36. For line reinforcements, strengthening line 25–37 gives the largest margin increase ($\Delta\zeta = 0.00631$), while strengthening lines 2–3, 25–26, 1–2, 3–4, and 1–39 reduces the damping margin even though the critical modal stiffness increases. Fig. 3 overlays these surrogate planning results with the Mix60 replacement map and the Phase-1 ANDES audit status. This is a damping-aware Braess-like reversal in the surrogate, not yet a full ANDES IBR topology claim.

D. SG-to-GFL/GFM Replacement Cases

To close the gap between the packaged synchronous benchmark and a true IBR benchmark, a case-generation script was added for direct SG-to-IBR replacement in ANDES. The static PV/Slack devices are retained so that the original power-flow dispatch is preserved. For each converted generator, the associated GENROU, governor, exciter, and PSS rows are removed from the derived case. Grid-following replacements use REGCP1, REECA1, REPCA1, and PLL1; grid-forming replacements use REGF1. The slack generator is kept synchronous in all pilot cases.

Table VI gives the first full-ANDES construction result, and Fig. 4 shows the exact bus-level replacement map for the stable Mix60 pilot. All five derived cases load, solve power flow, and run eigenanalysis. Four low-penetration pilot cases retain positive-real modes under the uncalibrated generic controller parameters; the 60% mixed GFL/GFM case has no positive-real eigenvalues under the same tolerance. The stable oscillatory damping ratios in the table are therefore not used as final validation of the estimator. Their immediate value is methodological: the workflow can now generate reproducible IEEE39 cases with active REGCP1/PLL1 and REGF1 dynamics, and the next validation step is to calibrate controller parameters and extract a physically matched reduced surrogate from the full ANDES state matrix.

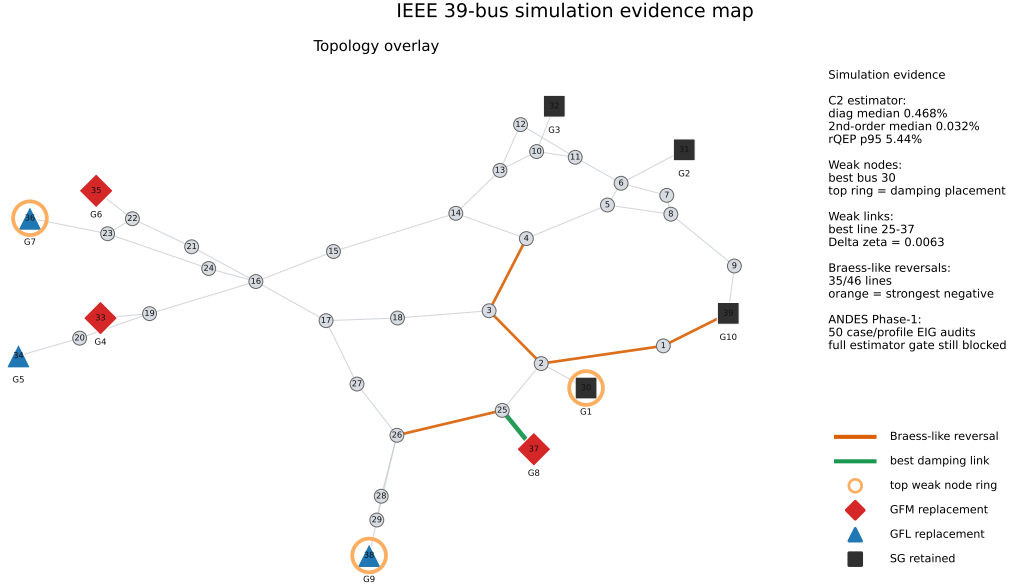


Fig. 3. IEEE 39-bus evidence map for the current draft. The topology overlay combines the Mix60 SG/GFL/GFM replacement map with IEEE39-derived surrogate results: orange lines are the strongest Braess-like damping reversals, the green line is the best damping-aware reinforcement, and orange rings mark the top weak-node damping-placement buses. The right panel lists the estimator and calibration-audit evidence used in the paper. This figure is a traceability map, not final validation against calibrated full-ANDES IBR poles.

TABLE VI

FULL ANDES IEEE39 SG-TO-IBR REPLACEMENT PILOT SIMULATIONS. ALL CASES SOLVED POWER FLOW AND EIGENANALYSIS. STABILITY IS ASSESSED FROM THE FULL LINEARIZED DAE WITH GENERIC UNCALIBRATED CONTROLLER PARAMETERS.

Case	GFL gens	GFM gens	GENROU left	Pos. modes	Max Re(s)	Stable	ζ_{osc} / Hz
GFL20	9,8	—	8	1	1.012	no	0.1148 / 13.7277
GFM20	—	9,8	8	6	4.291	no	0.1438 / 1.3910
Mix20	9	8	8	3	1.070	no	0.1370 / 13.7507
Mix40	9,7	8,6	6	2	1.072	no	0.1266 / 13.7856
Mix60	9,7,5	8,6,4	4	0	2.0×10^{-9}	yes	0.1236 / 13.7417

The frequency column in Table VI is computed as $|\text{Im}(s)|/(2\pi)$, so the entries near 13.7 Hz are not a rad/s-to-Hz unit error. A participation-factor audit shows that these high-frequency least-damped stable modes are dominated by control or auxiliary states, such as IEEEEST/IEEEEX1, REGF1, PLL1, or BusFreq states, rather than by a low-frequency inter-area electromechanical mode. This is consistent with the central warning of the paper: once IBR and controller states are active, the least-damped stable mode need not be the familiar low-frequency network mode.

E. Controller-Aware Rational Bridge by Contour Integration

The failed scalar bridge is not a failure of the full ANDES eigensolve; it is a failure of the coefficient model. In the packaged and derived cases, the physical GENROU.D entries are zero, while the observed modal damping is supplied by controller, exciter, governor, stabilizer, PLL, and inverter states. The appropriate reduced object is therefore not a constant-coefficient scalar QEP with a nodal D . It is a rational nonlinear eigenvalue problem (NEP) in which the condensed control states enter as a frequency-dependent self-energy.

Partition the full ANDES state matrix as

$$A = \begin{bmatrix} A_{ee} & A_{ec} \\ A_{ce} & A_{cc} \end{bmatrix}, \quad (42)$$

where x_e contains the retained network/interface states and x_c contains all condensed controller and auxiliary states. In second-order notation the bridge is

$$T(s) = s^2 M + s \Sigma(s) + \tilde{L}, \quad \Sigma(s) = D_0 + C(sI - A_{cc})^{-1} B. \quad (43)$$

The implementation solves the equivalent first-order Schur NEP

$$S_e(s) = sI - A_{ee} - A_{ec}(sI - A_{cc})^{-1} A_{ce}, \quad (44)$$

which avoids forcing an artificial $M, D_0, \tilde{L}$ factorization when grid-forming angle states and controller states are mixed. When the retained states are clean δ - ω pairs, (44) is the first-order form of (43).

IEEE 39-bus Mix60 SG-to-IBR replacement map

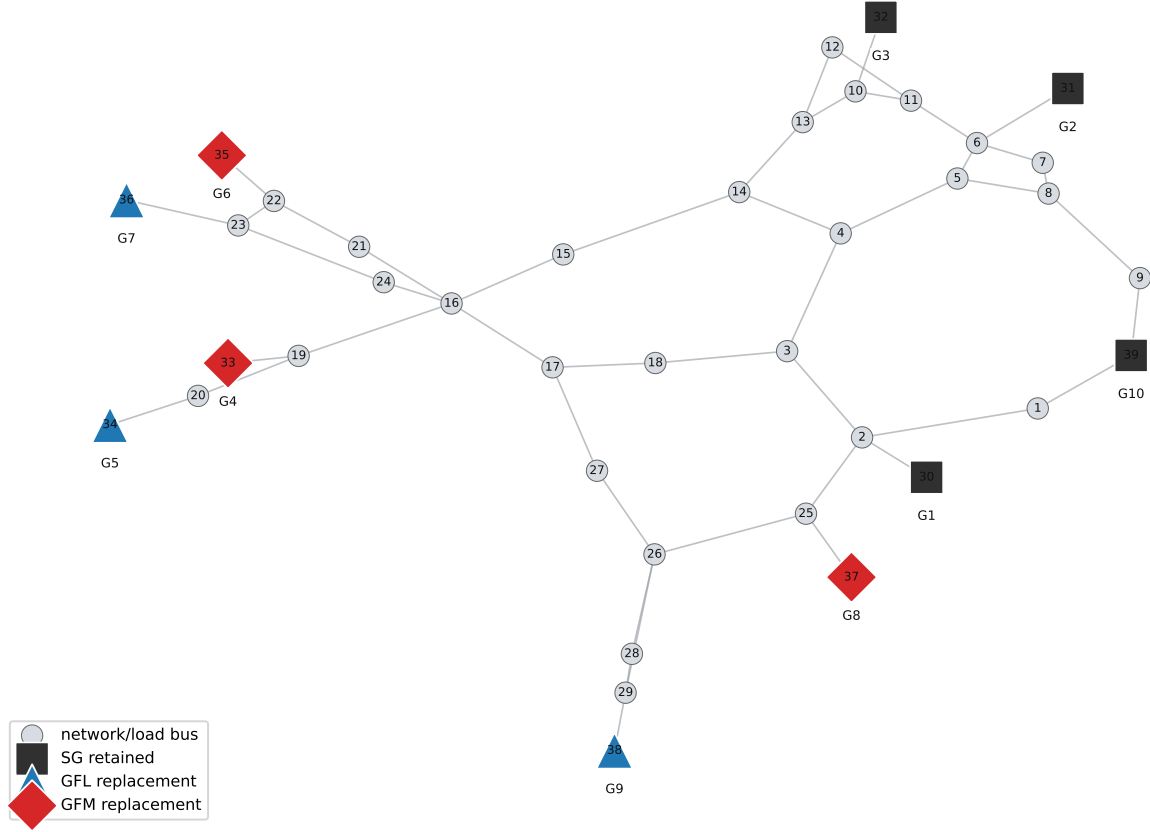


Fig. 4. IEEE 39-bus topology and the Mix60 SG-to-IBR replacement map used in the ANDES pilot. Generators 5, 7, and 9 are replaced by GFL devices at buses 34, 36, and 38; generators 4, 6, and 8 are replaced by GFM devices at buses 33, 35, and 37; generators 1, 2, 3, and 10 remain synchronous at buses 30, 31, 32, and 39. This figure documents the benchmark construction. It is not, by itself, estimator validation.

Zeros of (44) were extracted with Beyn contour integration. For a probing matrix $\hat{V}$, the moments are

$$A_0 = \frac{1}{2\pi j} \oint_{\Gamma} S_e(s)^{-1} \hat{V} ds, \quad (45)$$

$$A_1 = \frac{1}{2\pi j} \oint_{\Gamma} s S_e(s)^{-1} \hat{V} ds. \quad (46)$$

After the SVD of A_0 , the reduced matrix $U_K^* A_1 W_K \Sigma_K^{-1}$ gives the zeros inside Γ . A local polish was used only after the contour extraction: it starts from each Beyn Ritz value and solves for the minimum-modulus eigenvalue of $S_e(s)$ to vanish. The polish receives no ANDES pole, mode index, frequency, or damping target; ANDES poles are used only after extraction for validation.

Two contours were used. The *network-margin* contour covers $\text{Re}(s) \in [-4, 0.8]$ and $\text{Im}(s) \in [3.5, 12]$ rad/s. The *PLL-control* contour covers $\text{Re}(s) \in [-25, 0.8]$ and $\text{Im}(s) \in [60, 95]$ rad/s. Completeness was checked with the argument principle,

$$N_{\Gamma} = \frac{1}{2\pi j} \oint_{\Gamma} \text{tr} (S_e(s)^{-1} S_e'(s)) ds, \quad (47)$$

with the count corrected by the number of A_{cc} poles inside the contour. The extracted zero counts agree with (47): for no-PSS and base, the network-margin contour extracts 9 zeros versus argument counts 9.00006 and 9.00007; for Mix60, the network-margin contour extracts 4 zeros versus 4.00003, and the PLL-control contour extracts 3 zeros versus 2.999998 after adding the three A_{cc} poles inside the contour.

Table VII shows that the rational bridge reproduces the full ANDES critical poles in all three diagnostic cases. The important case is Mix60. The least-damped stable pole is not the 1.49 Hz electromechanical mode found by the fixed-point Schur iteration; it is a 13.7417 Hz control-family pole with $\zeta = 0.12357$. The NEP-Beyn bridge extracts that pole from the PLL-control contour with $\pi_c = 0.853$. The nearest condensed-control pole is $-10.5593 + j86.246$, at distance 0.215, and the reconstructed condensed states are led by PLL1 angle states. Thus the bridge identifies the mechanism that the scalar bridge and the fixed-point Schur iteration missed: the damping-margin mode can belong to the controller family.

Remark 2 (Fast-control limit): The static limit $\Sigma(0) = D_0 - CA_{cc}^{-1}B$ is a useful way to recover a constant-coefficient QEP only when A_{cc} is invertible. In the base and Mix60 audits,

TABLE VII
PHASE-0D CONTROLLER-AWARE RATIONAL BRIDGE VALIDATION. NEP-BEYN POLES ARE EXTRACTED BY CONTOUR INTEGRATION AND THEN COMPARED AGAINST THE FULL ANDES EIGENSOLVE. FREQUENCY AND DAMPING ERRORS ARE RELATIVE PERCENTAGES.

Case	ANDES critical pole	NEP-Beyn critical pole	Freq. error	Damping error	π_c / family
no-PSS	$-1.34635 + j8.61104$	$-1.34635 + j8.61104$	$4.86 \times 10^{-14}\%$	$6.29 \times 10^{-13}\%$	0.680 / I-network
base	$-1.34601 + j8.61148$	$-1.34601 + j8.61148$	$1.62 \times 10^{-13}\%$	$1.06 \times 10^{-12}\%$	0.653 / I-network
Mix60	$-10.7514 + j86.3419$	$-10.7514 + j86.3419$	$8.47 \times 10^{-11}\%$	$2.48 \times 10^{-9}\%$	0.853 / II-control

$sI - A_{cc}$ is singular or severely ill-conditioned at $s = 0$, consistent with integral controller states. Therefore $\Sigma(0)$ is not the robust validation object for these cases. The robust check is the contour NEP itself, evaluated against the full ANDES poles at the relevant nonzero critical frequencies.

F. NEP-Enabled Control-Resonant Reinforcement Reversal

The validated Schur NEP bridge also makes it possible to distinguish two different reasons why a line reinforcement can reduce damping ratio. The first is the geometric effect already visible in a scalar modal model: reinforcement can increase modal stiffness or frequency without increasing decay enough, so $\zeta = -\text{Re}(s)/|s|$ decreases. The second is a controller-mediated resonant effect: reinforcement can move a network-family mode toward a condensed-control pole, making the frequency-dependent self-energy dominate the pole sensitivity. The second mechanism is invisible to a scalar (L, M, D) bridge because it depends on the resolvent $(sI - A_{cc})^{-1}$.

For a line parameter w , the Schur-NEP pole sensitivity is

$$\frac{ds}{dw} = -\frac{y^*(\partial S_e(s)/\partial w)x}{y^*S'_e(s)x},$$

$$S'_e(s) = I + A_{cc}(sI - A_{cc})^{-2}A_{cc}. \quad (48)$$

where x and y are right and left null vectors of $S_e(s)$. In the audit below, the numerator is decomposed into three diagnostic parts: (A) a retained stiffness/frequency contribution from the retained $\omega-\delta$ block, (B) a retained-network residual that includes modal rotation within the retained coordinates, and (C) the condensed-control self-energy contribution. A reversal is called *resonant* only if the network-family damping decreases, the tracked mode moves closer to a matched A_{cc} pole, the nearest condensed-control pole is PLL-dominated, and the negative C term dominates the A and B terms.

Across Mix60, Mix60/no-PSS, and Mix60/soft-GFL audits, 348 line/mode records reduce the global margin and 232 reduce a tracked network-family mode. Forty records satisfy the strict resonant criteria above. All 40 have $|C| > 10 \max(|A|, |B|)$; the median dominance ratio is 1039 $\times$, so these cases are not geometric reversals mislabeled as resonant. The largest global effect in Table VIII remains small in absolute magnitude, and none of the 40 resonant cases makes the resonant network mode the global critical pole. The global margin remains set by a control-family pole in every resonant row. The claim is therefore deliberately limited: the bridge confirms a genuine, control-conditional, modal-local resonant mechanism, not a universal or dominant cause of margin collapse.

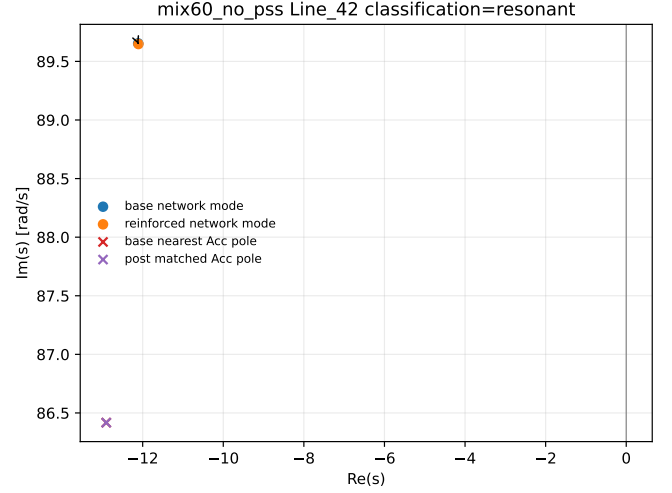


Fig. 5. Control-resonant reinforcement audit in the Schur-NEP plane. The validated NEP bridge tracks a network-family mode and nearby condensed-control poles. In the resonant case, strengthening Line 44 moves the network mode toward a PLL-dominated A_{cc} pole and the self-energy term dominates the damping decrease.

The soft-GFL control profile is the required negative control. For the same line/mode pairs that are resonant in Mix60/no-PSS, the median relative distance from the network mode to the nearest condensed-control pole increases from 0.0368 to 0.9167, while the denominator fraction associated with the frequency-dependent control term falls from 0.959 to 0.0789. Thus the mechanism turns off when the spectral condition predicted by the theory is removed. This distinction is actionable. A geometric reinforcement reversal is mitigated by changing modal damping or accepting the frequency-damping tradeoff; a resonant network-control reversal requires retuning the control pole or avoiding the reinforcement that pulls the network mode into that pole.

G. Hidden Control-Resonance Margin

The same NEP geometry gives a warning metric that is not visible in a static scalar bridge. A system may have all poles in the left half-plane and an acceptable present damping ratio, while a network-family mode lies close to a condensed-control pole. In that case, a small admissible topology or control perturbation can move the network-family mode into a frequency region where the resolvent $(sI - A_{cc})^{-1}$ and the derivative term $(sI - A_{cc})^{-2}$ dominate the pole sensitivity.

For a NEP zero s_k and condensed-control pole $\mu_j \in$

TABLE VIII

NEP-ENABLED CONTROL-RESONANT REINFORCEMENT AUDIT. THE EXAMPLE IS THE LARGEST GLOBAL-MARGIN RESONANT REVERSAL FOUND UNDER A 1% LINE-STRENGTHENING PERTURBATION. THE RESULT IS A MODAL-LOCAL MECHANISM: THE AFFECTED NETWORK MODE IS NOT THE GLOBAL DAMPING-MARGIN POLE IN THIS BENCHMARK.

Quantity	Mix60/no-PSS Line 44	Soft-GFL negative control	Interpretation
Line and buses	Line 44, 23–36	Same line and mode rank	Same physical reinforcement test.
Global margin change	$\Delta\zeta_{\min} = -2.750 \times 10^{-5}$	$\Delta\zeta_{\min} = +2.0 \times 10^{-6}$	The resonant case slightly lowers the system margin; the negative control does not.
Network-mode change	$\Delta\zeta_{\text{net}} = -3.581 \times 10^{-5}$	$\Delta\zeta_{\text{net}} = +4.6 \times 10^{-5}$	The mechanism acts on a network-family secondary mode.
Distance to matched A_{cc} pole	$3.332 \rightarrow 3.330$	$8.200 \rightarrow 8.199$	The resonant case is close to a condensed-control pole; soft-GFL moves the pole away.
Relative distance	0.0368	0.9167	Control proximity is about $25 \times$ smaller in the resonant case.
Control denominator fraction	0.959	0.0789	The resonant case is dominated by frequency-dependent control self-energy.
Dominant condensed states	PLL1 G9/G5/G7	Not PLL-dominated	The observed resonance is PLL-related.
A stiffness term	-1.1×10^{-9}	-2.98×10^{-4}	Geometric stiffness is not the resonant driver.
B retained-rotation term	≈ 0	-5.4×10^{-5}	Retained-network rotation is also not dominant.
C self-energy term	-3.583×10^{-3}	$+4.927 \times 10^{-3}$	The self-energy term sets the sign and disappears as a reversal under soft-GFL.

$\text{spec}(A_{cc})$, define the normalized control-pole distance

$$d_{kj}^{\text{ctrl}} = \frac{|s_k - \mu_j|}{|s_k| + |\mu_j|}. \quad (49)$$

Let

$$\phi_k^\Sigma = \frac{|y_k^* s_k \Sigma'(s_k) x_k|}{|y_k^* T_s(s_k) x_k| + \epsilon}, \quad (50)$$

where x_k and y_k are right and left null vectors of $T(s_k)$ and ϵ is a small numerical regularizer. A small d_{kj}^{ctrl} together with large ϕ_k^Σ indicates a hidden control-resonance margin: the present pole may look safe, but its sensitivity is already governed by nearby control dynamics.

The reason this metric is not visible in a constant-coefficient bridge is the resolvent bound. Let

$$\Delta_c(s) = \text{dist}(s, \text{spec}(A_{cc})). \quad (51)$$

If A_{cc} is diagonalizable with eigenvector condition number κ_c , then for $s \notin \text{spec}(A_{cc})$,

$$\begin{aligned} \|(sI - A_{cc})^{-1}\| &\leq \frac{\kappa_c}{\Delta_c(s)}, \\ \|\Sigma'(s)\| &\leq \frac{\|C\| \|B\| \kappa_c^2}{\Delta_c(s)^2}. \end{aligned} \quad (52)$$

Thus a pole may have acceptable ζ_k while its sensitivity denominator is dominated by $s\Sigma'(s_k)$ because the distance $\Delta_c(s_k)$ is small. This is the proposed *hidden* margin: it is a distance-to-control-resonance margin, not only a present damping-ratio margin.

For any admissible action parameter p , the exact local NEP sensitivity is

$$\begin{aligned} \frac{\partial s_k}{\partial p} &= -\frac{y_k^* T_p(s_k) x_k}{y_k^* T_s(s_k) x_k}, \\ T_s(s) &= 2sM + \Sigma(s) + s\Sigma'(s), \\ T_p(s) &= s^2 M_p + s\Sigma_p(s) + L_p. \end{aligned} \quad (53)$$

The induced damping-ratio derivative is

$$\frac{\partial \zeta_k}{\partial p} = -\frac{\text{Re}(s_{k,p}) \text{Re}(s_k) \text{Re}(\bar{s}_k s_{k,p})}{|s_k| |s_k|^3}, \quad s_{k,p} = \frac{\partial s_k}{\partial p}. \quad (54)$$

Equations (53)–(54) define a common sensitivity language for line, inertia, shunt, damping, and controller actions.

For a target damping ratio ζ_* , define the first-order damping action margin

$$m_p^\zeta(k) = \frac{\zeta_k - \zeta_*}{[-\partial \zeta_k / \partial p]_+}, \quad [a]_+ = \max(a, 0), \quad (55)$$

with $m_p^\zeta = \infty$ when the action does not decrease damping to first order. For a minimum allowed control-pole distance d_* , define the resonance-distance action margin

$$m_p^d(k, j) = \frac{d_{kj}^{\text{ctrl}} - d_*}{[-\partial d_{kj}^{\text{ctrl}} / \partial p]_+}. \quad (56)$$

The combined hidden safety margin is then

$$m_p^{\text{hid}}(k, j) = \min\{m_p^\zeta(k), m_p^d(k, j)\}, \quad (57)$$

and the system-level margin over an action set $\mathcal{P}$ is

$$m_{\mathcal{P}}^{\text{hid}} = \min_{k \in \mathcal{K}} \min_j \min_{p \in \mathcal{P}} m_p^{\text{hid}}(k, j), \quad (58)$$

restricted to tracked oscillatory network-family modes $\mathcal{K}$ or to all oscillatory zeros, depending on the claim. This is not a collapse certificate. It is a local damping-resonance vulnerability metric: it asks how much admissible topology or control movement is needed before a currently acceptable pole crosses the target damping margin.

This definition also gives controller-aware weak elements. A weak node is not defined by graph centrality alone, but by local pole movement through the rational self-energy. For a node action set $\mathcal{P}_i$,

$$\mathcal{W}_i^{\text{hid}} = \max_{k,j} \max_{p \in \mathcal{P}_i} \phi_k^\Sigma \left[-\frac{\partial \zeta_k}{\partial p} \right]_+ \left[-\frac{\partial d_{kj}^{\text{ctrl}}}{\partial p} \right]_+. \quad (59)$$

A large $\mathcal{W}_i^{\text{hid}}$ means that an admissible local action both reduces damping and pulls a pole toward a condensed-control resonance. The scalar sensitivities remain useful diagnostics,

$$W_i^D = -\frac{\partial \zeta_c}{\partial D_i}, \quad W_i^M = -\frac{\partial \zeta_c}{\partial M_i}, \quad W_i^\kappa = -\frac{\partial \zeta_c}{\partial \kappa_i}, \quad (60)$$

where κ_i denotes a local controller parameter such as PLL gain, droop, or virtual-inertia tuning. A weak link is analogously

$$W_e = -\frac{\partial \zeta_c}{\partial w_e}. \quad (61)$$

The sign matters: $W_e > 0$ means strengthening that link lowers damping to first order. A resolvent-aware weak link uses the same hidden score,

$$\mathcal{W}_e^{\text{hid}} = \max_{k,j} \phi_k^\Sigma \left[-\frac{\partial \zeta_k}{\partial w_e} \right]_+ \left[-\frac{\partial d_{kj}^{\text{ctrl}}}{\partial w_e} \right]_+. \quad (62)$$

A *hidden-resonant* weak link has $\mathcal{W}_e^{\text{hid}} > 0$ for a matched control pole. Thus the graph object being ranked is no longer only a line in L ; it is a line whose perturbation moves a pole through the rational self-energy geometry.

H. Instability-Flow Weak Nodes and Links

The previous definitions rank weak elements by local margin loss. The same NEP sensitivity can be expanded into an instability-flow interpretation: a disturbance enters through a physical node or line, excites retained network modes, couples through the rational self-energy to condensed control poles, and then moves the damping margin. This subsection derives that object step by step.

Assume first that A_{cc} is diagonalizable. Let

$$A_{cc} v_p = \mu_p v_p, \quad w_p^* A_{cc} = \mu_p w_p^*, \quad w_p^* v_q = \delta_{pq}. \quad (63)$$

Then

$$(sI - A_{cc})^{-1} = \sum_p \frac{v_p w_p^*}{s - \mu_p}, \quad (64)$$

and

$$\Sigma(s) = D_0 + \sum_p \frac{C v_p w_p^* B}{s - \mu_p}. \quad (65)$$

A retained-network pole s_k is therefore exposed to a condensed-control pole μ_p only when the poles are close and the input/output projections through B and C are nonzero. A sensitivity-weighted resonance score is

$$R_{kp}^{\text{ctrl}} = \frac{|y_k^* C v_p| |w_p^* B x_k|}{|s_k - \mu_p|^2 + \epsilon}. \quad (66)$$

The squared distance is used because $\Sigma'(s)$ contains $(s - \mu_p)^{-2}$; (66) is a pole-sensitivity weight, not only a distance score.

For a line $e = (i, j)$ with incidence vector $b_e = e_i - e_j$, let $G_e = b_e b_e^T$ denote the normalized stiffness direction. If a line perturbation also changes the operating point or controller condensation, the full local pole sensitivity is

$$\frac{\partial s_k}{\partial w_e} = -\frac{y_k^* (G_e + s_k \Sigma_{w_e}(s_k)) x_k}{y_k^* T_s(s_k) x_k}. \quad (67)$$

In a frozen-controller network screen, $\Sigma_{w_e} = 0$. In a symmetric graph-modal approximation with $x_k = y_k = q_k$,

$$q_k^T G_e q_k = (b_e^T q_k)^2, \quad (68)$$

so the graph-Fourier line participation in mode k is

$$\chi_{e,k} = |b_e^T q_k|^2. \quad (69)$$

For the full NEP, the analogous normalized participation is

$$\chi_{e,k}^{\text{NEP}} = \frac{|y_k^* G_e x_k|}{\|y_k\| \|G_e\| \|x_k\| + \epsilon}. \quad (70)$$

Equation (68) also explains modal propagation. For two graph modes q_k and q_ℓ ,

$$q_k^T G_e q_\ell = (b_e^T q_k)(b_e^T q_\ell). \quad (71)$$

Thus a line is not weak merely because it has high active-power loading. It is weak when it couples the active mode to nearby network modes or to control-pole resonances that reduce damping. A line-level instability current is

$$I_e^{\text{inst}} = \sum_{k,p} \chi_{e,k}^{\text{NEP}} R_{kp}^{\text{ctrl}} \left[-\frac{\partial \zeta_k}{\partial w_e} \right]_+. \quad (72)$$

The three factors have distinct roles: $\chi_{e,k}^{\text{NEP}}$ says the line moves the retained mode, R_{kp}^{ctrl} says the mode is exposed to a condensed-control pole, and the final factor keeps only line actions that lower damping to first order.

The associated modal resonance graph is

$$\mathcal{G}_{\text{res}} = (\mathcal{V}_{\text{net}} \cup \mathcal{V}_{\text{ctrl}}, \mathcal{E}_{\text{res}}), \quad \mathcal{V}_{\text{net}} = \{s_k\}, \quad \mathcal{V}_{\text{ctrl}} = \{\mu_p\}, \quad (73)$$

with edge weights R_{kp}^{ctrl} . This is not the bus graph. It is a mode-control graph that identifies which network-family poles are dynamically close to controller-family poles.

The same construction gives node scores. In a graph-modal approximation, node i participates in mode k through $|q_{k,i}|^2$. A resonance-exposure score is

$$I_i^{\text{res}} = \sum_{k,p} |q_{k,i}|^2 R_{kp}^{\text{ctrl}}. \quad (74)$$

For planning, exposure alone is not enough; an intervention must move a margin. With a local action set $\mathcal{A}_i$, define vulnerability and repairability as

$$V_i = \max_{a \in \mathcal{A}_i} \left[-\frac{\partial m_H}{\partial a} \right]_+, \quad (75)$$

$$R_i^{\text{repair}} = \max_{a \in \mathcal{A}_i} \frac{[\partial m_H / \partial a]_+}{\text{cost}(a) + \epsilon}. \quad (76)$$

The distinction is operational. A node can be a strong source of vulnerability while still being a poor repair location under cost or actuator limits.

Proposition 4 (Local weak-element certificate): Let $m_H(a)$ be differentiable at the current operating point for a tracked active pole, line, or control-pole pair. If an admissible action a satisfies $\partial m_H / \partial a < 0$, then there exists $\bar{\epsilon} > 0$ such that, for all $0 < \eta < \bar{\epsilon}$,

$$m_H(a + \eta) < m_H(a). \quad (77)$$

Proof. Taylor expansion gives

$$m_H(a + \eta) = m_H(a) + \eta \frac{\partial m_H}{\partial a} + o(\eta). \quad (78)$$

Because $\partial m_H / \partial a < 0$, the first-order term is negative. By the definition of $o(\eta)$, the remainder is smaller in magnitude than half of the negative first-order term for sufficiently small positive η . Hence $m_H(a + \eta) < m_H(a)$. $\square$

Proposition 4 is intentionally local. It proves that the proposed weak-node and weak-link scores are first-order margin-loss diagnostics. It does not prove that I_e^{inst} or V_i will outperform power-flow loading, betweenness, Fiedler participation, SCR, impedance, or damping-participation rankings in every grid. That is an empirical claim and must be tested against the full NEP or full ANDES eigensolve.

I. Protected-Resolvent Graph-Filter Shaping

The damping margin is a pole metric. A complementary view is needed when the question is not only where the least-damped pole is, but where a protected input can be amplified into a protected output. This gives an input–output graph-filter certificate. It is especially relevant in non-normal IBR models, where the forcing location and the response location can be spatially separated.

Let

$$G(j\omega) = C_o(j\omega I - A_{\text{sys}})^{-1} B_i \quad (79)$$

be a protected transfer matrix from disturbance channels B_i to monitored outputs C_o . Define the local protected gain

$$g_{\text{prot}} = \sigma_1(G(j\omega^*)), \quad \mathcal{S}_g = \log g_{\text{prot}}, \quad (80)$$

where ω^* is a simple local peak and σ_1 is simple. Let u^* and v^* be the corresponding left and right singular vectors. The two spatial fields

$$\phi^{\text{in}} = (j\omega^* I - A_{\text{sys}})^{-1} B_i v^*, \quad (81)$$

$$\phi^{\text{out}} = (j\omega^* I - A_{\text{sys}})^{-H} C_o^H u^* \quad (82)$$

are the input-forcing and output-response profiles. They are equal only in special normal or nearly collocated cases. Their mismatch is the spatial decoupling that ordinary graph centrality does not see.

Theorem 1 (Bi-orthogonal protected-resolvent sensitivity): Assume ω^* and $\sigma_1(G(j\omega^*))$ are locally simple and fixed to first order. For any real parameter p entering only through A_{sys} ,

$$\frac{\partial \mathcal{S}_g}{\partial p} = \frac{1}{g_{\text{prot}}} \text{Re} \left[(\phi^{\text{out}})^H \frac{\partial A_{\text{sys}}}{\partial p} \phi^{\text{in}} \right]. \quad (83)$$

Proof. Let $R = (j\omega^* I - A_{\text{sys}})^{-1}$. Since $dR = R(dA_{\text{sys}})R$, the first variation of $G = C_o R B_i$ is

$$dG = C_o R (dA_{\text{sys}}) R B_i. \quad (84)$$

For a simple singular value, $dg_{\text{prot}} = \text{Re}[(u^*)^H (dG) v^*]$. Substituting the definitions of ϕ^{in} and ϕ^{out} gives

$$dg_{\text{prot}} = \text{Re}[(\phi^{\text{out}})^H (dA_{\text{sys}}) \phi^{\text{in}}]. \quad (85)$$

Dividing by g_{prot} gives the derivative of $\log g_{\text{prot}}$. $\square$

The theorem supplies a common exchange rate for topology, damping, inertia, and controller tuning. It is not a substitute for the damping-ratio NEP sensitivity in (53); it answers a different question. The pole metric asks which action moves the critical pole. The protected-resolvent metric asks which action reduces the worst input–output amplification near the protected frequency band.

For local damping in a second-order state ordering $x = [\theta^T, \omega^T]^T$, the damping perturbation has the velocity block

$$\frac{\partial A_{\text{sys}}}{\partial D_i} = -\frac{1}{M_i} E_{\omega_i \omega_i}, \quad (86)$$

so (83) gives

$$\frac{\partial \mathcal{S}_g}{\partial D_i} = -\frac{1}{g_{\text{prot}} M_i} \text{Re} \left[\overline{\phi_{\omega_i}^{\text{out}}} \phi_{\omega_i}^{\text{in}} \right]. \quad (87)$$

This is the corrected spatial-decoupling rule. The tempting simplification $-|\phi_i^{\text{out}}|^2$ is valid only when input and output fields align. In a non-normal IBR grid, optimal damping placement may follow the bi-orthogonal product, not local inertia deficit or output magnitude alone.

For a line $e = (i, j)$ with incidence vector $b_e = e_i - e_j$, a swing block has

$$\frac{\partial A_{\text{sys}}}{\partial w_e} = \begin{bmatrix} 0 & 0 \\ -M^{-1} b_e b_e^T & 0 \end{bmatrix} \quad (88)$$

in the retained network coordinates. Therefore

$$\frac{\partial \mathcal{S}_g}{\partial w_e} = -\frac{1}{g_{\text{prot}}} \text{Re} \left[(\phi_{\omega}^{\text{out}})^H M^{-1} b_e b_e^T \phi_{\theta}^{\text{in}} \right]. \quad (89)$$

Equation (89) is a graph-filter weak-link score: it weights a line by the way it couples the protected forcing profile to the protected response profile. It is not the same as a Fiedler score, a power-flow loading score, or the pole-damping score W_e in (61).

Remark 3 (Relation to weak nodes and weak links): The weak-element scores now form a hierarchy. The pole score W_e asks whether strengthening line e lowers the damping ratio of a tracked pole. The hidden score $\mathcal{W}_e^{\text{hid}}$ asks whether the line also moves that pole toward a condensed-control resonance. The protected-resolvent score (89) asks whether the same line worsens protected input–output amplification. A robust planning claim should report which of these three certificates is active.

Theorem 2 (Flow-feasible dynamic Braess certificate): For an operating point with feasible base flows, a line reinforcement direction w_e is a first-order flow-feasible dynamic Braess candidate for the protected resolvent if

$$\frac{\partial \nu_c}{\partial w_e} > 0, \quad (90)$$

$$\frac{\partial \mathcal{S}_g}{\partial w_e} > 0, \quad (91)$$

$$|f_0 + \eta J_{f,e}| \leq \bar{f} \quad \text{for some } 0 < \eta \leq \bar{\eta}. \quad (92)$$

Then, for sufficiently small feasible $\eta > 0$, the reinforcement improves the stiffness proxy while increasing the protected amplification.

Algorithm 2 Flow-Feasible Protected-Resolvent Retrofitting

Require: A_{sys} , protected channels B_i, C_o , action set $\mathcal{A}$, flow limits $\bar{F}$, costs c .

- 1: Compute a local peak ω^* and $g_{\text{prot}} = \sigma_1(G(j\omega^*))$.
- 2: Extract singular vectors u^*, v^* and fields $\phi^{\text{in}}, \phi^{\text{out}}$ from (81)–(82).
- 3: For each bus action, compute (87) and the analogous inertia/controller derivatives.
- 4: For each line, compute (89) and the flow sensitivity $J_{f,e}$.
- 5: Solve the local retrofit LP (95).
- 6: Recompute the NEP or full ANDES eigensolve for the selected package.
- 7: If the active peak, pole family, or flow constraint switches, rebuild the local model and repeat.

Proof. By differentiability,

$$\nu_c(w_e + \eta) = \nu_c(w_e) + \eta \partial \nu_c / \partial w_e + o(\eta), \quad (93)$$

$$\mathcal{S}_g(w_e + \eta) = \mathcal{S}_g(w_e) + \eta \partial \mathcal{S}_g / \partial w_e + o(\eta). \quad (94)$$

The signs in (90)–(91) make the first-order changes positive for sufficiently small η . Condition (92) restricts the same perturbation to remain within the linearized flow limit. $\square$

The theorem deliberately uses “candidate.” It certifies a local conflict between a stiffness proxy and a protected dynamic certificate under a flow-feasible perturbation. It does not prove that the nonlinear AC dispatch remains feasible for a large switching action, nor that the damping-ratio margin must collapse.

The corresponding local retrofit problem is

$$\begin{aligned} \min_z \quad & c^T z \\ \text{s.t.} \quad & \mathcal{S}_{g,0} + \nabla \mathcal{S}_g^T z \leq \mathcal{S}_{g,*}, \\ & F_0 + J_F z \leq \bar{F}, \\ & z_{\min} \leq z \leq z_{\max}, \end{aligned} \quad (95)$$

where $z = (\Delta w, \Delta D, \Delta M, \Delta \kappa)$ collects topology, damping, inertia, and controller-retuning actions. Equation (95) is a local screening LP; the selected package must be revalidated by the NEP or the full ANDES eigensolve because the active singular vector, active pole, or active flow constraint can switch.

The scripts store the Mix60 topology diagram and supporting JSON/CSV artifacts under `outputs/ieee39_phase0_bridge`, the negative scalar bridge audit under `outputs/phase0_physical_bridge`, the contour-validated NEP audit under `outputs/phase0d_nep_contour_solver`, and the control-resonant reinforcement audit under `outputs/phase_braess_nep`. The negative scalar audit remains important: it shows why a nodal D extracted from `GENROU.D` cannot represent full ANDES damping. The Phase-0D result closes the next bridge gap by retaining controller states through the rational self-energy. It does not close the estimator-validation gap: the second-order correction and planning rankings still need registered comparison against full ANDES poles.

J. Phase-1 Controller Calibration Audit

A first calibration audit was then run on the five derived IBR cases. Ten documented profiles were tested: the original generic parameters; slower PLL gains; softened GFL voltage/reactive controls; softened GFM controls; combined soft profiles; and diagnostic variants that remove retained synchronous-machine IEEEEST/PSS or IEEEEX1+IEEEEST rows. Each profile was evaluated by full ANDES power flow and eigenanalysis, with participation factors recorded for the dominant positive mode and the least-damped stable oscillatory mode.

Table XI is a negative and useful result. Merely slowing or softening the IBR controllers does not stabilize the low-penetration cases. Diagnostic removal of retained synchronous-machine stabilizer/exciter rows stabilizes GFL20, Mix20, and Mix40, while GFM20 remains unstable even under the diagnostic removal; its mechanism is not yet isolated. Thus the remaining benchmark blocker is broader than IBR gain selection: a defensible full-ANDES IBR benchmark must choose or calibrate the retained synchronous-machine controls and the inverter controls together, then validate a reduced surrogate against the full calibrated Jacobian before estimator errors are reported.

K. Baselines Required for a Fair Test

Four baselines should be reported in every validation table. The first is the homogeneous damping formula, which is valid only when damping is proportional or can be represented by a scalar. The second is a Fiedler-only estimate, which tests whether low-frequency intuition is being confused with damping-margin estimation. The third is a frequency-selected reduced QEP, which tests whether reduction succeeds merely because it is low order or because it selects the correct low-damping subspace. The fourth is the full eigensolve, which is the reference and not a competitor.

These baselines prevent two common mistakes: crediting the adaptive estimator for a case where the diagonal all-mode formula already works, and crediting a graph-frequency proxy for a case where the mechanism is damping.

VI. PLANNING WITH THE DAMPING MARGIN

The estimator becomes useful when it ranks actions. Let a denote a planning action: conversion of a synchronous generator to an IBR, addition of virtual damping, addition of virtual inertia, or local controller retuning. Define

$$J(a) = \zeta_{\min}(L(a), M(a), D(a)), \quad (96)$$

and, when costs are available,

$$J_c(a) = \frac{\zeta_{\min}(a) - \zeta_{\min}(0)}{\text{cost}(a)}. \quad (97)$$

The action with highest J is the safest conversion; the action with highest J_c is the most cost-effective improvement.

TABLE IX
STATUS OF THE PROTECTED-RESOLVENT/FLOW-FEASIBLE EXTENSION.

Component	Mathematical status	Current evidence	Claim level
Bi-orthogonal sensitivity (83)	Direct consequence of singular-value perturbation of the resolvent.	Implemented in reduced-model coordination gates.	Valid formula; not new perturbation theory.
Spatial decoupling	General rule is a left-right product, not $-\ \phi_i\ ^2$.	Gamma gate shows S_ζ and S_g rankings agree only 21–29% for physical profiles.	Supported phenomenon; mechanism still open.
Coordinated packages	Local LP combines topology, damping, inertia, and controller actions.	Reduced-model gate: $w+D$ improves over the best single lever in 72.1–73.8% across physical Gamma profiles; $w+M$ in 67.2–73.8%; $D+M$ not robust.	Reduced-model planning candidate.
Nonlinear synergy	Would require a positive interaction beyond additive first-order value.	Gate found no robust nonlinear synergy: SynNorm never exceeds the registered 50% threshold.	Killed as headline.
Flow-feasible dynamic Braess	Theorem 2 gives local first-order certificate with flow constraint.	Needs explicit flow-feasibility runs on calibrated ANDES or DC/AC dispatch sensitivities.	Proposed next validation.

TABLE X
PHASE-0 VALIDATION GATES FOR THE FULL ANDES IEEE39 IBR BENCHMARK.

Gate	Status	Evidence and required next step
Baseline ANDES load/PFlow/EIG	pass	39 buses, 46 lines, and 160 eigenvalues; synchronous extraction benchmark only.
Positive-mode diagnostic	pass-diagnostic	Removing IEEEEST/PSS removes all positive-real modes; report as case-quality warning.
SG-to-GFL/GFM case generation	pass	Five derived cases solved power flow and eigenanalysis.
Generic IBR small-signal stability	partial	One of five cases is stable under generic gains; controller calibration is still required.
Controller-aware rational NEP bridge (contour-validated)	pass	Replaces the failed scalar (L, M, D) bridge. Beyn contour integration reproduces full ANDES critical poles in no-PSS, base, and Mix60; Mix60's 13.7 Hz PLL-family critical mode is captured.
Estimator and planning validation vs. full ANDES IBR poles	pending	The bridge is now validated and a resonant line-reinforcement mechanism audit has been completed. Experiments A–E must still compare the diagonal, second-order, reduced-QEP, conversion-ranking, and planning claims against full ANDES poles.

TABLE XI
PHASE-1 CONTROLLER CALIBRATION AUDIT. DIAGNOSTIC PROFILES REMOVE RETAINED SYNCHRONOUS-MACHINE CONTROLS AND ARE NOT FINAL TUNING RECOMMENDATIONS.

Case	Best controller-only profile	Best diagnostic profile	Mode audit
GFL20	Very-soft IBR controls: 1 positive mode, max $\text{Re}(s) = 1.011$.	No PSS plus soft IBR controls: stable, max $\text{Re}(s) = 4.0 \times 10^{-10}$.	Generic positive mode dominated by REECA1; high-frequency stable modes are control/auxiliary dominated.
GFM20	Baseline remains best: 6 positive modes, max $\text{Re}(s) = 4.291$.	No exciter/PSS: still 2 positive modes, max $\text{Re}(s) = 4.286$.	Remaining instability is not cleared by tested IBR or diagnostic retained-control-removal profiles; the mechanism is not yet isolated and requires a targeted GFM droop-network/retained-SG participation audit.
Mix20	Slow PLL: 3 positive modes, max $\text{Re}(s) = 1.066$.	No exciter/PSS: stable, max $\text{Re}(s) = 8.3 \times 10^{-8}$.	The least-damped stable mode near 13.75 Hz is high-frequency auxiliary/control dominated.
Mix40	Slow PLL: 2 positive modes, max $\text{Re}(s) = 1.065$.	No exciter/PSS: stable, max $\text{Re}(s) = 8.9 \times 10^{-8}$.	The stable high-frequency critical mode has leading PLL1 participation.
Mix60	Soft GFL controls: stable, max $\text{Re}(s) = 4.0 \times 10^{-10}$.	No PSS: stable, max $\text{Re}(s) = 2.0 \times 10^{-9}$.	The controller-only stable critical mode is dominated by REGF1 participation.

A. Interpretation of the Planning Score

The score is defined using the post-action margin rather than only a derivative at the current operating point. A derivative is local. A conversion from synchronous generation to an inverter can be a finite perturbation to inertia, damping, and control states. For small damping additions, a local formula is useful. For finite SG-to-IBR conversion, the estimator should be recomputed after each candidate action.

The distinction between absolute and incremental scores matters. A candidate action can produce a large incremental improvement but still leave the system below the target mar-

gin. Conversely, a candidate can yield modest improvement but preserve the margin during a conversion sequence. For decarbonization planning, the safest action can be written as

$$a^* = \arg \min_{a \in \mathcal{A}} [\zeta_* - \zeta_{\min}(a)]_+, \quad (98)$$

with cost normalization used after safety is satisfied.

B. Controller-Aware Graph Reconfiguration

Weak-node and weak-link scores are local objects. A graph reconfiguration problem asks how to combine them under a

TABLE XII

TRACEABILITY FROM THEORETICAL HYPOTHESES TO CURRENT SIMULATIONS. EACH ROW STATES WHAT IS SUPPORTED AND WHAT REMAINS OUTSIDE THE PRESENT EVIDENCE.

Theoretical claim	Simulation evidence	What the evidence supports	What it does not prove
Second-order rational-filter correction	IEEE39-derived surrogate: diagonal median error 0.468%, second order 0.032%, reduced QEP p95 5.44% versus second-order p95 9.91%.	Off-diagonal modal damping correction improves the cheap diagonal screen in median error; tail cases still require a QEP fallback.	Accuracy against full ANDES IBR poles through the Phase-0D bridge.
Controller-aware rational bridge	Phase-0D NEP-Beyn bridge reproduces full ANDES critical poles in no-PSS, base, and Mix60; Mix60 control-family critical mode has $\pi_c = 0.853$.	The full ANDES state-space bridge is valid for these diagnostic cases and separates network-margin from PLL-control contours.	Estimator accuracy, reversals, and planning rankings over this bridge.
Weak-node damping placement	In the representative surrogate, a 10% local damping increase gives the largest improvement at bus 30 ($\Delta\zeta = 0.0112$), followed by buses 38 and 36.	Damping-margin sensitivity gives an actionable weak-node ranking on the IEEE39 topology.	Superiority over all published placement indices in a full DAE benchmark.
Weak-link and Braess-like topology reversal	Line 25–37 gives the best reinforcement benefit ($\Delta\zeta = 0.00631$), while 35 of 46 line reinforcements raise critical modal stiffness but reduce $\zeta_{\min}$.	Frequency-only reinforcement can be damping-negative under non-proportional damping.	A practical topology-planning claim for calibrated IBR ANDES cases.
Control-resonant reinforcement mechanism	The NEP reinforcement audit finds 40 strict control-resonant records out of 348 global-margin-reversal records in Mix60/no-PSS; all 40 are dominated by the self-energy term C, with median $ C /\max(A , B) = 1039$.	The NEP bridge separates geometric stiffness effects from control self-energy; soft-GFL is a negative control with zero resonant reversals.	Confirmed mechanism in these audits, but modal-local and control-conditional; none of the 40 resonant network modes is the global critical pole.
Inertia sign reversal	In fixed- D_i/M_i ensembles, 5447 of 16229 perturbations reduce both the diagonal screen and the full-QEP margin; the strongest example matches the analytic sign by finite difference.	More inertia is not necessarily monotone-beneficial when modal rotation dominates the positive normalization term.	A universal inertia-placement law or a full-ANDES inertia-retuning result.
Control-state dominance of critical modes	The least-damped stable modes near 13.7 Hz in GFL/Mix pilots are dominated by IEEEEST/IEEEX1, REGF1, PLL1, or BusFreq states.	Once IBR and retained controller states are active, the least-damped stable mode need not be a low-frequency network mode.	That the current generic controllers are physically calibrated.
ANDES IBR benchmark construction	Five SG-to-GFL/GFM cases load, solve power flow, and run eigenanalysis; Phase-1 tested 50 case/profile audits; the scalar bridge fails but the controller-aware rational NEP bridge passes Phase 0D.	The reproducible benchmark harness exists, and the full-ANDES bridge for the diagnostic cases is validated.	Final estimator/planning validation against full ANDES poles over registered operating ranges.

TABLE XIII

PLANNING QUANTITIES AND INTENDED USE.

Quantity	Use
$\zeta_{\min}(a)$	Safety of finite conversion or reinforcement
$\Delta\zeta_{\min}/\text{cost}$	Cost-effective damping or inertia placement
$\partial\zeta_{\min}/\partial w_e$	Local line-reinforcement screening
$\partial v_k/\partial w_e$	Frequency mechanism, not final ranking
$q_k(i)^2$	Modal participation diagnostic

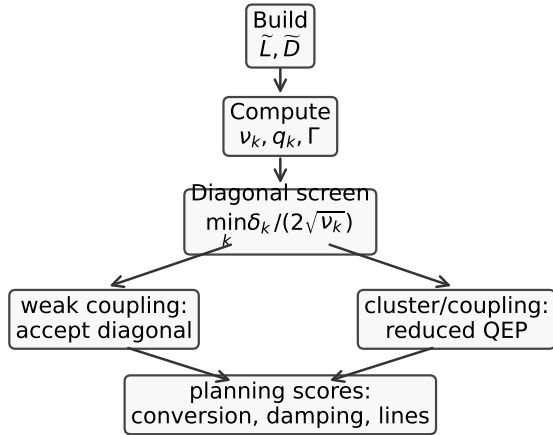


Fig. 6. Workflow of the adaptive estimator. The diagonal region is the first screen; second-order perturbation supplies the cheap correction; the reduced QEP is triggered by clustered low-damping modes or strong off-diagonal modal damping.

cost budget. The first step is a matrix taxonomy of the available levers. Let $b_{ij} = e_i - e_j$ and define

$$\mathcal{L}_{\text{all}} = \text{span}\{b_{ij}b_{ij}^T : 1 \leq i < j \leq n\}, \quad (99)$$

$$\mathcal{D}_{\text{sh}} = \{\text{diag}(g) : g \in \mathbb{R}^n\}. \quad (100)$$

Proposition 5 (Line-shunt action decomposition): As real vector spaces,

$$\text{Sym}(n) = \mathcal{L}_{\text{all}} \oplus \mathcal{D}_{\text{sh}}. \quad (101)$$

Proof. The line directions span the symmetric zero-row-sum Laplacian subspace, of dimension $n(n-1)/2$. The shunt space has dimension n . A diagonal matrix with zero row sums is zero, so the intersection is trivial and the dimensions add to $\dim \text{Sym}(n)$. $\square$

This algebra is standard; the use here is as a robustness-lever taxonomy. Given a symmetric perturbation X , its line part is the unique zero-row-sum matrix with the same off-diagonal entries as X , and the residual diagonal is the shunt part. If the existing edge set is $\mathcal{E}$, then

$$\mathcal{L}_{\text{all}} = \mathcal{L}_{\text{ex}} \oplus \mathcal{L}_{\text{new}}, \quad (102)$$

where $\mathcal{L}_{\text{ex}}$ is spanned by existing line directions and $\mathcal{L}_{\text{new}}$ by candidate new-line directions.

The physically useful consequence is that virtual inertia has a restricted structural location. Since $\tilde{L} = M^{-1/2}LM^{-1/2}$, a local inertia perturbation $E_i = e_ie_i^T$ gives

$$\frac{\partial \tilde{L}}{\partial M_i} = -\frac{1}{2M_i}(E_i\tilde{L} + \tilde{L}E_i). \quad (103)$$

The sign in (103) says that increasing inertia lowers normalized stiffness; the support statement below is independent of that sign.

Proposition 6 (Virtual-inertia lever location): For any bus i , the inertia-induced perturbation satisfies

$$\frac{\partial \tilde{L}}{\partial M_i} \in \mathcal{L}_{\text{ex}} \oplus \mathcal{D}_{\text{sh}}, \quad (104)$$

and has no component in $\mathcal{L}_{\text{new}}$.

Proof. The off-diagonal support of $E_i \tilde{L} + \tilde{L} E_i$ is inherited from the existing network support. Proposition 5 then gives a unique decomposition into existing-edge directions plus a diagonal residual; no new-edge direction is needed. $\square$

Proposition 6 does not make inertia and line reinforcement dynamically equivalent: they move modal stiffness with opposite signs in the conservative screen, and the shunt residual may be the part aligned with the critical mode. It gives a structural taxonomy: reinforce existing lines, add new topology, or add shunt-like virtual inertia/synchronous-condenser support.

The same taxonomy gives a modal classifier for whether topology can substitute for inertia. Let $X_i = -\partial \tilde{L} / \partial M_i = X_i^{\text{ex}} + \text{diag}(r_i)$ with $X_i^{\text{ex}} \in \mathcal{L}_{\text{ex}}$. For a normalized critical network-family mode q_c , define

$$s_i = 1 - \frac{|r_i^T q_c|}{\|r_i\| + \varepsilon_{\text{reg}}}. \quad (105)$$

A high s_i means that the shunt residual is nearly invisible to the critical mode, so existing-line actions are a plausible substitute. A low s_i means that the shunt component is aligned with the critical mode, so inertia or synchronous-condenser support is structurally harder to replace by line-only actions. The small $\varepsilon_{\text{reg}} > 0$ only handles the zero-residual case. This classifier is a design diagnostic, not yet a validated full-ANDES planner.

The scalar score in (105) is only the projection of a larger object. For every retained network-family mode q_k , define the node-mode substitutability spectrum

$$s_{i,k} = 1 - \frac{|r_i^T q_k|}{\|r_i\| \|q_k\| + \varepsilon_{\text{reg}}}, \quad S_{\text{sub}} = [s_{i,k}]_{i,k}. \quad (106)$$

The same bus can therefore be line-substitutable for one mode and equipment-requiring for another. This is the structural reason why one-number weak-node scores can lose the mechanism needed for planning: they say where a mode participates, but not whether the repair should be copper, new topology, or local shunt/inertia support. A structural random-graph audit reported in the project ledger found that mode-dependent node rankings disagreed in all 60 tested random graphs, that the distinction persisted among candidate critical modes, and that in about 98% of the tested graphs the top node for at least one candidate critical mode differed from the top node for the single selected critical mode. These numbers are not full-ANDES evidence; they motivate using S_{sub} as the object to validate, rather than collapsing it prematurely to a scalar.

The same spectrum can be projected onto planning categories. If a Fiedler partition divides the buses into two areas, then the existing-line space splits into intra-area and inter-area directions,

$$\mathcal{L}_{\text{ex}} = \mathcal{L}_{\text{intra}} \oplus \mathcal{L}_{\text{inter}}. \quad (107)$$

Writing $X_i^{\text{ex}} = X_i^{\text{intra}} + X_i^{\text{inter}}$ gives the mode-dependent inter-area share

$$\alpha_{i,k}^{\text{inter}} = \frac{|q_k^T X_i^{\text{inter}} q_k|}{|q_k^T X_i^{\text{ex}} q_k| + \varepsilon_{\text{reg}}}. \quad (108)$$

Large $\alpha_{i,k}^{\text{inter}}$ means that the line-substitutable part of the local inertia action mainly lives in corridors between areas; small $s_{i,k}$ with large inter-area share says that the corridor is structurally important but still cannot fully replace local equipment for that mode. In preliminary six-bus structural checks, this projection separated corridor buses from within-area buses. That observation is used only as a diagnostic template until it is repeated on calibrated IEEE39 cases.

There is also a complex-admittance version of the same taxonomy. Over the complex symmetric matrix space,

$$\text{Sym}_{\mathbb{C}}(n) = \mathcal{L}_{\text{all}}^{\mathbb{C}} \oplus \mathcal{D}_{\text{sh}}^{\mathbb{C}}, \quad (109)$$

where complex line directions represent series admittance changes and $\mathcal{D}_{\text{sh}}^{\mathbb{C}}$ represents shunt admittance. This is the Y -bus sister of Proposition 5: it supports a copper-versus-reactive-compensation taxonomy for voltage/reactive planning. It should not be confused with the inertia result in Proposition 6; inertia enters the dynamic normalization, while the complex Y -bus decomposition organizes line admittance and shunt compensation such as capacitors or STATCOMs.

The planning interpretation is a damping-aware hosting-capacity screen. For a proposed SG-to-IBR conversion at bus i , let $\mathcal{A}_i$ be the admissible local repair set containing existing-line reinforcement, candidate new lines, shunt/STATCOM or synchronous-condenser support, and controller retuning. A local damping-hosting capacity can be written as

$$H_i^{\text{damp}} = \sup_{\rho \in [0,1]} \{ \rho : \exists u \in \mathcal{A}_i, m_H(\rho, i, u) \geq m_*, c^T |u| \leq B \}. \quad (110)$$

Equation (110) is not a thermal or voltage hosting metric. It asks how much conversion can be hosted while preserving the composite phase-damping-control-resonance margin. The novelty, if it survives the literature and ANDES gates, would not be generic hosting capacity; it would be the mode-resolved decision of whether a low-inertia conversion is repairable by existing copper, requires new topology, or requires local equipment/control support.

Let u collect admissible changes in existing lines, candidate new lines, shunt/inertia support, damping, and controller settings:

$$u = [u_{\text{ex}}, u_{\text{new}}, u_{\text{sh}}, u_M, u_D, u_{\kappa}]^T. \quad (111)$$

The corresponding first-order matrix action has the structured form

$$\begin{aligned} \Delta \tilde{L}(u) = & \sum_{e \in \mathcal{E}} u_{\text{ex},e} G_e + \sum_{e \notin \mathcal{E}} u_{\text{new},e} G_e \\ & + \sum_i u_{\text{sh},i} E_i + \tilde{\Delta L}_M(u_M), \end{aligned} \quad (112)$$

where $G_e = b_e b_e^T$ in normalized coordinates and $\tilde{\Delta L}_M$ is the virtual-inertia perturbation in (103). Equation (112) is the matrix-geometric representation of the planning levers:

topology, new topology, shunt/inertia, damping, and control enter the same NEP but occupy different structured directions.

For a fixed tracked pole s_k , NEP sensitivity gives a first-order pole-motion map

$$\Delta s_k \approx J_k u, \quad J_k = \begin{bmatrix} \frac{\partial s_k}{\partial p_1}, \dots, \frac{\partial s_k}{\partial p_r} \end{bmatrix}. \quad (113)$$

In real pole coordinates, J_k maps feasible graph and control actions into a cone of possible pole motions in the complex plane. The damping-ratio gradient then gives

$$\Delta \zeta_k \approx g_k^T u, \quad g_k = \begin{bmatrix} \frac{\partial \zeta_k}{\partial p_1}, \dots, \frac{\partial \zeta_k}{\partial p_r} \end{bmatrix}^T. \quad (114)$$

Thus a local cost-constrained reconfiguration can be posed either as a max-margin problem,

$$\max_{u, t} \quad t \quad (115)$$

$$\text{s.t.} \quad \zeta_k + g_k^T u \geq t, \quad k \in \mathcal{K}, \\ c^T |u| \leq B, \quad u \in \mathcal{U}, \quad (116)$$

where $\mathcal{K}$ is the set of protected oscillatory modes, c is an action-cost vector, and $\mathcal{U}$ encodes engineering limits. A hidden-resonance constraint can be added by linearizing the control-pole distance:

$$d_{kj}^{\text{ctrl}} + h_{kj}^T u \geq d_{\min}, \quad h_{kj} = \nabla_u d_{kj}^{\text{ctrl}}. \quad (117)$$

or as a minimum-cost hidden-margin repair problem,

$$\min_u \quad c_{\text{ex}}^T |u_{\text{ex}}| + c_{\text{new}}^T |u_{\text{new}}| + c_{\text{sh}}^T |u_{\text{sh}}| \\ + c_M^T |u_M| + c_D^T |u_D| + c_{\kappa}^T |u_{\kappa}| \\ \text{s.t.} \quad \zeta_k + g_k^T u \geq \zeta_{\star}, \quad k \in \mathcal{K}, \\ d_{kj}^{\text{ctrl}} + h_{kj}^T u \geq d_{\star}, \quad (k, j) \in \mathcal{R}, \\ u \in \mathcal{U}. \quad (118)$$

Here $\mathcal{R}$ contains the network/control pole pairs that are close enough to matter. Equations (113)–(118) are not yet a validated planner. They define the matrix-geometric version of the proposed framework: topology and controller actions generate a pole-motion cone, damping-margin safety is a half-space in that cone, and hidden control-resonance safety is a distance constraint to condensed-control poles. This is the intended route from weak-element diagnosis to graph configuration optimization.

The feasible action set in (116)–(118) is often a polytope in the coordinates of (111), but the true margin objective is not a linear program. The global map

$$u \mapsto m_H(u) = \min\{m_{\phi}(u), m_{\zeta}(u), m_c(u)\} \quad (119)$$

is generally nonsmooth because the active mode, active line-angle constraint, or nearest control pole can change. Thus the safe use of the polytope is local: after fixing the active modes and pole pairs, first-order sensitivity gives the linearized problem

$$\max_{u \in \mathcal{P}} g_H^T u, \quad \mathcal{P} = \{u \in \mathcal{U} : c^T |u| \leq B\}, \quad (120)$$

where $g_H = \nabla_u m_H$ for the currently active margin. The optimum of (120) occurs at an extreme point of $\mathcal{P}$ when the

linearized model is valid, so the first-order repair often selects a pure lever or a sparse mixture. This is only a local screening statement. It should not be read as a global convexity theorem for damping-margin planning.

C. SG-to-IBR Conversion

A single conversion can be modeled at the reduced level by changing the inertia and damping assigned to the candidate bus:

$$M_i \leftarrow M_{\text{IBR}}, \quad D_i \leftarrow D_{\text{IBR}}. \quad (121)$$

The exact mapping depends on whether the replacement is grid-following or grid-forming. A grid-following inverter should not be treated as a perfect nodal damping element in final models; the approximation is a planning surrogate that must be validated against full IBR state-space models. The value of the estimator is that it screens many candidates cheaply before full eigensolve or time-domain analysis.

The baseline comparisons are essential. A Fiedler-participation rule ranks buses by $q_2(i)^2$. A frequency-participation rule ranks buses by participation in the highest-frequency mode. An inertia heuristic ranks buses by current M_i . None of these directly optimizes (10). The proposed rule ranks by post-conversion margin. If a simple proxy matches the exact margin in a benchmark, that result should be reported; if it fails, the failure is the justification for the spectral estimator.

D. Damping and Virtual-Inertia Placement

For small additions ΔD_i or ΔM_i , a first-order placement score can be derived from the diagonal region. If D_i changes while M and L are fixed,

$$\frac{\partial \delta_k}{\partial D_i} = \frac{q_{k,i}^2}{M_i}, \quad (122)$$

and hence

$$\frac{\partial \zeta_k}{\partial D_i} \approx \frac{q_{k,i}^2}{2M_i \sqrt{\nu_k}}. \quad (123)$$

This expression gives a transparent damping-placement score, but it must be applied to the mode that determines $\zeta_{\min}$ and corrected when modal coupling is strong. Virtual inertia is more delicate because it changes the normalization, modal frequencies, and damping geometry simultaneously. Therefore, inertia placement should be evaluated by recomputing the adaptive estimator after the candidate action rather than by relying solely on a closed-form scalar score.

For infinitesimal inertia actions, however, Appendix B gives the closed-form score

$$\frac{\partial \zeta_k}{\partial M_i} = \frac{1}{2\sqrt{\nu_k}} \frac{\partial \delta_k}{\partial M_i} + \frac{\delta_k q_{k,i}^2}{4M_i \sqrt{\nu_k}}, \quad (124)$$

with $\partial \delta_k / \partial M_i$ in (186). Unlike (123), this derivative is sign-indefinite: inertia changes both the graph frequency and the modal damping geometry.

A useful sign test follows when the local damping-per-inertia ratio $\bar{d}_i = D_i/M_i$ is held fixed during an infinitesimal

TABLE XIV
INERTIA-REVERSAL MONTE CARLO CHECK. A REVERSAL MEANS THAT INCREASING M_i LOWERS THE DAMPING MARGIN UNDER A FINITE-DIFFERENCE PERTURBATION.

Assumption	Tests	Full rev.	Diag+full rev.
Fixed D_i/M_i	16229	7694	5447
Fixed D_i	16229	11054	8425

inertia change. This removes the trivial effect in which fixed physical damping is diluted by larger inertia. Let

$$G_i = \frac{\partial \tilde{L}}{\partial M_i} = -\frac{E_i \tilde{L} + \tilde{L} E_i}{2M_i}, \quad (125)$$

where $E_i = e_i e_i^T$. For a simple mode k ,

$$\frac{\partial \nu_k}{\partial M_i} = -\frac{\nu_k q_{k,i}^2}{M_i}, \quad (126)$$

and the modal-damping rotation term is

$$\left(\frac{\partial \delta_k}{\partial M_i} \right)_{\text{rot}} = 2 \sum_{\ell \neq k} \frac{(q_\ell^T G_i q_k)(q_\ell^T \tilde{D} q_k)}{\nu_k - \nu_\ell}. \quad (127)$$

Thus the diagonal modal damping-ratio derivative is

$$\frac{\partial \zeta_k}{\partial M_i} = \frac{1}{2\sqrt{\nu_k}} \left(\frac{\partial \delta_k}{\partial M_i} \right)_{\text{rot}} + \frac{\delta_k q_{k,i}^2}{4M_i \sqrt{\nu_k}}. \quad (128)$$

The second term in (128) is positive, but the rotation term can be negative. Therefore, increasing inertia can reduce the diagonal damping-ratio screen whenever

$$\left(\frac{\partial \delta_k}{\partial M_i} \right)_{\text{rot}} < -\frac{\delta_k q_{k,i}^2}{2M_i}. \quad (129)$$

This is not a claim that inertia is harmful in general. It is a sign condition showing that, under non-proportional damping, inertia placement is not monotone in the damping margin.

Table XIV reports a reproducible random-graph audit. The fixed- D_i case is included as a control because it contains direct damping dilution. The fixed- D_i/M_i case is the nontrivial check: in the strongest five-node example, (128) predicts $\partial \zeta_k / \partial M_i = -0.0373$, the matched-mode finite difference gives -0.0397 , and the full-QEP damping-margin finite difference is also negative (-0.106). The larger full-QEP derivative is not a contradiction. Equation (128) is a diagonal matched-mode sensitivity, while the full-QEP margin also includes off-diagonal modal coupling and possible movement of the identity of the critical mode. In this example, the coupled system amplifies the same sign reversal detected by the diagonal formula. This supports the mechanism but remains a reduced-model result; it has not yet been validated on calibrated ANDES IBR dynamics.

E. Progressive Conversion

A practical transition is not a single conversion; it is a sequence. Let $\mathcal{C}$ be the set of synchronous generators still eligible for conversion. At step m , define

$$i_m = \arg \max_{i \in \mathcal{C}} \zeta_{\min}(M^{(m,i)}, D^{(m,i)}, L^{(m)}), \quad (130)$$

where $(M^{(m,i)}, D^{(m,i)}, L^{(m)})$ is the model after converting candidate i on top of the previous $m-1$ conversions. This greedy rule should be compared against a fixed merit order, random order, inertia order, and modal-participation order.

Progressive conversion is important because single-action ranking can overstate robustness. A bus that is safe to convert first may become unsafe after another conversion changes the critical mode. The benchmark should report the full margin trajectory versus IBR penetration, not only the first selected bus. If the greedy rule fails at a certain penetration level, that failure is a useful result: it identifies where local planning must be replaced by combinatorial or robust optimization.

VII. DAMPING-AWARE LINE REINFORCEMENT

The old line map is the Rayleigh sensitivity of the inertial-Laplacian eigenvalue:

$$\frac{\partial \nu_k}{\partial w_{ij}} = \left(\frac{q_{k,i}}{\sqrt{M_i}} - \frac{q_{k,j}}{\sqrt{M_j}} \right)^2. \quad (131)$$

This formula is useful, but it answers the wrong planning question if used alone. It says which line raises modal stiffness or frequency. It does not say which line raises damping margin. Equivalently, the unnormalized Laplacian perturbation is $\partial L / \partial w_{ij} = (e_i - e_j)(e_i - e_j)^T$, while the perturbation used in modal formulas is the inertia-normalized matrix

$$\frac{\partial \tilde{L}}{\partial w_{ij}} = M^{-1/2} (e_i - e_j)(e_i - e_j)^T M^{-1/2}. \quad (132)$$

The reason is immediate from the diagonal formula:

$$\zeta_k = \frac{\delta_k}{2\sqrt{\nu_k}}. \quad (133)$$

Differentiating with respect to a line weight w_e gives

$$\frac{\partial \zeta_k}{\partial w_e} = \frac{1}{2\sqrt{\nu_k}} \frac{\partial \delta_k}{\partial w_e} - \frac{\delta_k}{4\nu_k^{3/2}} \frac{\partial \nu_k}{\partial w_e}. \quad (134)$$

The second term is negative when reinforcement increases ν_k . Thus, a line can raise frequency while reducing damping ratio if the first term does not compensate. Fig. 7 illustrates the same mechanism in pole coordinates: increasing $|\omega|$ while holding $-\sigma$ fixed lowers $\zeta = -\sigma / \sqrt{\sigma^2 + \omega^2}$.

For a simple eigenvalue of $\tilde{L}$, the derivative of modal damping is

$$\frac{\partial \delta_k}{\partial w_e} = q_k^T \frac{\partial \tilde{D}}{\partial w_e} q_k + 2 \sum_{\ell \neq k} \frac{(q_\ell^T \frac{\partial \tilde{L}}{\partial w_e} q_k)(q_\ell^T \tilde{D} q_k)}{\nu_k - \nu_\ell}. \quad (135)$$

If line reinforcement affects only L , the first term is zero, but the second term can be nonzero because the eigenvectors rotate relative to the damping matrix. This is the precise mechanism by which non-proportional damping changes line-ranking decisions.

For the exact QEP, let s be a simple critical root with right and left eigenvectors v and y :

$$P(s)v = 0, \quad y^* P(s) = 0. \quad (136)$$

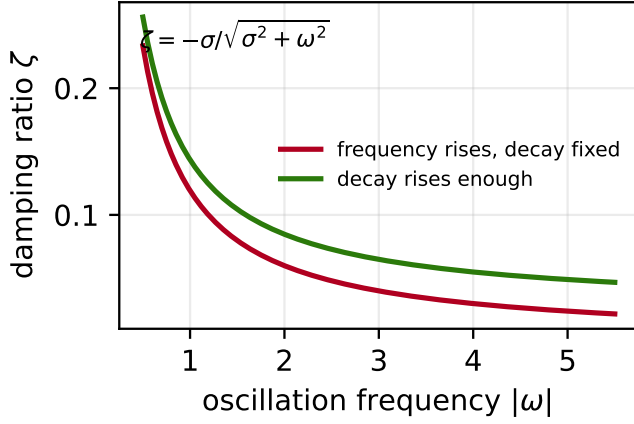


Fig. 7. Frequency–damping tradeoff. If a line reinforcement increases oscillation frequency without increasing decay rate enough, damping ratio decreases. This is not a topological Braess paradox; it is the definition of damping ratio.

The normalization of v and y is arbitrary because it cancels between numerator and denominator in the sensitivity expression; the only requirement is $y^*P'(s)v \neq 0$ for a simple root. For a scalar parameter ρ ,

$$\frac{ds}{d\rho} = -\frac{y^* \left(s \frac{\partial \tilde{D}}{\partial \rho} + \frac{\partial \tilde{L}}{\partial \rho} \right) v}{y^* (2sI + \tilde{D})v}, \quad (137)$$

assuming the coefficient of s^2 is fixed in the chosen coordinates. If $s = \sigma + j\omega$ and $ds/d\rho = a + jb$, then

$$\frac{d\zeta}{d\rho} = \frac{\sigma\omega b - \omega^2 a}{|s|^3}. \quad (138)$$

Equations (137)–(138) define the damping-aware line map. The frequency map (131) remains useful for mechanism interpretation, but the ranking variable is $\partial\zeta_{\min}/\partial w_e$ or a finite post-action $\Delta\zeta_{\min}$.

A. Finite-Difference Validation

Before using (137) operationally, the analytic sensitivity should be checked against finite differences. For edge $e = (i, j)$ with base weight w_e , define

$$\Delta_e^{\text{FD}}(\epsilon) = \frac{\zeta_{\min}(w_e(1 + \epsilon)) - \zeta_{\min}(w_e)}{\epsilon w_e}. \quad (139)$$

The perturbation should preserve Laplacian row sums:

$$\Delta L_e = \epsilon w_e (e_i - e_j)(e_i - e_j)^T. \quad (140)$$

The analytic sensitivity should converge to Δ_e^{FD} as ϵ decreases until roundoff dominates. If the identity of the critical eigenvalue changes during the perturbation, the line should be flagged as a nonsmooth ranking point rather than forced into a smooth derivative.

B. Why Frequency-Only Ranking Can Fail

The frequency map (131) is nonnegative in the scalar lossless model:

$$\frac{\partial \nu_k}{\partial w_e} \geq 0. \quad (141)$$

Thus reinforcement increases or preserves modal stiffness. The mistake is to equate stiffness with damping margin. For a lightly damped pole, $\omega \approx \sqrt{\nu_k}$ while σ is set by damping and modal rotation. If ω increases while σ is fixed, ζ decreases. If eigenvector rotation moves the mode toward lower-damped buses, σ can also become less negative. The corrected map therefore decomposes reinforcement into a frequency term and a modal-damping-rotation term.

VIII. BENCHMARK VALIDATION PROTOCOL

The decisive validation target is the IEEE 39-bus New England system in ANDES [20]. A final submission should not rely only on random graphs or low-order examples. The intended validation pipeline is as follows.

A. System Setup

The base case should be initialized with a solved power flow. Synchronous machines should use documented dynamic models available in ANDES, such as GENROU or the closest supported equivalent. IBR penetration should be introduced by replacing a fraction ρ of synchronous capacity with documented inverter models. Grid-following replacements should include PLL dynamics where available. Grid-forming replacements should use droop or virtual-synchronous-machine models where available. Controller gains should be documented and should not be hand-tuned to make the proposed method look better.

B. Experiments

Experiment A compares margin estimators. For each penetration level ρ , compute the full small-signal poles using ANDES, then compare the homogeneous formula, Fiedler-only formula, diagonal all-mode estimate, and adaptive reduced QEP.

Experiment B tests the coupling diagnostic. The expected outcome is not that the diagonal estimate always fails or always succeeds. The expected outcome is that the diagnostic predicts when it can be trusted.

Experiment C ranks SG-to-IBR conversions. Each candidate conversion is evaluated by the adaptive estimator and by full eigensolve. Baselines include Fiedler participation, highest-frequency participation, and inertia-only heuristics.

Experiment D tests multi-node conversion. Greedy conversion by post-action damping margin is compared against random conversion and heuristic merit orders. Failures must be reported, especially if interactions make the greedy rule unreliable.

Experiment E evaluates line reinforcement. Rank lines by $\partial\nu/\partial w$, finite-difference $\Delta\zeta_{\min}$, and analytic QEP sensitivity. The key test is whether frequency-only reinforcement disagrees with damping-aware reinforcement and whether the analytic sensitivity predicts the full eigensolve.

TABLE XV
PRE-REGISTERED ANDES VALIDATION MATRIX.

Experiment	Metric
Estimator accuracy	Median and 95th-percentile error vs full eigensolve
Coupling trigger	Precision/recall of diagonal-failure detection
Conversion ranking	Top-1 and top-3 agreement with full eigensolve
Multi-conversion	Margin trajectory vs penetration level
Line sensitivity	Correlation and top- k overlap with finite differences
Surrogate fidelity	Scalar model poles vs full ANDES IBR poles

C. Extracting Reduced Quantities From ANDES

The ANDES bridge must be documented carefully. The full ANDES state matrix contains device states, algebraic variables, controllers, and possibly load or network dynamics. The reduced spectral model requires L , M , and D or their best physically justified equivalents.

The extraction should proceed in two levels. Level 1 constructs the second-order surrogate. The active-power Jacobian with respect to bus angles gives the stiffness matrix corresponding to L under the assumptions of (2). Synchronous-machine inertia and virtual inertia terms define M_i . Mechanical damping, droop, and equivalent local damping define D_i only when the mapping is physically meaningful. Level 2 compares that surrogate against the full ANDES linearization with IBR controller states. Failure at Level 2 would not invalidate the mathematics of the estimator; it would delimit the regime where a scalar L, M, D surrogate is insufficient.

D. Success Criteria

The method should be judged by pre-registered criteria:

- median and 95th-percentile error of margin estimates,
- fraction of cases where the adaptive estimator identifies the correct critical mode or cluster,
- top- k accuracy of conversion ranking,
- agreement between analytic and finite-difference line sensitivities,
- failure cases and their mechanisms.

If the adaptive estimator fails at high IBR penetration, that result should be included. The value of the paper is a calibrated method, not a universal claim.

E. Decision Gates for Claim Strength

The final benchmark should assign each contribution a claim level before submission. The highest level is an empirical contribution: the method is validated against full ANDES poles over the stated operating range and beats the registered baselines. The middle level is a theoretical or diagnostic contribution: the mathematics is correct and the mechanism is visible, but benchmark advantage is narrow or case-dependent. The lowest level is a cautionary result: the proposed idea identifies a failure mode of simpler reasoning, but it is not strong enough to headline the paper.

TABLE XVI
CLAIM LEVELS USED AFTER BENCHMARK VALIDATION.

Level	Meaning
Empirical contribution	Verified against full benchmark poles and stronger than registered baselines
Diagnostic contribution	Mathematically correct and mechanism-clear, but benchmark advantage is case-dependent
Cautionary result	Useful for avoiding a wrong inference, but not a standalone planning method
Rejected claim	Control comparison or mechanism check fails

For C1, the empirical level requires that the modal points in the (ν, δ) plane predict safe and unsafe cases in the commuting or weakly coupled regime, and that the same plot visually explains at least one non-proportional failure. If the plot is only exact in toy proportional cases, C1 should be framed as theory background plus visualization, not as a benchmark contribution.

For C2, the empirical level requires that the adaptive estimator reduce error or computational burden relative to full eigensolve and relative to the diagonal screen in the cases where the coupling diagnostic fires. If the diagonal all-mode formula is already accurate on every IEEE 39 case, the reduced QEP should be demoted to a safeguard for degenerate or artificially clustered modes. This would still be a valid result, but it would change the title and contribution ordering.

For C3, the empirical level requires low regret, not only high correlation. A conversion ranking that is usually correlated with the full eigensolve but occasionally chooses an action with a much lower margin is not operationally acceptable. The correct fallback claim would be that the estimator provides a screening list for full eigensolve, not a standalone planner.

For C4, the empirical level requires two checks. First, analytic sensitivity must match finite differences when the critical pole is smooth. Second, damping-aware ranking must differ materially from frequency-only ranking in at least some realistic operating points. If both rankings agree on IEEE 39, the result should be stated as a theoretical correction that prevents a known category error rather than as evidence of a frequent planning reversal.

F. Negative Controls

The benchmark should include controls that are expected to fail. These controls are not straw men; they are the reason the proposed method is needed. A Fiedler-only estimate should fail whenever the damping-critical mode is not the lowest nonzero graph frequency. A homogeneous damping formula should fail when IBR conversion creates large variation in D_i/M_i . A frequency-only line map should fail whenever reinforcement raises ν_k while modal damping does not rise enough to preserve ζ_k . A low-frequency QEP should fail when the least damped mode is outside the retained frequency band.

Reporting these controls prevents a common source of false positives. If the proposed estimator beats only an obviously wrong control but not the diagonal all-mode screen, then the central claim is not the reduced QEP. If the damping-aware line map beats frequency sensitivity but not finite-difference

full eigensolve, then the analytic implementation is wrong or the pole is nonsmooth. Each negative control therefore has a paired diagnostic that identifies whether the failure is caused by the baseline assumption, the proposed approximation, or the reduced surrogate itself.

IX. CLAIM CALIBRATION AND EVIDENCE MAP

The proposed paper has to survive a strict novelty and mechanism audit. This section states how each contribution should be defended and what evidence would be required in a final submission.

A. C1 Is a Region Framework, Not a New Scalar Formula

The formula $\zeta_k = \delta_k / (2\sqrt{\nu_k})$ in the commuting case is a standard modal result. The contribution is not the formula. The contribution is the use of the (ν, δ) plane as a planning coordinate for heterogeneous IBR damping, together with the statement that this coordinate is exact only in the commuting case and diagnostic otherwise. The consensus-region analogy is useful because it changes the question from “which graph eigenvalue is critical?” to “which modal points lie outside the admissible damping region?” This avoids the common mistake of treating λ_2 as a damping margin.

Fig. 1 therefore includes two visual checks. The commuting case shows where the region is an exact margin certificate. The non-proportional panel shows the failure mode: a diagonal modal point can lie above the target curve while the coupled QEP pole falls below the margin. This makes C1 a boundary-aware framework rather than a standalone novelty claim.

B. C2 Is a Mode-Selection Rule, Not Generic Model Reduction

Reduced QEPs are well known. The estimator’s claim is the selection logic. Low-frequency truncation is natural in inter-area oscillation analysis, but the damping margin is the minimum damping ratio over all oscillatory modes. A high-frequency mode can be less damped than a Fiedler mode if δ_k is small enough. Conversely, a low-frequency mode can be safe if its modal damping is large. The reduced model should therefore be built from low estimated damping and strong modal coupling, not from frequency alone.

The evidence needed for this claim is an ablation table. Let ζ_{full} be the reference. Compare:

$$e_{\text{hom}} = |\zeta_{\text{hom}} - \zeta_{\text{full}}| / \zeta_{\text{full}}, \quad (142)$$

$$e_{\text{Fiedler}} = |\zeta_2 - \zeta_{\text{full}}| / \zeta_{\text{full}}, \quad (143)$$

$$e_{\text{freqQEP}} = |\zeta_{\text{lowfreqQEP}} - \zeta_{\text{full}}| / \zeta_{\text{full}}, \quad (144)$$

$$e_{\text{adapt}} = |\zeta_{\text{adapt}} - \zeta_{\text{full}}| / \zeta_{\text{full}}. \quad (145)$$

The adaptive estimator is justified only if e_{adapt} improves the cases where simple baselines fail and if the diagnostic explains the improvement.

Algorithm 3 Damping-aware line ranking

```

1: Input:  $L, M, D$ , edge set  $\mathcal{E}$ , perturbation size  $\epsilon$ 
2: Compute  $\zeta_{\min}$  and the critical pole  $s$ 
3: for each edge  $e = (i, j) \in \mathcal{E}$  do
4:   Build  $\partial L / \partial w_e = (e_i - e_j)(e_i - e_j)^T$ 
5:   Evaluate analytic  $ds/dw_e$  from (137)
6:   Convert  $ds/dw_e$  to  $d\zeta/dw_e$  using (138)
7:   Optionally compute finite-difference  $\Delta_e^{\text{FD}}(\epsilon)$ 
8: end for
9: Rank edges by  $d\zeta_{\min}/dw_e$  or finite  $\Delta\zeta_{\min}$ 
10: Report  $\partial\nu/\partial w_e$  only as a mechanism diagnostic

```

C. C3 Is a Planning Rule With a Full-Eigensolve Reference

The conversion and placement contribution should be judged by ranking accuracy, not only by correlation. A high correlation can hide top-action mistakes. Let $r_{\text{est}}(i)$ and $r_{\text{full}}(i)$ be the estimated and true ranks of candidate action i . Useful metrics include top-1 hit rate, top-3 overlap, Kendall rank correlation, and regret:

$$\text{regret} = \zeta_{\min}(a_{\text{full}}^*) - \zeta_{\min}(a_{\text{est}}^*). \quad (146)$$

Regret is especially important for planning. If several actions are nearly tied, a wrong top-1 selection may have negligible regret. If the wrong action collapses the margin, the ranking method is not acceptable even with high average correlation.

D. C4 Is a Damping Sensitivity, Not a Braess Claim

The line-reinforcement contribution is easiest to overstate. The paper should not say that reinforcement creates a new dynamic Braess paradox unless a topological feedback mechanism is demonstrated. The defensible claim is simpler and stronger: frequency-only line ranking can be wrong because damping ratio depends on both decay and frequency. The exact QEP sensitivity gives the derivative of the pole, and (138) converts it to a damping-ratio derivative.

The NEP reinforcement audit refines this boundary. Most observed reinforcement reversals remain geometric or non-resonant damping-ratio effects, but a minority satisfy a stricter network-control resonance test: reinforcement moves a network-family mode toward a PLL-dominated condensed-control pole and the self-energy term dominates the damping decrease. This is a confirmed mechanism in the Mix60/no-PSS audits, not a universal topology law and not the dominant source of global margin loss.

The final line-reinforcement experiment should report three lists for each operating point: top lines by $\partial\nu/\partial w$, top lines by analytic $\partial\zeta/\partial w$, and top lines by finite-difference $\Delta\zeta$. A disagreement between the first and the latter two is the evidence for C4. Agreement between analytic sensitivity and finite difference is the evidence that the formula is implemented correctly.

X. DISCUSSION

The proposed framework has three important limits. First, the scalar swing-equation Laplacian is not a full dq-domain inverter-network model. It captures the electromechanical and

TABLE XVII
CONTRIBUTION STATUS AND REQUIRED VALIDATION.

Component	What is established	What is not established	Required next check
Damping region	Exact for commuting damping; first-order screen for weak non-proportional damping	Novelty of the region language for IBR planning	Literature search and IEEE 39 modal plots
Adaptive estimator	Synthetic tests separate diagonal-safe and QEP-needed regimes	Universal diagnostic threshold	ANDES full eigensolve comparison
Conversion ranking	Planning score is well-defined by post-action $\zeta_{\min}$	Multi-node interaction robustness	Greedy conversion sweep on IEEE 39
Damping-aware line map	Mechanism and QEP sensitivity formulas are derived	Analytic implementation vs finite differences	Line perturbation sensitivity in ANDES

Algorithm 4 Greedy SG-to-IBR conversion screening

- 1: Input: base model (L, M, D) , candidates $\mathcal{C}$, target penetration
- 2: Initialize converted set $\mathcal{S} = \emptyset$
- 3: **while** penetration target not reached **do**
- 4: **for** each $i \in \mathcal{C} \setminus \mathcal{S}$ **do**
- 5: Apply candidate conversion i to obtain (L_i, M_i, D_i)
- 6: Estimate $\hat{\zeta}_i$ with Algorithm 1
- 7: **end for**
- 8: Convert $i^* = \arg \max_i \hat{\zeta}_i$
- 9: Append i^* to $\mathcal{S}$ and recompute the operating point if required
- 10: **end while**
- 11: Validate the selected sequence against the full eigensolve

TABLE XVIII
EVIDENCE REQUIRED BEFORE FINAL SUBMISSION.

Claim	Required evidence
C1 region	Modal plots showing safe/unsafe boundary and failures under coupling
C2 estimator	Ablation against homogeneous, Fiedler, frequency-QEP, and full eigensolve
C3 planning	Top- k ranking, regret, and multi-conversion trajectory
C4 line map	Analytic-vs-finite-difference sensitivity and disagreement with frequency-only ranking

stiffness geometry used by many spectral power-network studies, but high-order inverter states, current loops, limiters, and dynamic line models can change the relevant modes. The ANDES validation stage is therefore not optional.

Second, the second-order correction is not a universal replacement for a reduced or full eigensolve. It improves the diagonal screen in weak-to-moderate non-proportional regimes and gives an interpretable coupling decomposition, but it can worsen tail errors in strongly coupled or near-degenerate cases. Those cases are exactly why the reduced QEP remains in the workflow.

Third, the line-reinforcement result separates two mechanisms that should not be conflated. If reinforcement raises frequency and lowers damping ratio because decay does not increase enough, the mechanism is ordinary damping-ratio geometry. The Schur-NEP audit confirms a second, narrower mechanism: a network-family mode can be pulled toward

a PLL-dominated condensed-control pole, so the self-energy term drives the damping decrease. This resonant mechanism is control-conditional, appears in 40/348 global-margin-reversal audit records in the Mix60/no-PSS audit, and is absent under the soft-GFL negative control. The paper should therefore say that a control-resonant reinforcement reversal exists and is actionable, not that it dominates line-reinforcement planning or collapses the global margin in this benchmark.

The H-infinity disturbance view and switched-topology Lyapunov view are valuable extensions, but they should not be central in this manuscript. The H-infinity view asks which disturbance shapes excite the reduced system. It is complementary to damping margin but risks reviving poorly defined “weak pathway” claims if not controlled carefully. The switched-topology view requires common or multiple Lyapunov functions and dwell-time assumptions; it is a larger thesis direction rather than a clean first paper.

A. Originality Boundaries

The paper’s novelty boundary should be stated plainly. Modal swing-equation formulas under proportional damping, Rayleigh eigenvalue sensitivity, QEP sensitivity, Schur complements, contour NEP solvers, broad stability-region ideas, and eigenvalue perturbation are known. The candidate contribution is not new mathematics in isolation. It is a power-systems contribution: identify the controller-aware rational self-energy as the damping-margin bridge when static nodal damping fails, use it to separate network-family from control-family criticality, and then use second-order perturbation as an interpretable estimator component whose full-ANDES advantage must still be tested against registered baselines.

This boundary matters for peer review. A reviewer can object if the paper appears to claim that $\zeta_k = \delta_k / (2\sqrt{\nu_k})$ or generic QEP sensitivity is new. The defensible claim is that the paper uses the diagonal relation as the zero-order network margin, derives the second-order control-coupling correction for the critical margin pole, and shows when this cheap correction should yield to a reduced QEP. Originality should be judged by the formulation, the control comparisons, and the benchmark results, not by whether every mathematical identity is new.

In final form, each contribution should therefore be labeled as follows. The unifying framework is the spectral-attribution ledger (1): network geometry enters through $\bar{L}$, inertia through M , and controller dynamics through the rational self-energy

$\Sigma(s)$. The strongest full-ANDES empirical contribution is the controller-aware rational NEP bridge, which reproduces the critical poles and enables mechanism audits that a static scalar bridge cannot perform. The central reduced-model mathematical contribution is the second-order margin correction and rational graph-filter interpretation, verified first in the surrogate QEP and reduced PLL model. The control-resonant reinforcement reversal is a bridge-enabled mechanism, but its current evidence is modal-local and control-conditional. The inertia sign reversal is a reduced-model sign result. The joint inertia-control cross term is a valid local perturbation term, but the first Phase-E0 ANDES gate found no measurable singleton GENROU-PLL case; it must not be treated as an empirical headline unless a physically justified virtual-inertia/control case passes a new gate. Planning and line-reinforcement rankings are applications; they become headline contributions only if ANDES validation shows lower regret or materially different rankings than registered controls.

B. Falsification Criteria

Three outcomes would weaken the paper and should be reported if observed. First, if the second-order correction is no more accurate than the diagonal all-mode formula on IEEE 39 across relevant IBR penetrations, it should be presented only as an interpretive decomposition. Second, if the reduced PLL or scalar L, M, D surrogate does not track full ANDES IBR poles, the contribution should be reframed as a reduced-model screening method rather than a full IBR stability estimator. Third, if damping-aware line ranking does not differ materially from frequency-only ranking on the benchmark, C4 should be reduced to a theoretical caution rather than a headline application.

These criteria make the manuscript stronger. They state before the benchmark what evidence would consolidate, weaken, or refocus the claims.

C. Final Claim-Readiness Audit

The current repository includes a final claim audit generated from the versioned validation outputs. Table XIX summarizes the status at the close of this draft. The central mathematical and reduced-model claims are supported, and the controller-aware full-ANDES bridge is now validated on three diagnostic cases. The estimator and planning claims over that bridge are not yet supported. This distinction is the intended submission boundary of the manuscript.

XI. CONCLUSION

This paper proposed a controller-aware rational self-energy bridge for damping-margin analysis in inverter-dominated grids. The exact Schur NEP retains the frequency-dependent effect of condensed controller and auxiliary states, while the lower-order spectral model organizes the retained network dynamics through (ν_k, δ_k) : inertial-Laplacian stiffness and modal damping. In non-proportional IBR regimes, off-diagonal modal damping moves the critical pole away from

this diagonal network prediction. The closed-form second-order correction quantifies that movement and gives a network-vs-control decomposition of the damping margin.

The same framework yields planning tools, but those tools are secondary until estimator validation is completed over the controller-aware bridge. SG-to-IBR conversion and damping or inertia placement can be ranked by post-action damping margin rather than by Fiedler participation or connectivity. Line reinforcement should be ranked by damping-margin sensitivity, not by frequency sensitivity alone. The hidden control-resonance margin extends this logic: weak nodes and weak links are elements whose local actions can move poles toward a damping threshold or toward a condensed-control resonance. The resolvent bound explains why this margin is hidden from static scalar reductions: the sensitivity can grow as the network-family pole approaches $\text{spec}(A_{cc})$. The instability-flow construction makes this statement operational by decomposing weak elements into modal participation, control-pole resonance exposure, and first-order damping loss. The protected-resolvent extension adds a second certificate: instead of tracking only pole damping, it tracks protected input-output amplification and reveals when forcing and response are spatially decoupled in a non-normal IBR model. This gives a principled way to compare topology, damping, inertia, and controller-retuning actions in one local exchange-rate language, but current evidence is reduced-model and must not be overstated as a full-ANDES flow-feasible planner. In matrix-geometric terms, admissible topology and control actions generate a pole-motion cone, and local graph reconfiguration becomes a cost-constrained problem over damping-margin, protected-gain, flow, and control-distance constraints. The mode-resolved substitutability spectrum further refines weak-node language: a bus is not simply weak or strong, because it may be copper-substitutable for one mode and equipment-requiring for another. This gives a possible damping-aware hosting-capacity interpretation for SG-to-IBR conversion, but it remains a proposed application bridge until novelty and full-ANDES validation gates are closed. These applications require explicit comparison against existing damping-ratio placement, QEP-sensitivity, impedance/SCR, hosting-capacity, and topology-planning methods before they can become headline claims.

The paper is therefore complete at a stronger methodological-draft level than before: the derivation, reduced-model evidence, IEEE39-derived surrogate checks, ANDES baseline diagnosis, SG-to-GFL/GFM case construction, Phase-1 calibration audit, and Phase-0D controller-aware rational NEP bridge are all documented. The bridge result is the key full-ANDES advance: Beyn contour integration reproduces the full ANDES critical poles in no-PSS, base, and Mix60 cases, including the Mix60 PLL-family critical mode that scalar and fixed-point bridges miss. It also enables a control-resonant reinforcement audit: a minority of Mix60/no-PSS line-reinforcement reversals are driven by controller self-energy as network modes move toward PLL-dominated condensed-control poles, while the mechanism disappears under the soft-GFL negative control. This is a confirmed modal-local mechanism, not a global-

TABLE XIX
FINAL CLAIM-READINESS AUDIT FOR THE CURRENT DRAFT.

Claim	Verified evidence	Control or comparison	Current status
Damping region	Exact in commuting damping; paired figure shows non-proportional screen failure.	Commuting diagonal modal equations versus coupled QEP behavior.	Theoretical/diagnostic; not a standalone novelty claim.
Second-order correction	IEEE39-derived surrogate median error improves from 0.468% to 0.032%; reduced PLL model improves median error from 2.16% to 0.09%.	Full surrogate QEP, diagonal all-mode screen, and reduced-QEP fallback.	Verified on reduced models; full ANDES IBR validation pending.
Adaptive workflow	Surrogate tails show why fallback is needed: second-order p95 9.91%, reduced-QEP p95 5.44%, adaptive p95 7.43%.	Always-diagonal, always-second-order, and reduced-QEP baselines.	Diagnostic workflow; no universal trigger theorem claimed.
Controller-aware rational bridge	NEP-Beyn contour solver reproduces full ANDES critical poles in no-PSS, base, and Mix60, including a 13.7 Hz PLL-family critical mode with $\pi_c = 0.853$.	Full ANDES eigensolve; argument-principle counts match extracted contour zeros.	Bridge verified for diagnostic cases; estimator accuracy experiments next.
SG-to-IBR conversion	Five derived ANDES IEEE39 cases load, solve power flow, and run eigenanalysis with active GFL/GFM devices; Phase-1 tested 50 case/profile controller audits.	Device inventory, power-flow/eigenanalysis success, controller-only profiles, and diagnostic removal of retained SG controls.	Case-construction and calibration audit; final conversion-ranking validation over the NEP bridge remains pending.
Mode-resolved lever substitutability	Line-shunt decomposition and virtual-inertia support result are algebraic; project ledger reports a 60-graph structural audit where node rankings depend on the mode and scalar collapse loses information in about 98% of graphs.	Compared against scalar weak-node ranking and single-critical-mode projection.	Structural planning diagnostic; algebra is classical, novelty is the node-mode copper-versus-equipment taxonomy and it still needs adversarial novelty search plus full-ANDES validation.
Damping-aware hosting capacity	Formulated as a composite-margin feasibility problem over SG-to-IBR conversion level, repair action, budget, and hidden phase-damping-control margin.	Distinguishes thermal/voltage hosting capacity from damping-margin hosting and requires comparison against Fiedler, inertia-only, and damping-index baselines.	Proposed application bridge; not yet an empirical hosting-capacity claim.
Weak links and reinforcement	In the IEEE39-derived surrogate, 35/46 reinforcements raise critical modal stiffness while reducing damping margin.	Frequency-only reinforcement versus finite post-action damping margin.	Surrogate planning diagnostic; not yet a full ANDES topology claim.
Control-resonant reinforcement reversal	Schur-NEP sensitivity audit identifies 40/348 global-margin-reversal audit records as strict resonant cases in Mix60/no-PSS; the largest global case is Line 44 (23-36) with $\Delta\zeta_{\min} = -2.75 \times 10^{-5}$ and C-term dominance.	Soft-GFL negative control gives zero resonant cases; for matching line/mode pairs, relative pole distance increases from 0.0368 to 0.9167 and the control denominator fraction falls from 0.959 to 0.0789.	Mechanism confirmed, control-conditional, minority of reversals, and modal-local in this benchmark; not a dominant global-margin-collapse claim.
Hidden control-resonance margin	Defined by normalized distance from a tracked NEP zero to condensed-control poles, the self-energy denominator fraction, the resolvent bound (52), and the combined damping/distance action margin (57).	Uses the same NEP sensitivity and soft-GFL negative control logic as the resonant reinforcement audit.	New diagnostic and repair formulation; not yet validated as a system-level collapse predictor or planning certificate.
Instability-flow weak elements	Derived from NEP pole sensitivity, control-pole resolvent expansion, graph/modal line participation, and the local weak-element certificate in Proposition 4.	Must be compared against power-flow loading, betweenness, Fiedler participation, SCR/impedance, and damping-participation rankings.	Mathematically local first-order diagnostic; superiority as a practical weak-node/weak-link predictor remains an empirical claim.
Protected-resolvent graph-filter shaping	The bi-orthogonal sensitivity (83) gives a common local exchange rate for topology, damping, inertia, and controller retuning; line and damping specializations are (87) and (89).	Gamma robustness gate: physical protection profiles give only 21–29% agreement between S_ζ and S_g rankings; reduced coordination gate supports $w + D$ and $w + M$ packages but kills strong nonlinear synergy as a headline.	Valid input-output extension and reduced-model planning candidate; flow-feasible dynamic Braess and retrofit LP still require full ANDES/dispatch validation.
Local action polytope	Feasible topology, shunt, inertia, damping, and controller changes can be represented in line-shunt coordinates; local first-order repair reduces to a linearized problem over a polytope.	Explicitly controlled against the false stronger claim that $\zeta_{\min}$ or m_H is globally convex or smooth.	Geometry language only; safe as local screening, not a global optimization theorem.
Inertia sign reversal	With fixed D_i/M_i , 5447/16229 perturbations reduce both the diagonal modal screen and the full-QEP damping margin.	Fixed- D_i/M_i check versus fixed- D_i direct damping-dilution control.	Reduced-model sign diagnostic; not yet a full ANDES IBR claim.
Joint inertia-damping term	Local perturbation formula shows the cross term between $E^{(D)}$ and $E^{(L)}$; initial Phase-E0 ANDES tests found 0/13 measurable singleton GENROU-PLL cross cases in Mix60/no-PSS.	Reduces to the fixed-inertia second-order correction when $\Delta M = 0$; E0 compares additive versus additive-plus-cross predictions.	Theoretical local mechanism; not yet an empirical headline or finite-conversion estimator.

margin-collapse claim. The manuscript is still not complete as a final IEEE Transactions empirical submission. The next version must run the estimator and planning experiments over the validated bridge with registered controls. Until that validation is complete, the estimator and planning claims should be read as a mathematically grounded framework verified on surrogate QEP and reduced PLL models, plus a now-validated full-ANDES bridge and one bridge-enabled mechanism audit, not as final practical-grid planning evidence.

APPENDIX A STEP-BY-STEP DERIVATION LEDGER

This appendix records the algebraic chain behind the main formulas. It is included to make clear which results are exact, which are perturbative, and which are only diagnostic decompositions.

A. From the Swing Model to the Modal QEP

Start from the homogeneous form of (2),

$$M\ddot{\theta} + D\dot{\theta} + L\theta = 0. \quad (147)$$

Use the inertia-normalized coordinate $\vartheta = M^{1/2}\theta$, so that $\theta = M^{-1/2}\vartheta$. Since M is constant,

$$\dot{\theta} = M^{-1/2}\dot{\vartheta}, \quad \ddot{\theta} = M^{-1/2}\ddot{\vartheta}. \quad (148)$$

Substituting and premultiplying by $M^{-1/2}$ gives

$$\ddot{\vartheta} + M^{-1/2}DM^{-1/2}\dot{\vartheta} + M^{-1/2}LM^{-1/2}\vartheta = 0, \quad (149)$$

which is (4). Thus the normalized stiffness and damping operators are

$$\tilde{L} = M^{-1/2}LM^{-1/2}, \quad \tilde{D} = M^{-1/2}DM^{-1/2}. \quad (150)$$

Taking a modal trial solution $\vartheta(t) = ve^{st}$ gives

$$(s^2I + s\tilde{D} + \tilde{L})v = 0, \quad (151)$$

which is the QEP (8). This step is exact for the second-order surrogate.

B. Exact Diagonal Region

Let $Q^T\tilde{L}Q = \Lambda = \text{diag}(\nu_k)$ and $x = Q^T\vartheta$. If $\tilde{L}$ and $\tilde{D}$ commute on the nonzero-angle subspace, then $Q^T\tilde{D}Q = \Delta = \text{diag}(\delta_k)$. The modal equations decouple:

$$\ddot{x}_k + \delta_k\dot{x}_k + \nu_k x_k = 0. \quad (152)$$

The scalar characteristic polynomial is

$$p_k(s) = s^2 + \delta_k s + \nu_k. \quad (153)$$

For an underdamped root,

$$s_k = -\frac{\delta_k}{2} \pm j\sqrt{\nu_k - \frac{\delta_k^2}{4}}, \quad (154)$$

and its magnitude is $|s_k| = \sqrt{\nu_k}$. Therefore

$$\zeta_k = \frac{-\text{Re}(s_k)}{|s_k|} = \frac{\delta_k}{2\sqrt{\nu_k}}. \quad (155)$$

A target ζ_* is met if and only if

$$\delta_k \geq 2\zeta_*\sqrt{\nu_k} \quad (156)$$

for every nonzero mode. This proves the damping-region formula in the commuting case and also shows why it becomes only a screen when $Q^T\tilde{D}Q$ has off-diagonal terms.

C. Second-Order Critical-Pole Correction

Write the modal damping matrix as

$$\Gamma = Q^T\tilde{D}Q = \Delta + \epsilon E, \quad (157)$$

where $\Delta = \text{diag}(\delta_k)$, E has zero diagonal, and ϵ is a bookkeeping parameter set to one after the expansion. In modal coordinates, the QEP is

$$(s^2I + s\Delta + \Lambda + \epsilon sE)z = 0. \quad (158)$$

Choose the diagonal critical mode c and partition $z = (z_c, z_R)$. With

$$p_\ell(s) = s^2 + \delta_\ell s + \nu_\ell, \quad (159)$$

the partitioned equations are

$$p_c(s)z_c + \epsilon s E_{cR} z_R = 0, \quad (160)$$

$$P_R(s)z_R + \epsilon s E_{Rc} z_c + \epsilon s E_{RR} z_R = 0, \quad (161)$$

where $P_R(s) = \text{diag}(p_\ell(s))_{\ell \neq c}$. Since the desired correction is second order in E , the term $\epsilon s E_{RR} z_R$ in (161) contributes only third-order and higher terms to the scalar critical equation. Thus

$$z_R = -\epsilon s P_R(s)^{-1} E_{Rc} z_c + O(\epsilon^2). \quad (162)$$

Substituting into (160) gives the scalar Schur complement

$$p_c(s) - \epsilon^2 s^2 E_{cR} P_R(s)^{-1} E_{Rc} = O(\epsilon^3). \quad (163)$$

Let $s = s_0 + \Delta s_c$, where $p_c(s_0) = 0$ and $\Delta s_c = O(\epsilon^2)$. Taylor expansion gives

$$p_c(s_0 + \Delta s_c) = (2s_0 + \delta_c)\Delta s_c + O(\epsilon^4). \quad (164)$$

Evaluating the second-order Schur term in (163) at s_0 therefore yields

$$(2s_0 + \delta_c)\Delta s_c - \epsilon^2 s_0^2 E_{cR} P_R(s_0)^{-1} E_{Rc} = 0. \quad (165)$$

Setting $\epsilon = 1$ gives

$$\Delta s_c = \frac{s_0^2}{2s_0 + \delta_c} \sum_{\ell \neq c} \frac{E_{c\ell} E_{\ell c}}{s_0^2 + \delta_\ell s_0 + \nu_\ell}, \quad (166)$$

which is (23). The derivation also states the failure mode: if some denominator $p_\ell(s_0)$ is small, the perturbation is not safely separated and the reduced QEP should be used.

D. Rational Filter Identity

The previous result can be written as a filter identity by collecting the modal denominators into

$$H(\Lambda; s_0) = \text{diag}(p_\ell(s_0)^{-1})_{\ell \neq c}. \quad (167)$$

Then

$$E_{cR} P_R(s_0)^{-1} E_{Rc} = [E H(\Lambda; s_0) E]_{cc}, \quad (168)$$

with the critical row and column omitted in the middle operator. Hence

$$\Delta s_c = \frac{s_0^2}{2s_0 + \delta_c} [E H(\Lambda; s_0) E]_{cc}. \quad (169)$$

This is why the correction is called a rational graph filter of the pair $(\tilde{L}, \tilde{D})$. The filter is rational in the modal frequencies and pole s_0 , and it uses $\tilde{D}$ through both diagonal damping and off-diagonal coupling. A polynomial of $\tilde{L}$ alone cannot generate this term because it remains diagonal in the Q basis.

E. Schur-NEP Sensitivity and Damping-Ratio Derivative

For the full ANDES bridge, the reduced object is not a polynomial QEP but the Schur NEP (44). Let $s(w)$ be a simple zero with right and left null vectors x and y :

$$S_e(s(w), w)x(w) = 0, \quad y(w)^* S_e(s(w), w) = 0. \quad (170)$$

Differentiate with respect to a line parameter w :

$$\left(S_s \frac{ds}{dw} + S_w \right) x + S_e \frac{dx}{dw} = 0. \quad (171)$$

Premultiplying by y^* removes the eigenvector derivative because $y^* S_e = 0$:

$$y^* S_s x \frac{ds}{dw} + y^* S_w x = 0. \quad (172)$$

Therefore

$$\frac{ds}{dw} = -\frac{y^* S_w x}{y^* S_s x}, \quad (173)$$

which is (48). Since

$$S_e(s) = sI - A_{ee} - A_{ec}(sI - A_{cc})^{-1}A_{ce}, \quad (174)$$

and $d(sI - A_{cc})^{-1}/ds = -(sI - A_{cc})^{-2}$,

$$S_s = I + A_{ec}(sI - A_{cc})^{-2}A_{ce}. \quad (175)$$

The term with $(sI - A_{cc})^{-2}$ is the frequency-dependent control signature used in the control-resonant reinforcement audit. If s approaches an eigenvalue of A_{cc} , this term can dominate the denominator and amplify the effect of controller self-energy.

Finally, convert pole motion to damping-ratio motion. Let $s = a + jb$ and $r = |s|$. Since $\zeta = -a/r$,

$$d\zeta = -\frac{da}{r} + \frac{a}{r^2}dr, \quad dr = \frac{\text{Re}(\bar{s} ds)}{r}. \quad (176)$$

Thus

$$d\zeta = -\frac{\text{Re}(ds)}{|s|} + \frac{\text{Re}(s)}{|s|^3} \text{Re}(\bar{s} ds). \quad (177)$$

Every line-sensitivity number in the NEP audit follows this chain: perturb the line, compute ds/dw from the Schur NEP, and convert ds/dw to $d\zeta/dw$ using (177).

APPENDIX B

DERIVATION OF MODAL DAMPING SENSITIVITY

For a simple eigenpair of $\tilde{L}$, standard symmetric eigenvalue perturbation gives

$$\frac{\partial \nu_k}{\partial \rho} = q_k^T \frac{\partial \tilde{L}}{\partial \rho} q_k \quad (178)$$

and

$$\frac{\partial q_k}{\partial \rho} = \sum_{\ell \neq k} q_\ell \frac{q_\ell^T (\partial \tilde{L} / \partial \rho) q_k}{\nu_k - \nu_\ell}, \quad (179)$$

up to the usual normalization convention $q_k^T \partial q_k / \partial \rho = 0$. Since $\delta_k = q_k^T \tilde{D} q_k$,

$$\frac{\partial \delta_k}{\partial \rho} = q_k^T \frac{\partial \tilde{D}}{\partial \rho} q_k + 2 \sum_{\ell \neq k} \frac{(q_\ell^T (\partial \tilde{L} / \partial \rho) q_k)(q_\ell^T \tilde{D} q_k)}{\nu_k - \nu_\ell}. \quad (180)$$

Substitution into $\zeta_k = \delta_k / (2\sqrt{\nu_k})$ yields (134). The formula is ill-conditioned near repeated eigenvalues, which is precisely the case where the reduced QEP should replace the diagonal mode-by-mode sensitivity.

APPENDIX C

DERIVATION OF INERTIA SENSITIVITY

Let the inertia at bus i be the perturbation variable. Since

$$\tilde{L} = M^{-1/2} L M^{-1/2}, \quad \tilde{D} = M^{-1/2} D M^{-1/2}, \quad (181)$$

inertia enters both normalized operators. With $E_i = e_i e_i^T$,

$$\frac{\partial M^{-1/2}}{\partial M_i} = -\frac{1}{2} M_i^{-3/2} E_i, \quad (182)$$

and hence

$$\frac{\partial \tilde{L}}{\partial M_i} = -\frac{1}{2M_i} (E_i \tilde{L} + \tilde{L} E_i), \quad (183)$$

$$\frac{\partial \tilde{D}}{\partial M_i} = -\frac{1}{2M_i} (E_i \tilde{D} + \tilde{D} E_i), \quad (184)$$

when physical D is held fixed. For a simple eigenpair (ν_k, q_k) of $\tilde{L}$,

$$\frac{\partial \nu_k}{\partial M_i} = q_k^T \frac{\partial \tilde{L}}{\partial M_i} q_k = -\frac{\nu_k}{M_i} q_{k,i}^2. \quad (185)$$

The modal damping coefficient is $\delta_k = q_k^T \tilde{D} q_k$. Its derivative has a direct normalized-damping term and a modal-rotation term:

$$\begin{aligned} \frac{\partial \delta_k}{\partial M_i} &= q_k^T \frac{\partial \tilde{D}}{\partial M_i} q_k + 2 \sum_{\ell \neq k} \frac{(q_\ell^T (\partial \tilde{L} / \partial M_i) q_k) (q_\ell^T \tilde{D} q_k)}{\nu_k - \nu_\ell} \\ &= -\frac{q_{k,i}}{M_i} (\tilde{D} q_k)_i - \frac{1}{M_i} \sum_{\ell \neq k} \frac{(\nu_k + \nu_\ell) q_{\ell,i} q_{k,i}}{\nu_k - \nu_\ell} (q_\ell^T \tilde{D} q_k). \end{aligned} \quad (186)$$

Therefore,

$$\frac{\partial \zeta_k}{\partial M_i} = \frac{1}{2\sqrt{\nu_k}} \frac{\partial \delta_k}{\partial M_i} - \frac{\delta_k}{4\nu_k^{3/2}} \frac{\partial \nu_k}{\partial M_i}. \quad (187)$$

Substitution of (185) gives

$$\frac{\partial \zeta_k}{\partial M_i} = \frac{1}{2\sqrt{\nu_k}} \frac{\partial \delta_k}{\partial M_i} + \frac{\delta_k q_{k,i}^2}{4M_i \sqrt{\nu_k}}. \quad (188)$$

The last term is the frequency-normalization effect and is positive. The first term is sign-indefinite because inertia changes both normalized damping and modal orientation. If, instead, the local damping-per-inertia ratio D_i/M_i is held fixed, the direct term $q_k^T (\partial \tilde{D} / \partial M_i) q_k$ is removed and only the modal-rotation contribution remains. This is the nontrivial sign test used in Table XIV.

APPENDIX D

REPRODUCIBILITY CHECKLIST

Every figure and table in a final submission should record: random seed, system case, dynamic models, operating point, IBR penetration level, estimator settings, and git commit. Synthetic experiments should be separated from benchmark experiments. The final repository should include scripts that regenerate all figures from raw cases.

APPENDIX E

IMPLEMENTATION DETAILS FOR SYNTHETIC TESTS

The synthetic tests in this draft can be reproduced with the following structure. Generate a connected weighted graph by first sampling a random spanning tree and then adding edges with probability p . Draw inertias M_i and damping-per-inertia ratios D_i/M_i from log-uniform distributions. Build L , compute $\tilde{L}$ and $\tilde{D}$, and compare

$$\zeta_{\text{full}} = \min_{s_j(A)} -\text{Re}(s_j)/|s_j|, \quad (189)$$

$$\zeta_{\text{diag}} = \min_k \delta_k / (2\sqrt{\nu_k}), \quad (190)$$

$$\zeta_{\text{reqp}} = \min_{s_j(P_r)} -\text{Re}(s_j)/|s_j|. \quad (191)$$

The clustered test can be produced with a nearly complete graph with almost equal edge weights. This creates repeated or nearly repeated eigenvalues and exposes the instability of mode-by-mode diagonal reasoning.

For line sensitivity, use a perturbation $\epsilon \in [10^{-5}, 10^{-3}]$ and verify that the finite-difference value stabilizes over at least one decade of ϵ . If the critical mode switches, report the switch rather than averaging incompatible derivatives. For conversion ranking, log the full candidate list, not only the winner; top- k agreement is more informative than top-1 agreement when several actions have nearly equal margins.

APPENDIX F

LITERATURE SEARCH PROTOCOL

Before the word “novel” appears in a final submission, the exact claims must be searched. The search should not ask whether damping ratios, QEPs, or line sensitivities exist; they do. The search should ask whether the specific combination exists:

- damping-region or consensus-region geometry for non-proportionally damped power-grid QEPs,
- all-mode diagonal damping screen plus coupling-triggered reduced QEP,
- damping-selected QEP reduction for IBR planning,
- SG-to-IBR conversion ranking by reduced damping margin,
- damping-aware line reinforcement using $d\zeta_{\min}/dw_e$ rather than $d\nu/dw_e$.

The minimum search set should include IEEE Xplore, Google Scholar, arXiv, and recent proceedings in power systems and control. Useful query strings include:

- “non-proportional damping power network quadratic eigenvalue damping ratio”,
- “reduced QEP damping ratio inverter power system”,
- “spectral graph damping margin inverter based resources”,
- “power system damping ratio line sensitivity eigenvalue sensitivity”,
- “consensus region damping ratio power network Laplacian”.

Each close paper should be classified into one of three categories: direct overlap, adjacent method, or background. A direct-overlap paper would force a reframe from novelty

to validation/application. An adjacent-method paper should be cited and used to sharpen the distinction. A background paper supports the framing but does not reduce the contribution.

The final related-work section should include a claim table. For each claimed contribution, the table should identify the nearest prior art, the exact difference, and the evidence supplied in the manuscript. This prevents broad novelty claims and makes the contribution reviewer-checkable.

APPENDIX G

EXPECTED DATA PRODUCTS FOR SUBMISSION

A submission-ready package should include more than the PDF. It should include a reproducible data trail:

- raw case files for every network tested,
- scripts that generate L, M, D from each case,
- scripts that run the full eigensolve, diagonal estimate, and reduced QEP,
- scripts for conversion ranking and line-sensitivity ranking,
- CSV files for every figure and table,
- figure-generation scripts with fixed seeds,
- a manifest recording software versions and git commit.

For the IEEE 39-bus study, the primary tables should have the following fields. The estimator table should include penetration level, number of dynamic states, true critical pole, true $\zeta_{\min}$, diagonal estimate, reduced-QEP estimate, adaptive trigger, and relative error. The conversion table should include candidate bus, device type, post-conversion margin by full eigensolve, post-conversion margin by estimator, rank difference, and regret. The line table should include edge index, base weight, $\partial\nu/\partial w$, analytic $\partial\zeta/\partial w$, finite-difference $\Delta\zeta$, and whether the critical pole switched.

The final paper should not hide failed cases in supplementary material. If a high-IBR case violates the estimator assumptions, it belongs in the main discussion. The practical value of the method is knowing when a cheap spectral screen is reliable and when a full model is needed.

APPENDIX H

REVIEWER CHECKLIST FOR THE FINAL VERSION

The final version should be checked against four reviewer questions before submission. First, does each numerical result identify its control? For the estimator, the controls are the homogeneous formula, the Fiedler-only formula, the diagonal all-mode formula, a frequency-selected reduced QEP, and the full eigensolve. For planning, the controls are random selection, inertia-only selection, Fiedler participation, and frequency-only sensitivity. A result without one of these controls should not be described as evidence for superiority.

Second, does each positive result identify its mechanism? A large error reduction can occur because damping is low, because the model is nearly proportional, because the critical mode is accidentally included in the reduced subspace, or because the benchmark is too easy. The manuscript should state which mechanism is active. For example, if the diagonal estimator is accurate, the correct explanation may be weak damping and separated modes, not an inherently new spectral

law. If the reduced QEP improves the estimate, the explanation should be modal clustering or off-diagonal damping coupling, not generic model reduction.

Third, does each claimed contribution have an originality status? The labels should be conservative: verified and likely original after literature search, verified but known, or not yet verified. The formulas for proportional modal damping, Rayleigh line sensitivity, and QEP eigenvalue sensitivity are known tools. The candidate originality lies in their damping-region integration for IBR planning, the coupling-triggered estimator, and the damping-aware use of line sensitivity. If the literature search finds direct overlap, the paper should reframe the contribution as validation, synthesis, or application rather than novelty.

Fourth, does the paper report the negative cases? If the diagonal screen beats the reduced QEP on most benchmark cases, that is important. If damping-aware line ranking agrees with frequency ranking on IEEE 39, that is important. If the scalar surrogate fails to track the full IBR model at high penetration, that is important. These outcomes do not make the project useless; they define the validity region. A careful negative result is better than an exaggerated positive claim.

The recommended final-submission decision is therefore simple. Submit only after the IEEE 39-bus validation supplies at least one empirically supported headline claim. If C2 is strongest, title the paper around the adaptive estimator. If C4 is strongest, title it around damping-aware planning maps. If validation shows mostly boundaries and failure modes, target the paper as a rigorous methodological note or master-thesis chapter rather than a full Transactions claim.

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